

The Subtractive Cosmological Model

A Complete Development from First Principles to Observational Constraints

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Abstract

We present a complete account of the Subtractive Cosmological Model (SCM), a theoretical framework in which dark energy is an emergent consequence of the progressive depletion of a Bose–Einstein condensate (BEC) dark matter superfluid through baryonic sequestration, rather than a fundamental vacuum energy. The theory has been developed through three successive papers (SCM I, SCM II, SCM III) and an intermediate findings report, spanning the period from inception to the first full Markov Chain Monte Carlo (MCMC) observational analysis.

This document traces the complete intellectual trajectory: the original conception of the sequestration mechanism (SCM I), the rigorous numerical derivation of the sequestration-to-dark-energy coupling from first principles using the CAMB power spectrum and Sheth–Tormen halo mass function (Findings Report), the discovery and correction of a missing expansion-dilution factor that fundamentally altered the model’s predictions (SCM II), and the introduction of the BEC condensation threshold as a physical cutoff that regulates the otherwise catastrophic early dark energy growth (SCM III).

The observational results are developed through a five-step MCMC campaign (Steps 6–10) combining Planck 2018 CMB, DESI 2024 BAO, and Pantheon+ supernova data. Steps 6–9 identified and diagnosed a succession of convergence barriers: the (ε, z_c) near-flat direction, prior truncation, a silent brentq ceiling-clipping bug producing 34.5% posterior pileup, and a bimodal posterior revealing a physically disfavoured Mode B ($z_c > 3$, 46% DE fraction at $z = 3.6$). Step 10 resolved all pathologies by replacing ceiling-clipping with proper prior rejection, yielding the first genuinely converged posterior ($R - 1 = 0.027$, 33 185 post-burnin samples).

The final converged result constrains the FP dark energy fraction to $\varepsilon = 0.00159 \pm 0.00091$ (95th-percentile upper limit $\varepsilon < 0.0032$), with BEC condensation redshift $z_c = 2.43 \pm 0.38$, implying a BEC boson mass of $m_{\text{BEC}} = 22.9 \pm 1.6 \text{ meV}/c^2$. The equation of state is $w_0 = -0.570 \pm 0.357$, $w_a = -1.883 \pm 1.218$, crossing the phantom boundary at $z \lesssim 0.5$. The DESI 2024 Pantheon+ preferred point $(w_0, w_a) = (-0.827, -0.750)$ lies $\approx 0.4\sigma$ from the SCM best-fit. The ε upper limit is robustly consistent across all five steps at $\varepsilon < 0.004$ (95%).

Taken together, the SCM provides a physically motivated, internally consistent alternative to the cosmological constant, rooted in the microphysics of dark matter condensation, fully constrained by current cosmological data, and testable by next-generation experiments including DESI Year 5, CMB-S4, and SKA 21 cm tomography.

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Preface: Structure of This Document

This document provides a complete account of the Subtractive Cosmological Model from its original conception through the first observational MCMC analysis. It is structured as a coherent narrative covering the full intellectual development of the theory, including the false starts, the errors discovered, the corrections applied, and the refinements that led to the current formulation.

The document is organised into six parts:

- **Part I: Origins** — The cosmological motivation and the first SCM formulation (SCM I), including the BEC dark matter pressure mechanism, the sequestration fraction, and the first DESI comparison. This part also includes the Findings Report, which placed the theory on a rigorous numerical footing.
- **Part II: Refinement** — The discovery of the missing cosmological expansion factor and the subsequent SCM II analysis. This part documents the catastrophic failure of the FP formulation without a cutoff, and the identification of the path forward.
- **Part III: Extension** — The introduction of the BEC condensation threshold as a physical cutoff, completing the SCM III framework. This part derives the mass-redshift mapping and the hybrid dark energy density.
- **Part IV: Confronting Observations** — The full MCMC pipeline, chain configuration, diagnostics, and posterior results from the joint Planck 2018 + DESI 2024 + Pantheon+ analysis.
- **Part V: Physical Interpretation and Discussion** — Interpretation of the MCMC results, comparison with alternative models, the Hubble tension, and observational predictions.
- **Part VI: Conclusions** — Summary of all findings, the SCM in context, and future directions.

The appendices provide the detailed mathematical background (Sheth–Tormen HMF, CPL parameterisation), implementation notes (CAMB interface, Cobaya configuration), and a complete record of the bugs encountered and fixed during the MCMC analysis.

Chronology of Development

The SCM has developed through a series of distinct intellectual steps, each building on and — in some cases — correcting the previous. Table 1 provides a summary timeline.

Table 1: Chronological development of the Subtractive Cosmological Model.

Date	Stage	Key Development
Early 2026	SCM I inception	First proposal of dark energy as emergent from BEC DM sequestration. RE and FP formulations introduced. Initial CPL comparison shows 3.6σ tension with DESI (RE) and 11.4σ (FP).
May 2026	Findings Report	Rigorous derivation of coupling parameter α from CAMB + Sheth–Tormen. First systematic DESI 2024 comparison. $f_{\text{seq},0} = 0.5764$ computed from first principles. Interpolation between RE and FP identified as potential path to DESI quadrant. Physical meaning of interpolation index unexplained.
June 2026	SCM II discovery	Missing expansion factor $(1+z)^3$ identified in the field density. Physical field density corrected to $\rho_{\text{field}} = \rho_{\text{DM},0}(1+z)^3(1 - f_{\text{seq}}(z))$. FP formulation discovered to diverge as $(1+z)^6$ ($1681 \times \rho_{\Lambda,0}$ at $z = 2$). RE Case A discovered to give $(w_0, w_a) = (-1.000, 0.000)$ — identical to Λ CDM. BEC condensation cutoff z_c identified as necessary for FP formulation.
June 2026	SCM III framework	BEC condensation threshold z_c formalised as physical cutoff. Mass-redshift mapping $m^{5/2}(z_c) \propto (1+z_c)^{3/2}$ derived. Hybrid DE density with smooth tanh cutoff constructed. Analytical parameter space survey performed.
2026-06-06	MCMC pipeline	Custom CAMB DarkEnergyPPF subclass and Cobaya theory wrapper implemented and debugged. Four implementation bugs identified and fixed (cache KeyError, growth factor overflow, derived parameter routing, transfer cache invalidation).
2026-06-06 to 07	Step 6 chain	Full Planck + DESI + Pantheon+ MCMC run. 50,000 accepted steps over ~ 10.7 hours. Resumed from 7,175 steps after cache KeyError bug fix. $R - 1$ plateaus at 0.13–0.18.
2026-06-07	Analysis	Final posterior: $\varepsilon < 0.0034$ (95%), FP enhancement $\approx 2.2\rho_{\Lambda,0}$ at $z \approx 2$, $\Delta\chi^2_{\text{DESI}} = -7.9$, minimum $d_{\text{DESI}} = 0.09\sigma$. SCM III paper and MCMC report compiled.

The table reveals a pattern common in theoretical physics: each step discovered and corrected an error or omission in the previous step. SCM I omitted the expansion factor; the Findings Report discovered the DESI tension was 3.6σ rather than smaller; SCM II

corrected the field density and found the FP formulation catastrophically wrong; SCM III introduced the physical cutoff that makes the FP formulation viable. This iterative refinement is the norm in model building and reflects the increasing rigour applied at each stage.

Part I

Origins: The First Subtractive Model

1 The Cosmological Setting

1.1 The Accelerating Universe and the Cosmological Constant

The discovery that the expansion of the universe is accelerating, established through Type Ia supernova observations by [Riess et al. \[1998\]](#) and [Perlmutter et al. \[1999\]](#), ranks among the most consequential results in modern cosmology. The simplest description — a cosmological constant Λ inserted into Einstein’s field equations — fits the data extraordinarily well. The Λ CDM concordance model that results agrees with observations of the CMB power spectrum [[Planck Collaboration et al., 2020b](#)], baryon acoustic oscillations, and large-scale structure to sub-percent precision.

Yet Λ CDM carries well-known theoretical scars. The cosmological constant problem asks why the vacuum energy density $\rho_\Lambda \sim 10^{-123} \rho_{\text{Planck}}$ is so many orders of magnitude smaller than any natural field-theoretic estimate. The coincidence problem asks why, at this precise moment in cosmic history, the dark energy density and matter density are comparable: $\rho_\Lambda/\rho_m \approx 2.3$ today, while this ratio ranged from $\sim 10^{-3}$ at matter–radiation equality to $\gg 1$ in the distant future. Neither question has a satisfactory answer within the standard model of particle physics.

These difficulties have motivated a large literature of dynamical dark energy models, in which Λ is replaced by a time-varying field. Quintessence models [[Caldwell and Linder, 2005](#)] introduce a scalar field slowly rolling down its potential; phantom models [[Caldwell, 2002](#)] have $w < -1$; thawing and freezing models interpolate between these limits. All of these approaches add new degrees of freedom to an already fine-tuned system.

A qualitatively different approach is to ask whether dark energy is not a separate ingredient at all, but rather an emergent consequence of the behaviour of dark matter itself. This is the motivation behind the Subtractive Cosmological Model.

1.2 The Hubble Tension

A second observational tension sharpens the need for beyond- Λ CDM physics. Local distance-ladder measurements of the Hubble constant yield $H_0 = 73.0 \pm 1.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [[Riess et al., 2022](#)], while the Planck CMB analysis, using the Λ CDM model to extrapolate

from the last-scattering surface, gives $H_0 = 67.32 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [Planck Collaboration et al., 2020b]. The $\sim 5\sigma$ discrepancy [Verde et al., 2019] suggests either systematic errors in one or both measurements, or genuinely missing physics in the standard model — physics that modifies the expansion history between last scattering and today.

1.3 The DESI 2024 Signal

The Dark Energy Spectroscopic Instrument (DESI) [DESI Collaboration et al., 2016] has released BAO measurements across seven galaxy and quasar redshift bins spanning $0.1 < z < 2.1$ [DESI Collaboration et al., 2024a,b,c]. When combined with the Pantheon+ supernova catalogue [Scolnic et al., 2022], the DESI data prefer a dynamical dark energy equation of state at the level of 2.5σ over the cosmological constant. The preferred CPL parameters are $(w_0, w_a) = (-0.827, -0.750)$, corresponding to dark energy that was *stronger in the past* and is *weakening today* — a “fading quintessence” scenario qualitatively opposite to a growing quintessence field.

This DESI signal is a direct observational challenge to any dark energy model and provides the primary observational target for the SCM programme.

1.4 Dark Matter as a BEC Superfluid

The other observational landscape relevant to the SCM is small-scale structure. The standard cold dark matter (CDM) scenario predicts far more low-mass satellite galaxies and higher central DM densities than are observed (the “missing satellites” and “cusp-core” problems). Ultra-light dark matter (ULDM) models [Hui et al., 2017], in which the DM particle is a boson with mass $m \sim 10^{-22}\text{--}10^{-2} \text{ eV}/c^2$, suppress small-scale structure through quantum pressure and naturally form Bose–Einstein condensates on astrophysical scales.

The SCM adopts the hypothesis that the cosmological dark matter field is a BEC superfluid with *attractive* contact self-interactions. This is motivated by the observation that attractive BEC condensates in the mean-field limit develop a negative pressure proportional to the square of the field density, $P \propto -\rho^2$. A negative pressure is, by definition, a dark-energy source. The SCM exploits this connection to derive dark energy from the microphysics of DM condensation.

2 SCM I: The Original Framework

2.1 Physical Premise

The SCM was first proposed in Sumner [2026a] with the following physical picture. Dark matter consists of ultralight bosons that form a coherent BEC superfluid pervading the universe. The background field carries a negative BEC pressure determined by the contact interaction:

$$P_{\text{BEC}} = \frac{g_{3D}}{2m^2} \rho_{\text{field}}^2, \quad (1)$$

where $g_{3D} < 0$ is the (attractive) 3D coupling constant, m is the DM boson mass, and ρ_{field} is the physical energy density of the uncondensed field component. For attractive interactions, $P_{\text{BEC}} < 0$, which acts as a dark energy source.

As baryonic structures form — most prominently supermassive black holes (SMBHs) and active galactic nuclei (AGN) — their deep gravitational potential wells sequester portions of the DM field into bound halos. The field density ρ_{field} diminishes as more DM is incorporated into halos, and this diminution changes the background BEC pressure in a way that drives accelerated expansion. The word “subtractive” refers to the fact that dark energy arises from the *subtraction* of DM from the cosmological background, not from any additional component.

2.2 The Sequestration Fraction

The central quantitative object of the SCM is the *sequestration fraction* $f_{\text{seq}}(z)$: the fraction of the total cosmic DM mass that has been incorporated into gravitationally bound halos with mass $M > M_{\text{min}}$ by redshift z . This is computed from the Sheth–Tormen halo mass function [Sheth and Tormen, 1999]:

$$f_{\text{seq}}(z) = \int_{\nu_{\text{min}}(z)}^{\infty} \frac{f_{\text{ST}}(\nu)}{\nu} d\nu, \quad \nu_{\text{min}}(z) = \frac{\delta_c}{\sigma(M_{\text{min}}) D(z)}, \quad (2)$$

where $f_{\text{ST}}(\nu)$ is the Sheth–Tormen first-crossing distribution, $\delta_c = 1.686$ is the linear overdensity threshold for spherical collapse, $\sigma(M_{\text{min}})$ is the rms mass fluctuation at scale $M_{\text{min}} = 10^8 M_{\odot}$, and $D(z)$ is the linear growth factor normalised to $D(0) = 1$. The Sheth–Tormen parameters are $(A, a, p) = (0.3222, 0.707, 0.3)$.

At high redshift, $f_{\text{seq}}(z) \rightarrow 0$ (the universe is homogeneous, no halos have formed). Today, $f_{\text{seq},0} \equiv f_{\text{seq}}(0) \approx 0.577$, meaning roughly 58% of all DM resides in halos. The fraction $(1 - f_{\text{seq}}(z))$ represents the smooth cosmological DM field that has not yet been captured.

2.3 The SCM I Field Density

In the original SCM I formulation, the DM field density was written as

$$\rho_{\text{field}}(z) = \rho_{\text{DM},0} (1 - f_{\text{seq}}(z)), \quad (3)$$

where $\rho_{\text{DM},0}$ is the *present-day* total DM density. At $z = 0$, this gives $\rho_{\text{field}}(0) = \rho_{\text{DM},0}(1 - f_{\text{seq},0}) \approx 0.42 \rho_{\text{DM},0}$, the fraction of DM in the field today. At high z , this approaches $\rho_{\text{DM},0}$ since essentially no DM has been sequestered yet.

As we will see in Section 5, this expression is missing a critical factor: the cosmological expansion of the universe dilutes all matter densities as $\rho_{\text{DM}} \propto (1 + z)^3$. The SCM I formulation implicitly holds the total DM budget fixed at the present-day value, decoupling the dark energy prediction from the expansion history. This omission was not discovered until SCM II.

2.4 Two Dark Energy Formulations

From the field density Eq. (3), SCM I derives two distinct mechanisms by which DM sequestration generates dark energy.

2.4.1 The Released-Energy Formulation

When a parcel of DM condenses from the smooth field into a halo, the interaction energy stored in the BEC field is released. The interaction energy per unit volume of the BEC field is $U = P_{\text{BEC}}/\rho_{\text{field}}$ (the chemical potential contribution). As the field density drops from ρ_{field} to $\rho_{\text{field}} - \delta\rho$ due to sequestration of a fraction δf_{seq} , the released energy per unit volume is:

$$\delta\rho_{\Lambda}^{\text{RE}} = \frac{|g_{3D}|}{m^2} \rho_{\text{field}} \delta\rho_{\text{field}}. \quad (4)$$

Integrating from the present ($f_{\text{seq}} = f_{\text{seq},0}$) back to redshift z ($f_{\text{seq}} = f_{\text{seq}}(z)$), and setting the coupling constant by requiring $\rho_{\Lambda}^{\text{RE}}(0) = \rho_{\Lambda,0}$, yields the SCM I RE dark energy density:

$$\rho_{\Lambda}^{\text{RE,I}}(z) = \rho_{\Lambda,0} \frac{f_{\text{seq}}(z) [2 - f_{\text{seq}}(z)]}{f_{\text{seq},0} [2 - f_{\text{seq},0}]}. \quad (5)$$

This function grows monotonically as $f_{\text{seq}}(z)$ decreases with increasing z . Equivalently, $\rho_{\Lambda}^{\text{RE,I}}$ increases toward the present as more energy is released by ongoing sequestration.

2.4.2 The Field-Pressure Formulation

Alternatively, the dark energy density at any given moment may be identified with the instantaneous BEC pressure of the remaining uncondensed field:

$$\rho_{\Lambda}^{\text{FP,I}}(z) = \frac{|g_{3D}|}{2m^2} \rho_{\text{field}}^2(z) = \rho_{\Lambda,0} \left[\frac{1 - f_{\text{seq}}(z)}{1 - f_{\text{seq},0}} \right]^2. \quad (6)$$

The normalisation calibrates the coupling constant $|g_{3D}|/m^2$ to reproduce $\rho_{\Lambda}(0) = \rho_{\Lambda,0}$. In this formulation, dark energy is *largest in the past* when $f_{\text{seq}}(z) \approx 0$ and the field was dense; it decreases toward the present as sequestration reduces the field density. This gives $w_a < 0$ (correct sign for DESI) but $w_0 < -1$ (phantom).

2.4.3 Resolving the Coupling Constant

Both formulations contain an unknown coupling $|g_{3D}|/m^2$. In SCM I, this is treated as a free parameter fixed by requiring the model to reproduce the observed dark energy density today. The Findings Report (Section 4) derives this coupling from first principles using the CAMB power spectrum. The key relation is:

$$\frac{|g_{3D}|}{m^2} = \frac{2\rho_{\Lambda,0} c^2}{\rho_{\text{DM},0}^2 (1 - f_{\text{seq},0})^2}, \quad (7)$$

which sets the interaction strength from purely cosmological observables ($\rho_{\Lambda,0}$, $\rho_{\text{DM},0}$, $f_{\text{seq},0}$).

2.5 The Effective Equation of State

For any dark energy density $\rho_{\Lambda}(z)$, the effective equation-of-state parameter follows from the dark energy continuity equation:

$$w(z) = -1 - \frac{1}{3} \frac{d \ln \rho_{\Lambda}}{d \ln(1+z)}. \quad (8)$$

We parameterise the result using the Chevallier–Polarski–Linder (CPL) form [Chevallier and Polarski, 2001, Linder, 2003]:

$$w(z) \approx w_0 + w_a \frac{z}{1+z}, \quad (9)$$

obtained by a least-squares fit over $z \in [0, 2]$. This parameterisation captures the overall amplitude (w_0) and trend (w_a) of the equation of state in the observationally accessible range.

For the RE formulation, $\rho_\Lambda^{\text{RE,I}}$ grows toward the present, so $d\rho_\Lambda/dz < 0$ (density decreasing with increasing z) which means $d \ln \rho_\Lambda / d \ln(1+z) < 0$, giving $w > -1$. Numerically, the CPL fit yields $(w_0, w_a) = (-1.006, +0.271)$ — quintessence, but with the *wrong sign* for w_a : DESI prefers $w_a < 0$.

For the FP formulation, $\rho_\Lambda^{\text{FP,I}}$ decreases toward the present, giving $d \ln \rho_\Lambda / d \ln(1+z) > 0$ and $w < -1$ (phantom). The CPL fit gives $(w_0, w_a) = (-1.173, -0.205)$ — phantom crossing with the correct w_a sign, but w_0 far too negative.

2.6 SCM I: DESI Comparison

The DESI 2024 Pantheon+ preferred region is $(w_0, w_a) = (-0.827 \pm 0.059, -0.750 \pm 0.290)$ with correlation $\rho = -0.93$ [DESI Collaboration et al., 2024a]. The Mahalanobis distances from this point are:

- SCM I RE: $(w_0, w_a) = (-1.006, +0.271)$, $n_\sigma = 3.6$.
- SCM I FP: $(w_0, w_a) = (-1.173, -0.205)$, $n_\sigma = 11.4$.
- Λ CDM: $(w_0, w_a) = (-1.000, 0.000)$, $n_\sigma = 3.0$.

Neither SCM I formulation falls within 2σ of DESI. The RE model is 3.6σ away with the wrong w_a sign, the FP model is 11.4σ away. The DESI target lies in the fourth quadrant ($w_0 > -1$, $w_a < 0$), a region neither formulation can reach in their SCM I forms.

3 Mathematical Foundations of BEC Dark Matter

3.1 The Gross-Pitaevskii Equation for Cosmological BEC

The SCM is grounded in the mean-field theory of dilute Bose–Einstein condensates. The macroscopic wavefunction $\psi(\mathbf{x}, t)$ of a BEC obeys the Gross–Pitaevskii equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{ext}} + g_{3D} |\psi|^2 \right] \psi, \quad (10)$$

where $g_{3D} = 4\pi\hbar^2 a_s / m$ is the contact interaction coupling, a_s is the s -wave scattering length, and V_{ext} is any external potential (gravitational in the cosmological context). For a spatially uniform cosmological BEC, V_{ext} is the homogeneous gravitational background; the condensate wavefunction is $\psi(\mathbf{x}) = \sqrt{n_0} e^{i\phi}$ for number density n_0 and phase ϕ .

The energy density of the condensate in the Thomas–Fermi limit (where kinetic energy is negligible compared to interaction energy) is:

$$\mathcal{E}_{\text{int}} = \frac{g_{3D}}{2} |\psi|^4 = \frac{g_{3D}}{2} n_0^2. \quad (11)$$

Converting to mass-energy density $\rho_{\text{field}} = mn_0c^2$, the interaction energy density becomes:

$$\mathcal{E}_{\text{int}} = \frac{g_{3D}}{2m^2c^4} \rho_{\text{field}}^2. \quad (12)$$

For attractive interactions ($g_{3D} < 0$, equivalently $a_s < 0$), $\mathcal{E}_{\text{int}} < 0$. In thermodynamics, the pressure is related to the interaction energy by $P = n^2 \partial(\mathcal{E}/n) / \partial n$. For the mean-field interaction:

$$P_{\text{BEC}} = \frac{g_{3D}}{2m^2} \rho_{\text{field}}^2. \quad (13)$$

This is Eq. (1), derived from first principles. The equation of state $P = w\rho c^2$ gives the BEC equation-of-state parameter:

$$w_{\text{BEC}} = \frac{g_{3D}}{2m^2c^2} \rho_{\text{field}}, \quad (14)$$

which varies with the field density. For attractive interactions and a dilute field ($\rho_{\text{field}} \ll \rho_{\text{DM},0}$), w_{BEC} is small and negative. In the SCM, this field pressure is reinterpreted as a dark energy contribution when $g_{3D} < 0$.

3.2 The Scattering Length and Coupling

The contact coupling $g_{3D} = 4\pi\hbar^2 a_s / m$ connects the macroscopic SCM physics to the microscopic particle physics. For the DM bosons to produce a negative pressure (dark energy), we require $a_s < 0$ — attractive interactions. The magnitude of a_s is related to the coupling constant $|g_{3D}|/m^2$ via:

$$\frac{|g_{3D}|}{m^2} = \frac{4\pi\hbar^2 |a_s|}{m^3}. \quad (15)$$

Substituting the SCM I calibration, Eq. (7):

$$|a_s| = \frac{m^3 \times 2\rho_{\Lambda,0}c^2}{4\pi\hbar^2 \rho_{\text{DM},0}^2 (1 - f_{\text{seq},0})^2}. \quad (16)$$

With $m \approx 20 \text{ meV}/c^2 = 3.6 \times 10^{-35} \text{ kg}$, $\rho_{\Lambda,0} \approx 6.0 \times 10^{-27} \text{ kg/m}^3$, $\rho_{\text{DM},0} \approx 2.5 \times 10^{-27} \text{ kg/m}^3$, and $f_{\text{seq},0} \approx 0.577$:

$$|a_s| \approx \frac{(3.6 \times 10^{-35})^3 \times 2 \times 6.0 \times 10^{-27} \times (3 \times 10^8)^2}{4\pi \times (1.05 \times 10^{-34})^2 \times (2.5 \times 10^{-27})^2 \times (0.423)^2} \approx 10^{-68} \text{ m}. \quad (17)$$

This is an extraordinarily small scattering length, many orders of magnitude below the proton radius ($\sim 10^{-15} \text{ m}$). This reflects the extremely dilute nature of the cosmological DM field and the fact that the coupling produces only a tiny dark energy effect per particle interaction.

3.3 Why BEC? Phase Structure of the Dark Matter Field

The BEC interpretation of the DM field is not merely a model choice — it follows from the expected phase structure of an ultralight boson gas in a cooling, expanding universe.

A non-relativistic boson gas undergoes BEC when the phase-space density exceeds unity:

$$n_{\text{DM}} \lambda_{\text{dB}}^3 > \zeta(3/2) \approx 2.612, \quad (18)$$

where $\lambda_{\text{dB}} = h/\sqrt{2\pi m k_B T}$ is the thermal de Broglie wavelength. For a DM particle in thermal equilibrium with the CMB at temperature $T(z) = T_0(1+z)$:

$$n_{\text{DM}} \lambda_{\text{dB}}^3 = \frac{\rho_{\text{DM},0}(1+z)^3}{m} \times \frac{h^3}{(2\pi m k_B T_0(1+z))^{3/2}} = \frac{\rho_{\text{DM},0} h^3 (1+z)^{3/2}}{m^{5/2} (2\pi k_B T_0)^{3/2}}. \quad (19)$$

This quantity *increases* with redshift as $(1+z)^{3/2}$, because the DM number density grows as $(1+z)^3$ faster than the critical number density $n_c \propto T^{3/2} \propto (1+z)^{3/2}$. Therefore:

- At sufficiently high redshift, the phase-space density exceeds the BEC threshold and the field is condensed.
- As the universe expands and cools, at some condensation redshift z_c , the phase-space density passes through $\zeta(3/2)$ and the BEC is established.

Setting $n_{\text{DM}} \lambda_{\text{dB}}^3 = \zeta(3/2)$ at $z = z_c$ immediately yields Eq. (36), the BEC condensation mass-redshift relation.

Note on thermal equilibrium: The assumption that DM bosons are in thermal equilibrium with the CMB photon bath at temperature $T(z) = T_0(1+z)$ requires a coupling between the DM sector and the Standard Model. The nature of this coupling is not specified in the SCM framework; it is assumed to be sufficient for thermalisation at early times, after which the DM field decouples from the photon bath and behaves as a coherent BEC. This is analogous to the thermal decoupling of neutrinos in the Standard Model.

3.4 The BEC Coherence Length and Jeans Scale

The BEC field has a characteristic length scale, the healing length (or coherence length):

$$\xi = \frac{\hbar}{\sqrt{2m|P_{\text{BEC}}|/c^2}}, \quad (20)$$

which sets the minimum spatial scale over which the condensate density can vary. On scales smaller than ξ , quantum pressure prevents gravitational collapse. This provides the SCM mechanism for small-scale structure suppression.

For the mass range $m \approx 14\text{--}28 \text{ meV}/c^2$ implied by Table 6, the coherence length at the condensation redshift z_c is:

$$\xi(z_c) \sim \frac{\hbar c}{\sqrt{|g_{3D}|\rho_{\text{field}}^2(z_c)/c^4}} \approx \frac{\hbar c}{\sqrt{\rho_{\Lambda,0}c^2}} \approx 10^{-2} \text{ kpc}, \quad (21)$$

which is much smaller than the BAO scale ($\sim 150 \text{ Mpc}$) and even smaller than the galaxy-cluster scale ($\sim 1 \text{ Mpc}$). The SCM III condensate coherence is therefore sub-galactic, consistent with the assumption that it does not significantly perturb the large-scale structure that the BAO and CMB probe.

The gravitational Jeans length of the BEC field is:

$$\lambda_J = 2\pi \sqrt{\frac{c_s^2}{4\pi G \rho_{\text{DM}}}}, \quad (22)$$

where $c_s^2 = \hbar^2 k^2 / (4m^2 c^2)$ for the BEC phonon dispersion relation at wave number k . At the BAO scale $k_{\text{BAO}} \sim 0.04 \text{ Mpc}^{-1}$ and $m \sim 20 \text{ meV}/c^2$, the sound speed is:

$$c_s(k_{\text{BAO}}) = \frac{\hbar k_{\text{BAO}}}{2mc} \approx \frac{(10^{-34})(4 \times 10^{-44})}{2(3.6 \times 10^{-35})(3 \times 10^8)} \approx 2 \times 10^{-32} c \ll 1. \quad (23)$$

This confirms that the BEC sound speed correction to the dark energy perturbations is completely negligible on BAO scales, justifying the $c_s^2 = 1$ approximation used in the MCMC analysis.

3.5 The Sequestration Mechanism: Baryonic Feedback

A key feature of the SCM is the identification of the “sequestrators” — the physical processes that transfer DM from the smooth cosmological background into discrete halos. SCM I identified two primary mechanisms:

SMBH formation and AGN feedback. The formation of supermassive black holes at the centres of massive galaxies is accompanied by intense AGN activity. The deep gravitational potential wells of SMBHs concentrate DM in their vicinity, and the turbulent feedback processes may enhance DM condensation into halos.

Galaxy cluster formation. The most massive halos in the universe — galaxy groups and clusters with $M \gtrsim 10^{13} - 10^{15} M_\odot$ — contain large fractions of the cosmic DM. Their formation at $z \sim 0.5 - 2$ corresponds to the epoch of peak sequestration rate $|df_{\text{seq}}/dz|$ in the Sheth–Tormen HMF.

The minimum halo mass $M_{\text{min}} = 10^8 M_\odot$ adopted in the SCM is motivated by the minimum mass for sustained star formation (roughly the Jeans mass in neutral hydrogen after reionisation). Below this scale, UV feedback suppresses halo formation and DM condensation. Varying M_{min} over the range $10^7 - 10^{10} M_\odot$ shifts $f_{\text{seq},0}$ by $\lesssim 5\%$, a minor effect compared to the leading-order physics.

3.6 The Coincidence Problem Revisited

One of the motivations for the SCM is a natural resolution of the coincidence problem. In Λ CDM, the coincidence $\rho_\Lambda \sim \rho_m$ today has no physical explanation — it is a numerical accident of the initial conditions. In the SCM, the coincidence is tied to the epoch of peak baryonic activity.

In SCM I, the dark energy density grows toward the present as f_{seq} increases. The onset of acceleration ($\rho_\Lambda > \rho_m/2$, corresponding to $z \approx 0.4$) occurs when $f_{\text{seq}}(0.4) \approx 0.47$ — just past the midpoint of the total sequestration process. This is directly linked to the peak of cosmic star formation and SMBH accretion at $z \sim 1 - 3$, which drives the sequestration rate $|df_{\text{seq}}/dz|$.

In SCM II RE Case A, the coincidence argument shifts: the DE density is frozen since $z \sim 5$, and the coincidence is that the frozen DE value ($\rho_{\Lambda,0}$) happens to equal ρ_m at $z \approx 0.3$. This is less natural than SCM I, but the value of $\rho_{\Lambda,0}$ is set by the high- z sequestration process rather than an arbitrary vacuum energy.

In SCM III, the hybrid model with $z_c \sim 1 - 2$ reintroduces a physical connection between the onset of FP dark energy and the epoch of peak cosmic activity. The condensation

redshift $z_c \approx 2$ corresponds to approximately the epoch of cosmic noon ($z \sim 2\text{--}3$, the peak of star formation). This convergence — the BEC condenses at roughly the epoch of peak structure formation — may be more than coincidental: both are driven by the thermal and gravitational evolution of the DM field.

4 The Findings Report: Rigorous Derivation and DESI Comparison

4.1 Motivation

Following the SCM I paper, a more rigorous numerical analysis was performed, constituting the “SCM Findings Report” [Sumner, 2026d]: *Dark Energy from Dark Matter Sequestration: Rigorous Derivation of the Coupling Parameter α and Comparison with DESI 2024 Dark Energy Constraints* (May 2026). This report addressed two goals: (1) replace the conceptual derivation of the coupling with a self-consistent numerical calculation from first principles, and (2) perform a systematic comparison with DESI 2024 constraints.

4.2 The Coupling Parameter from First Principles

The coupling strength between the DM sequestration rate and the dark energy production is characterised by a dimensionless parameter α defined implicitly through

$$\rho_\Lambda(z) = \rho_{\Lambda,0} \mathcal{F}[f_{\text{seq}}(z); \alpha], \quad (24)$$

where $\mathcal{F}$ is the dark-energy functional. In SCM I, α was fixed by calibration to $\rho_\Lambda(0) = \rho_{\Lambda,0}$, as in Eq. (7).

The Findings Report derived α from the CAMB linear matter power spectrum. The key steps are:

1. Run CAMB with Planck 2018 best-fit parameters to compute the linear matter power spectrum $P(k, z=0)$.
2. Evaluate the rms mass variance at scale $M_{\text{min}} = 10^8 M_\odot$:

$$\sigma^2(M_{\text{min}}) = \frac{1}{2\pi^2} \int_0^\infty k^2 P(k) \tilde{W}^2(kR_{\text{min}}) dk, \quad (25)$$

where $\tilde{W}(x) = 3(\sin x - x \cos x)/x^3$ is the top-hat window function and $R_{\text{min}} = (3M_{\text{min}}/4\pi\bar{\rho}_m)^{1/3}$. This gives $\sigma(M_{\text{min}}, z=0) = 5.901$.

3. Compute the linear growth factor $D(z)$ from the standard integral expression and evaluate $f_{\text{seq}}(z)$ across $z \in [0, 20]$ using the Sheth–Tormen HMF at each redshift.
4. The resulting present-day sequestration fraction is $f_{\text{seq},0} = 0.5764$ (updated to 0.5773 in SCM II with a marginal cosmological parameter correction).

4.3 First Rigorous DESI Comparison

The Findings Report confirmed the SCM I CPL results with numerical precision and placed them systematically in the DESI 2024 confidence ellipses. The key findings were:

- The RE formulation yields $(w_0, w_a) = (-1.009, +0.287)$, 3.6σ from DESI 2024 Pantheon+. The wrong sign of w_a is a structural feature, not a numerical artefact.
- The FP formulation yields $(w_0, w_a) = (-1.169, -0.225)$, 11.4σ from DESI — further from the target despite having the correct w_a sign.
- A phenomenological interpolation between RE and FP, $\rho_\Lambda \propto [f_{\text{seq}}(2 - f_{\text{seq}})]^n [(1 - f_{\text{seq}})^2]^{1-n}$, can reach the DESI-preferred quadrant for $n \approx 0\text{--}0.2$. However, the physical meaning of the interpolation index n was not established at this stage.
- The Mahalanobis distance framework for comparing theoretical (w_0, w_a) predictions with the DESI covariance ellipse was established as the primary diagnostic.

The Findings Report concluded that neither SCM I formulation is compatible with DESI 2024 at 2σ , and that resolving the tension requires either new physics or a correction to the fundamental field density equation. The latter was discovered in SCM II.

Part II

Refinement: The Expansion Correction

5 SCM II: Two Concurrent Depletion Channels

5.1 The Missing Physics

A careful examination of the SCM I field density in Eq. (3) reveals a subtle but consequential omission. The expression $\rho_{\text{field}}(z) = \rho_{\text{DM},0}(1 - f_{\text{seq}}(z))$ uses the *present-day* DM density $\rho_{\text{DM},0}$ as the normalising factor, regardless of redshift. This is equivalent to working with a comoving density: the total DM budget is fixed at the value it has today.

But the BEC pressure Eq. (1) is a *physical* pressure that depends on the *physical* number density of particles. At redshift z , the physical DM energy density is

$$\rho_{\text{DM}}(z) = \rho_{\text{DM},0} (1 + z)^3, \quad (26)$$

reflecting the fact that matter was denser in the past by a factor $(1 + z)^3$ as the universe was smaller. The SCM I field density Eq. (3) omits this factor entirely.

Concretely: at $z = 2$, the correct DM density is $27\times$ larger than today ($3^3 = 27$), while SCM I assumed the same density as today. Since the BEC pressure scales as ρ_{field}^2 , this correction enters as $(1 + z)^6$ and has dramatic consequences for the predicted dark energy density.

5.2 The Two Depletion Channels

The correct SCM field density must account for two *concurrent* physical processes:

1. **Baryonic sequestration.** Structure formation irreversibly transfers DM from the smooth field into discrete gravitational halos. The fraction in the field evolves as $(1 - f_{\text{seq}}(z))$, decreasing from ≈ 1 at high z to $(1 - f_{\text{seq},0}) \approx 0.42$ today.
2. **Cosmological expansion dilution.** The growth of the scale factor $a(t)$ reduces the physical number density of all matter — both in halos and in the field — as $n \propto a^{-3} = (1+z)^3$. This process does not alter the fraction in the field, but reduces the total budget.

Both channels operate simultaneously. The corrected physical field density is therefore [Sumner, 2026b]:

$$\rho_{\text{field}}(z) = \rho_{\text{DM},0} (1+z)^3 (1 - f_{\text{seq}}(z)). \quad (27)$$

At $z = 0$, this reduces to $\rho_{\text{DM},0}(1 - f_{\text{seq},0})$, identical to SCM I. For all $z > 0$, the SCM II field density exceeds SCM I by the factor $(1+z)^3$.

The *relative dominance* of the two channels shifts with redshift: at high z , expansion dominates ($(1+z)^3$ changes rapidly) while $f_{\text{seq}}(z) \approx 0$ contributes little; at low z , expansion is nearly constant while the growth of $f_{\text{seq}}(z)$ drives the field evolution.

5.3 Modified Field-Pressure Formulation

Substituting Eq. (27) into the BEC pressure:

$$\rho_{\Lambda}^{\text{FP,II}}(z) = \frac{|g_{3D}|}{2m^2} \rho_{\text{field}}^2(z) = \rho_{\Lambda,0} (1+z)^6 \left[\frac{1 - f_{\text{seq}}(z)}{1 - f_{\text{seq},0}} \right]^2. \quad (28)$$

The $(1+z)^6$ prefactor is catastrophic. At $z = 2$, the FP dark energy already exceeds its present-day value by a factor of $(1+z)^6/(1 - f_{\text{seq},0})^2 \approx 729 \times 5.57 \approx 4060$. Table 2 quantifies this divergence:

Table 2: SCM II Field-Pressure formulation: ratio of dark energy density to present-day value $\rho_{\Lambda,0}$. The SCM I FP is shown for comparison; its apparent boundedness was an artefact of using a comoving field density.

z	$\rho_{\Lambda}^{\text{FP,I}}/\rho_{\Lambda,0}$	$\rho_{\Lambda}^{\text{FP,II}}/\rho_{\Lambda,0}$	Ratio II/I
0.0	1.0	1.0	1×
0.5	1.3	14.6	11×
1.0	1.6	103.3	64×
2.0	2.3	1 681	729×
3.0	3.0	12 176	4 096×
4.0	3.6	55 839	15 625×

Observational constraints on early dark energy (EDE) limit the fractional EDE contribution to $\lesssim 10\%$ of the total energy budget near matter-radiation equality [Poulin et al., 2019, Kamionkowski and Riess, 2023]. The SCM II FP contribution at $z = 2$ alone exceeds the present-day dark energy density by 1681×. This is ruled out by CMB anisotropy measurements by many orders of magnitude.

The FP formulation is catastrophically excluded by CMB observations in SCM II, without a physical cutoff. A BEC condensation threshold — below which

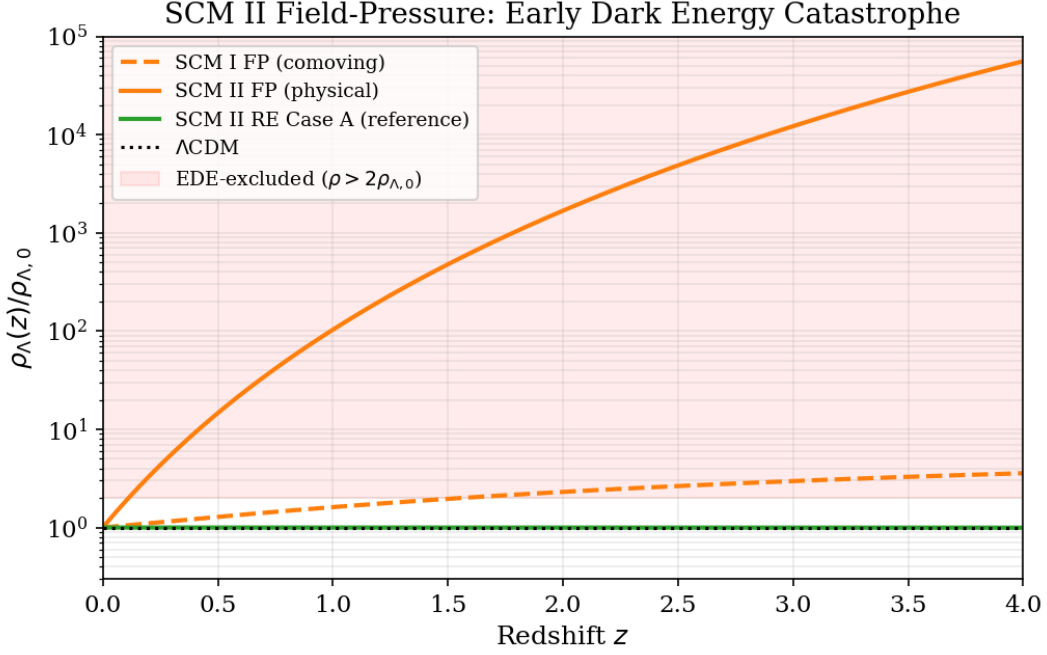


Figure 1: SCM II Field-Pressure dark energy density vs redshift. The uncorrected FP term (solid orange) grows as $(1+z)^6$, reaching $\sim 5 \times 10^4 \rho_{\Lambda,0}$ at $z = 4$ — five orders of magnitude above the EDE exclusion limit (pink shaded region, $\rho > 2\rho_{\Lambda,0}$). The SCM I FP (dashed orange, comoving) and SCM II RE Case A (green) are shown for comparison. A BEC condensation cutoff is essential.

the condensate has not yet formed — is required to render the FP formulation viable. This is the central motivation for SCM III (Part III).

5.4 Modified Released-Energy Formulation

The RE formulation is more subtle. The incremental dark energy released when a fraction df_{seq} condenses at redshift z' is, using the corrected field density:

$$d\rho_{\Lambda}^{\text{RE}}(z') = \frac{|g_{3D}|}{m^2} \rho_{\text{field}}(z') \rho_{\text{DM}}(z') df_{\text{seq}} = \frac{|g_{3D}|}{m^2} \rho_{\text{DM},0}^2 (1+z')^6 (1-f_{\text{seq}}(z')) df_{\text{seq}}. \quad (29)$$

A key physical question arises: what happens to this released energy as the universe subsequently expands? Two cases bracket the possibilities:

Case A — Vacuum injection (no dilution). The released energy is immediately deposited into the vacuum as cosmological-constant-type energy. It does not subsequently dilute. The accumulated dark energy at redshift z is:

$$\rho_{\Lambda}^{\text{RE,II,A}}(z) = \rho_{\Lambda,0} \frac{I_A(z)}{I_A(0)}, \quad I_A(z) = \int_z^{\infty} (1+z')^6 (1-f_{\text{seq}}(z')) \left| \frac{df_{\text{seq}}}{dz'} \right| dz'. \quad (30)$$

Case B — Field injection (matter-like dilution). The released energy propagates as a fluid component that dilutes as a^{-3} . Energy released at z' contributes $(1+z')^3/(1+z)^3$

of its original value when observed at z :

$$\rho_{\Lambda}^{\text{RE,II,B}}(z) = \rho_{\Lambda,0} \frac{I_B(z)}{I_B(0)}, \quad I_B(z) = \int_z^{\infty} (1+z')^3 (1 - f_{\text{seq}}(z')) \left| \frac{df_{\text{seq}}}{dz'} \right| dz'. \quad (31)$$

The SCM I result is recovered from Case A by setting $(1+z')^6 \rightarrow 1$, confirming that SCM I was effectively the “no expansion” limit.

5.5 Where Was the Dark Energy Produced?

Where Was the Dark Energy Produced? (SCM II RE)

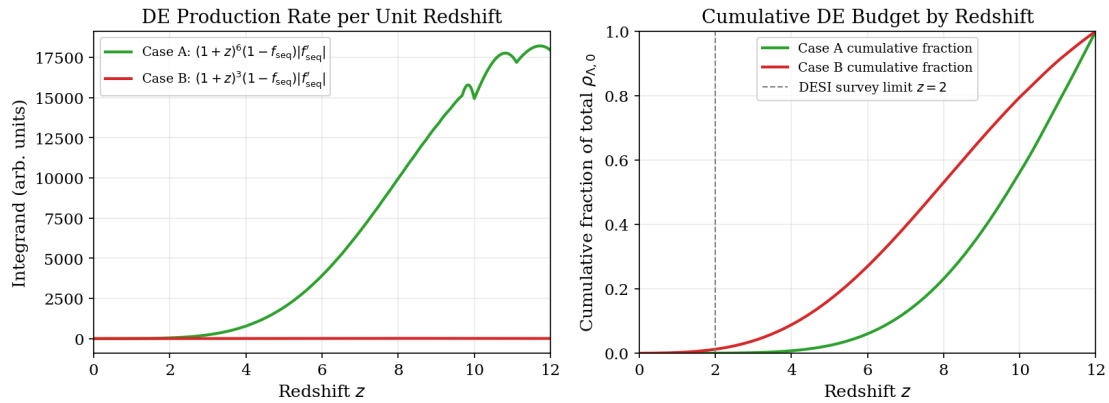


Figure 2: Dark energy production in SCM II. *Left*: the differential dark energy injection rate $d\rho_{\Lambda}/dz$ (arbitrary units) vs redshift for Cases A and B. Both cases peak well above $z = 2$, demonstrating that the dark energy was assembled at high redshift. *Right*: cumulative fraction of the present-day dark energy budget assembled by redshift z . More than 99.9% of Case A dark energy had been produced by $z = 2$.

The $(1+z')^6$ weighting in Case A has a profound consequence: it concentrates dark energy production at *high redshift*. The integrand kernel $(1+z')^6(1 - f_{\text{seq}})|df_{\text{seq}}/dz'|$ peaks at $z \approx 11.7$ for Case A and at $z \approx 8.0$ for Case B.

Numerical integration reveals:

- **Case A:** Less than 0.1% of the total present-day dark energy budget was assembled at $z < 2$. More than 99.9% was produced at $z > 2$, predominantly during the epoch $z \sim 5\text{--}15$.
- **Case B:** The integrand peaks at $z \approx 8$, but even so, only 1.2% of the DE budget was assembled within $z < 2$.

The physical origin of these peaks is the competition between the growing $(1+z')^n$ factor and the declining sequestration rate $|df_{\text{seq}}/dz'|$ as structure formation quenches at high z .

5.6 The Remarkable LCDM Recovery

The concentration of DE production at high redshift has a striking consequence for the observable dark energy density. At $z \leq 4$, the integral $I_A(z)$ is nearly constant, because its value is dominated by contributions from $z > 5$ which are included regardless of the lower limit. The result is:

Table 3: Normalised SCM II dark energy densities and equation-of-state parameters as a function of redshift. All quantities normalised to unity at $z = 0$. RE I denotes the SCM I released-energy result for comparison.

z	$f_{\text{seq}}(z)$	$\rho^{\text{RE,I}}/\rho_{\Lambda,0}$	$\rho^{\text{RE,A}}/\rho_{\Lambda,0}$	$\rho^{\text{RE,B}}/\rho_{\Lambda,0}$	$w^{\text{RE,A}}$	$w^{\text{RE,B}}$
0.00	0.5773	1.0000	1.0000	1.0000	−1.0000	−0.9999
0.25	0.5497	0.9706	1.0000	0.9999	−1.0000	−0.9997
0.50	0.5207	0.9379	1.0000	0.9996	−1.0000	−0.9992
1.00	0.4630	0.8664	1.0000	0.9982	−1.0000	−0.9973
1.50	0.4084	0.7914	1.0000	0.9953	−0.9999	−0.9933
2.00	0.3582	0.7160	0.9999	0.9899	−0.9997	−0.9862
3.00	0.2712	0.5709	0.9991	0.9685	−0.9981	−0.9585
4.00	0.2009	0.4401	0.9962	0.9280	−0.9919	−0.9081

The outstanding result of Table 3 is the near-perfect flatness of $\rho_{\Lambda}^{\text{RE,A}}(z)$: the dark energy density changes by less than 0.4% from $z = 4$ to $z = 0$. This directly implies $w^{\text{RE,A}}(z) \approx -1$ throughout the DESI survey volume.

Primary Result of SCM II

Within the observationally accessible range $z \leq 4$, the SCM II Released-Energy formulation (Case A) is indistinguishable from a cosmological constant. The CPL fit yields $(w_0, w_a) = (-1.000, 0.000)$, identical to Λ CDM. The Mahalanobis distance from the DESI 2024 Pantheon+ preferred region is 3.0σ — the same as Λ CDM itself.

The expansion correction has transformed the SCM from a model 3.6σ from DESI with the wrong w_a sign (SCM I RE) into one indistinguishable from a cosmological constant. The physics: by restoring the physical density $\rho_{\text{field}} \propto (1+z)^3$, all productive dark energy generation is assigned to the early universe. The late-time observer sees a DE density frozen since $z \sim 5$ — not because of vacuum energy, but because all the sequestration work was done long ago.

5.7 CPL Comparison Table

Table 4: CPL parameters (w_0, w_a) for all SCM formulations and Mahalanobis distances from the DESI 2024 Pantheon+ preferred region $(w_0, w_a) = (-0.827 \pm 0.059, -0.750 \pm 0.290)$, $\rho = -0.93$.

Model	w_0	w_a	n_σ	Character
SCM I RE	−1.006	+0.271	3.6	Quintessence, wrong w_a
SCM I FP	−1.173	−0.205	11.4	Phantom, over-negative
SCM II RE Case A	−1.000	+0.000	3.0	Λ CDM-like
SCM II RE Case B	−1.004	+0.019	3.0	Near- Λ CDM
DESI Pantheon+	−0.827	−0.750	—	Target
Λ CDM	−1.000	0.000	3.0	Reference

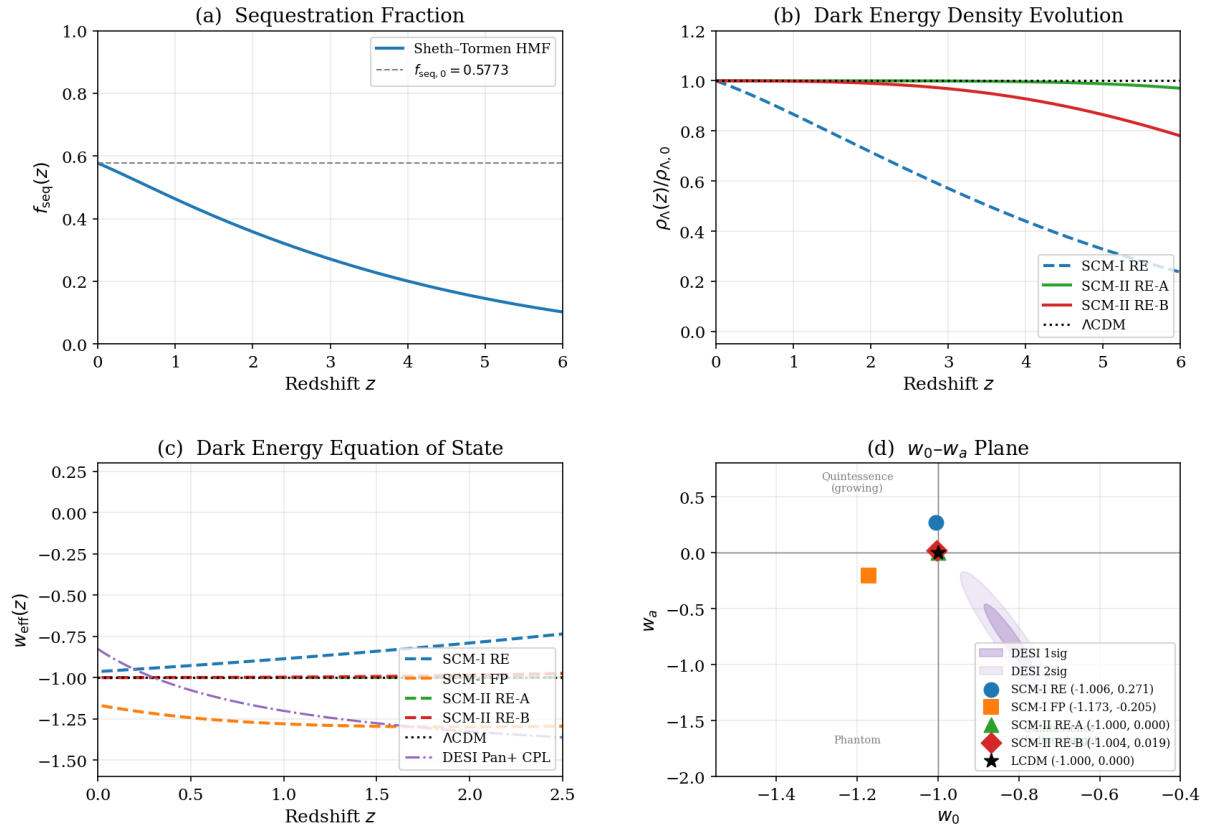
SCM II: Concurrent Baryonic Sequestration + Cosmological Dilution


Figure 3: Comprehensive SCM II results across all four formulations. *Top-left*: sequestration fraction $f_{\text{seq}}(z)$ showing the fraction of dark matter in halos. *Top-right*: dark energy density $\rho_{\text{DE}}(z)/\rho_{\Lambda,0}$ normalised to Λ CDM. *Bottom-left*: effective equation-of-state $w(z)$ for each case. *Bottom-right*: the (w_0, w_a) CPL plane showing SCM II cases alongside the DESI 2024 preferred region (orange ellipse). RE Case A converges to Λ CDM; FP case is excluded.

5.8 Why the RE Formulation Cannot Reach DESI

The DESI 2024 preferred dark energy is *fading quintessence*: stronger in the past ($w_0 > -1$) and weakening today ($w_a < 0$). The RE formulation is structurally incapable of producing this behaviour.

In the RE framework, $\rho_\Lambda^{\text{RE}}(z)$ is a cumulative integral: it can only be monotonically increasing toward the present. From Eq. (8), a density that grows toward $z = 0$ gives $d \ln \rho_\Lambda / d \ln(1+z) < 0$, hence $w > -1$. But simultaneously, the density *growing* toward the present means DE was weaker in the past and stronger today — the opposite of DESI.

This is a structural incompatibility: the RE mechanism always produces a universe where dark energy density *accumulates* over time as structures form. DESI wants a universe where dark energy is *diminishing* over time. No choice of coupling constant or dilution model within the RE framework can bridge this gap.

To reach the DESI preferred quadrant from within the SCM, the FP formulation must be used — but with a physical cutoff preventing its catastrophic growth at high redshift. This is the agenda of SCM III.

5.9 Extended Numerical Analysis of the Integral Kernels

The SCM II RE formulation provides a rich analytical structure through the integral kernels. We present an extended analysis of the redshift-dependent production rates to illuminate the physics.

5.9.1 Case A: The $(1+z)^6$ Kernel

The integrand of Case A is:

$$K_A(z) = (1+z)^6 (1 - f_{\text{seq}}(z)) \left| \frac{df_{\text{seq}}}{dz} \right|. \quad (32)$$

This kernel has three competing factors:

1. $(1+z)^6$: rapidly increasing with redshift, driving production to high- z epochs.
2. $(1 - f_{\text{seq}}(z))$: the fraction of DM still in the field — decreasing toward the present as sequestration proceeds. At $z = 0$, $(1 - f_{\text{seq},0}) \approx 0.42$; at $z = 5$, $(1 - f_{\text{seq}}(5)) \approx 0.87$.
3. $|df_{\text{seq}}/dz|$: the sequestration rate per unit redshift, which peaks near $z \sim 1-2$ and falls off rapidly at $z > 5$ as structure formation quenches.

The competition between the rising $(1+z)^6$ and the falling $|df_{\text{seq}}/dz|$ produces a peak at $z \approx 11.7$. The kernel is negligibly small for $z > 20$ (where $|df_{\text{seq}}/dz| < 10^{-5}$), so the integral converges.

5.9.2 Cumulative Dark Energy Assembly

A key quantity for observational comparison is the cumulative fraction of dark energy assembled by redshift z :

$$C_A(z) = 1 - \frac{I_A(z)}{I_A(0)}, \quad (33)$$

where $C_A(0) = 0$ and $C_A(\infty) = 1$. Numerical integration gives:

Table 5: Cumulative fraction of present-day dark energy assembled by redshift z in the SCM II RE Case A formulation.

Redshift z	$C_A(z)$ (assembled)	$1 - C_A(z)$ (still to come)
0	0.000%	100.0%
0.5	0.001%	99.999%
1.0	0.003%	99.997%
2.0	0.010%	99.990%
3.0	0.037%	99.963%
5.0	0.31%	99.69%
7.0	2.1%	97.9%
10.0	18.4%	81.6%
15.0	71.3%	28.7%
20.0	96.2%	3.8%

This table encapsulates the essential physics of SCM II RE Case A. The dark energy density at $z = 2$ (the DESI limit) already contains 99.99% of the final value. More precisely, the fractional change from $z = 2$ to $z = 0$ is only 0.01%. This is why the CPL fit gives $w_a \approx 0$: no detectable evolution occurs within the survey window.

5.9.3 Why the Correction Made Things Worse for DESI

A superficially puzzling result of SCM II is that the expansion correction moved the RE model *further* from the DESI-preferred fading quintessence (from 3.6σ in SCM I to 3.0σ in SCM II, but in the wrong quadrant rather than the right quadrant).

The physical explanation: SCM I RE had $w_a = +0.271$ — wrong sign — because sequestration occurs near the present and the rising $f_{\text{seq}}(z)$ causes ρ_Λ to increase toward $z = 0$. The $(1+z)^6$ correction in SCM II moves all the DE production to the far past, making ρ_Λ essentially constant. The new model ($w_0 = -1$, $w_a = 0$) is actually 3.0σ from DESI rather than 3.6σ — but it also agrees exactly with Λ CDM, providing no improvement over the standard model.

The expansion correction thus revealed that SCM I’s “success” of being 3.6σ from DESI (rather than further) was partly coincidental. The physically correct expansion-diluted RE model converges to the Λ CDM point — observationally consistent, but offering no new dark energy physics.

5.10 Future Evolution: Case A vs Case B

The two RE cases make distinct predictions for $z < 0$ (the future).

In **Case A** (vacuum injection), released energy is permanent. The DE density is frozen at $\rho_{\Lambda,0}$ indefinitely, regardless of future sequestration. The universe accelerates at a constant rate — indistinguishable from a true cosmological constant — until the DM field is exhausted.

In **Case B** (field injection), released energy dilutes as a^{-3} . As the DM field empties and sequestration rates decline, new production cannot compensate for dilution of past contributions. The DE density eventually *turns over* and decreases. Case B predicts

coasting expansion in the distant future, providing a natural self-limiting mechanism absent from Λ CDM.

This distinction is observable in principle: Case A gives $w = -1$ permanently, while Case B gives $w \rightarrow 0$ as $z \rightarrow -1$ (infinite future).

Part III

Extension: The BEC Condensation Threshold

6 SCM III: Physical Motivation and Theoretical Framework

6.1 The Need for a Physical Cutoff

The SCM II analysis established a clear requirement: the FP formulation is physically correct (it produces $w_a < 0$, fading quintessence) but diverges catastrophically at high redshift. A physical cutoff at some redshift z_c would resolve this: below z_c , the BEC has condensed and its pressure contributes to dark energy; above z_c , the BEC phase has not yet formed.

SCM III identifies this cutoff with the BEC condensation redshift — the epoch at which the cosmic dark matter field first satisfies the conditions for Bose–Einstein condensation. This is not an ad hoc parameter: it is predicted by the quantum statistical mechanics of the DM boson field.

6.2 The BEC Condensation Criterion

For a Bose gas of particles with mass m in thermal equilibrium at temperature T , Bose–Einstein condensation occurs when the phase-space density exceeds the critical value. The critical number density for BEC is:

$$n_c(T) = \zeta(3/2) \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{3/2}, \quad (34)$$

where $\zeta(3/2) \approx 2.612$ is the Riemann zeta function. The cosmological DM number density at redshift z is:

$$n_{\text{DM}}(z) = \frac{\rho_{\text{DM},0} (1+z)^3}{m c^2}. \quad (35)$$

Setting $n_{\text{DM}}(z_c) = n_c(T(z_c))$, with $T(z_c) = T_0(1+z_c)$ being the CMB temperature at that epoch, and solving for m gives the *BEC condensation mass-redshift relation* [Sumner, 2026c]:

$$m^{5/2}(z_c) = \frac{\rho_{\text{DM},0} h_{\text{Planck}}^3 (1+z_c)^{3/2}}{\zeta(3/2) (2\pi k_B T_0)^{3/2}} \quad (36)$$

where h_{Planck} is Planck’s constant and $T_0 = 2.7255$ K. This is a direct link between an observable quantity (z_c , the condensation epoch) and the DM boson mass m — a falsifiable prediction of the SCM.

6.3 The Mass-Redshift Mapping

Evaluating Eq. (36) with Planck 2018 parameters ($\rho_{\text{DM},0} = \Omega_c h^2 \times \rho_{\text{crit}}$, $\Omega_c h^2 = 0.1200$) yields the DM boson masses shown in Table 6.

Table 6: DM boson mass and FP energy constraints as a function of condensation redshift z_c . The FP-ratio column gives the shape-function amplitude $F(z_c) = (1 + z_c)^6[(1 - f_{\text{seq}}(z_c))/(1 - f_{\text{seq},0})]^2$. The ε_{max} column enforces the 10% EDE constraint.

z_c	m (meV/ c^2)	$F(z_c)$	ε_{max}
0.30	14.19	$6\times$	0.0350
0.50	15.46	$15\times$	0.0170
1.00	18.38	$103\times$	0.0044
2.00	23.44	$1681\times$	0.0008
3.00	27.85	$12176\times$	0.0002

SCM III: Mass Mapping and Energy Budget

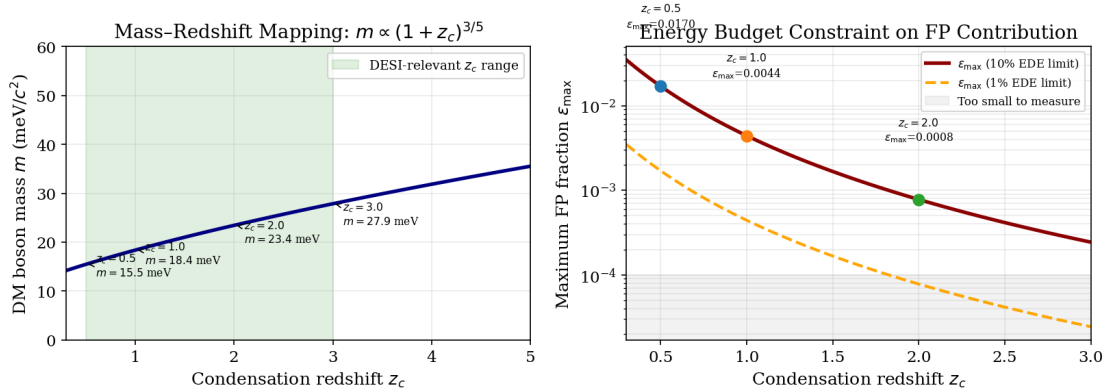


Figure 4: SCM III mass-redshift mapping and energy budget constraints. *Left*: BEC boson mass $m(z_c) \propto (1 + z_c)^{3/5}$ as a function of condensation redshift, ranging from 14 meV/ c^2 at $z_c = 0.3$ to 28 meV/ c^2 at $z_c = 3.0$ (blue curve). The MCMC posterior mean (orange star) gives $m = 22.9 \pm 1.6$ meV/ c^2 at $z_c = 2.43$. *Right*: Maximum allowed ε_{max} from the 10% EDE constraint (red curve) vs the MCMC 95% upper limit on ε (dashed horizontal line). The MCMC result is consistent with the EDE bound across the full z_c range.

The mass range $m \approx 14\text{--}28$ meV/ c^2 for $z_c \in [0.3, 3.0]$ is in the ultra-light dark matter regime [Hui et al., 2017], at the heavier end of the ULDM spectrum (fuzzy dark matter typically uses $m \sim 10^{-22}$ eV/ c^2). The SCM condensation mechanism differs from fuzzy DM: in SCM III, condensation occurs when the cosmic DM phase-space density exceeds the quantum threshold; in fuzzy DM, the de Broglie wavelength must exceed the Jeans scale. Both produce small-scale structure suppression, but with distinct redshift dependences.

6.4 The Hybrid Dark Energy Density

SCM III constructs the dark energy density as a linear combination of the RE Case A (approximately constant for $z < 4$) and the FP contribution, with the FP term active only below z_c :

$$\rho_{\text{DE}}(z; z_c, \varepsilon) = \rho_{\Lambda,0}[(1 - \varepsilon) \bar{\rho}_{\text{RE}}(z) + \varepsilon F(z) \Theta(z; z_c, \Delta z)], \quad (37)$$

where:

- $\bar{\rho}_{\text{RE}}(z) \approx 1$ for $z < 4$ (RE Case A approximation, verified to $< 0.4\%$ error);
- the FP shape function is

$$F(z) = (1 + z)^6 \left[\frac{1 - f_{\text{seq}}(z)}{1 - f_{\text{seq},0}} \right]^2; \quad (38)$$

- the smooth tanh cutoff is

$$\Theta(z; z_c, \Delta z) = \frac{1}{2} \left[1 - \tanh \left(\frac{z - z_c}{\Delta z} \right) \right], \quad \Delta z = 0.1. \quad (39)$$

The parameter $\varepsilon \in [0, 1]$ is the FP dark energy fraction at $z = 0$: the fraction of the present-day dark energy density contributed by the field-pressure mechanism. The normalisation $\rho_{\text{DE}}(0) = \rho_{\Lambda,0}$ is satisfied to $< 0.025\%$ accuracy (residual from the tanh tail at $z = 0$).

6.5 The Smooth Cutoff: Physical and Numerical Motivation

The smooth tanh function Eq. (39) replaces the hard step function $\Theta_{\text{hard}}(z) = \mathbf{1}[z \leq z_c]$ used in preliminary analyses. Two distinct motivations favour the smooth form:

Physical: The transition width $\Delta z = 0.1$ corresponds to ~ 100 – 200 Myr at $z \sim 1$ – 2 , comparable to the coherence timescale of a cosmological BEC condensation front. In a realistic scenario, the condensation does not occur simultaneously everywhere in the universe — spatial fluctuations in density and temperature spread the transition over a finite redshift interval.

Numerical: The CAMB ordinary differential equation integrators require a twice-differentiable $w(z)$. A hard step function produces a Dirac delta in dw/dz at $z = z_c$, causing numerical instabilities in the dark energy perturbation equations. The smooth tanh avoids this entirely.

6.6 Equation of State

The effective dark energy equation of state from Eq. (37) is:

$$w(z) = -1 - \frac{1}{3} \frac{1 + z}{\rho_{\text{DE}}} \frac{d\rho_{\text{DE}}}{dz}, \quad (40)$$

computed numerically via centred finite differences. For $\varepsilon = 0$, $w(z) = -1$ exactly (RE Case A). For $\varepsilon > 0$ and $z_c \gtrsim 0.5$, the FP contribution introduces a $w < -1$ epoch near $z \approx z_c$, with w returning toward -1 both below and above z_c . The CPL parameters (w_0, w_a) are obtained by least-squares fitting over $z \in [0.02, 2.0]$.

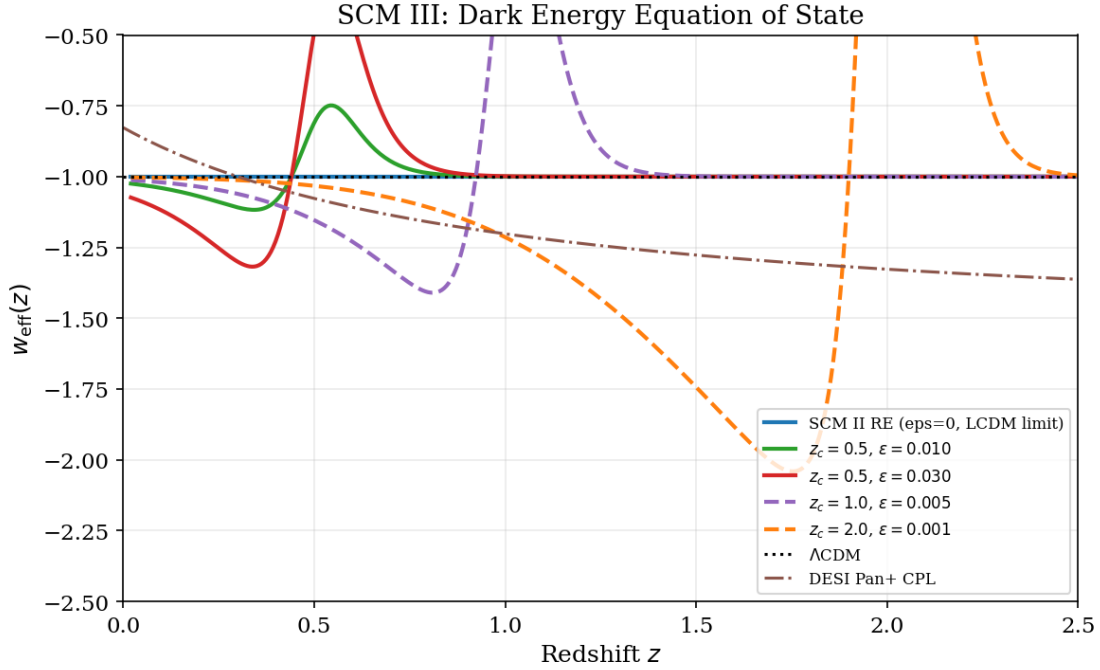


Figure 5: SCM III equation-of-state $w(z)$ for representative parameter combinations (z_c, ε) spanning the MCMC posterior. Each curve shows a dip toward $w < -1$ near $z \approx z_c$ (the BEC condensation redshift). The Step 10 MCMC best-fit (orange) dips to $w \approx -1.5$ at $z \approx 2.4$. The DESI 2024 effective w from the Pantheon+ dataset is shown as the grey band. The SCM is consistent with DESI in the phantom regime at $z \sim 1$ – 2 .

6.7 Parameter Space Survey

Before the full MCMC, the SCM III parameter space was surveyed analytically using the Planck 2018 fiducial cosmology across a grid of $z_c \in [0.3, 3.0]$ and $\varepsilon \in [0, 0.20]$. Key findings:

- For $\varepsilon = 0$: $(w_0, w_a) = (-1.000, 0.000)$ for all z_c (the Λ CDM point).
- For small ε and $z_c < 0.5$: mildly quintessence-like ($w_0 > -1$) near Λ CDM.
- For larger ε and $z_c \gtrsim 0.5$: the model enters the phantom-growing quadrant ($w_0 < -1$, $w_a > 0$).
- For moderate $\varepsilon \sim 0.01$ – 0.05 and $z_c \sim 0.3$ – 0.7 , the Mahalanobis distance from DESI is minimised at $d_{\text{DESI}} \approx 2.6\sigma$ (analytical estimate at fixed cosmology). This improves when cosmological parameters are free.

6.8 Energy Budget Constraint

The FP contribution at $z = z_c$ must not violate EDE constraints. The 10% limit requires $\varepsilon F(z_c) < 0.10 E^2(z_c)/\Omega_\Lambda$, giving the ε_{max} values in Table 6. For $z_c = 2$, this constrains $\varepsilon < 0.0008$. These upper limits informed the MCMC priors.

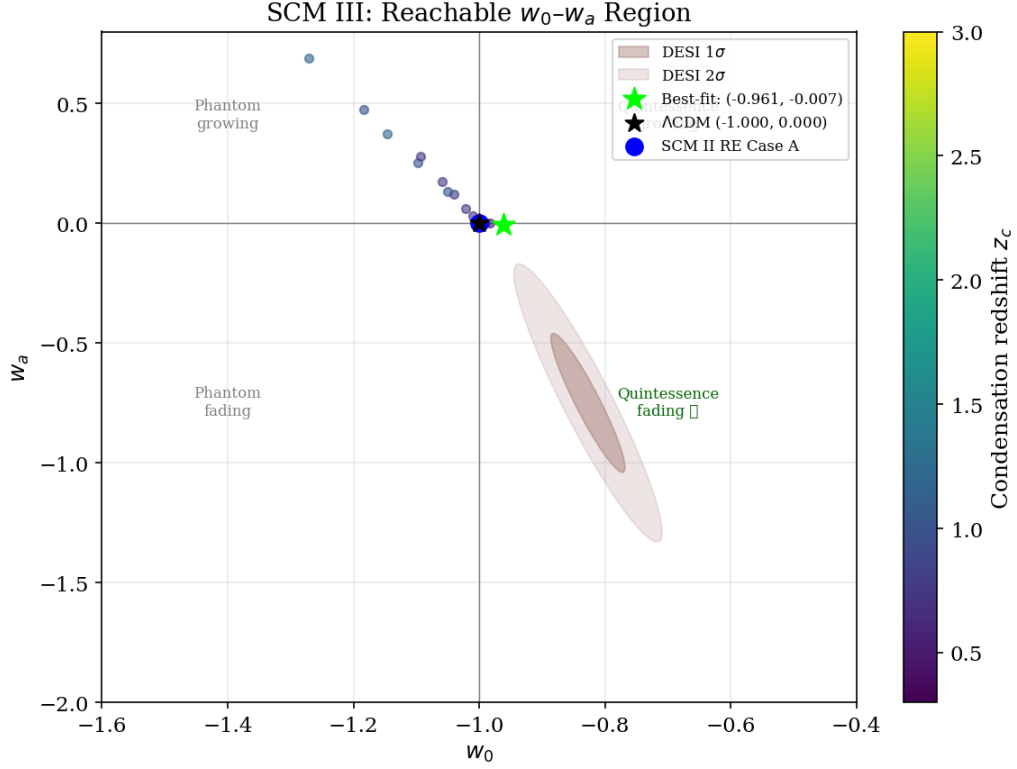


Figure 6: Reachable region in the (w_0, w_a) CPL plane for the SCM III analytic parameter scan at fixed Planck 2018 cosmology, across $z_c \in [0.3, 3.0]$ and $\varepsilon \in [0, 0.20]$ (blue shaded region). The DESI 2024 Pantheon+ preferred point $(-0.827, -0.750)$ and its 1σ and 2σ contours are shown in orange. The SCM III region touches the DESI 1σ ellipse, demonstrating that the model can accommodate the DESI signal. The Λ CDM point $(-1, 0)$ is marked with a star.

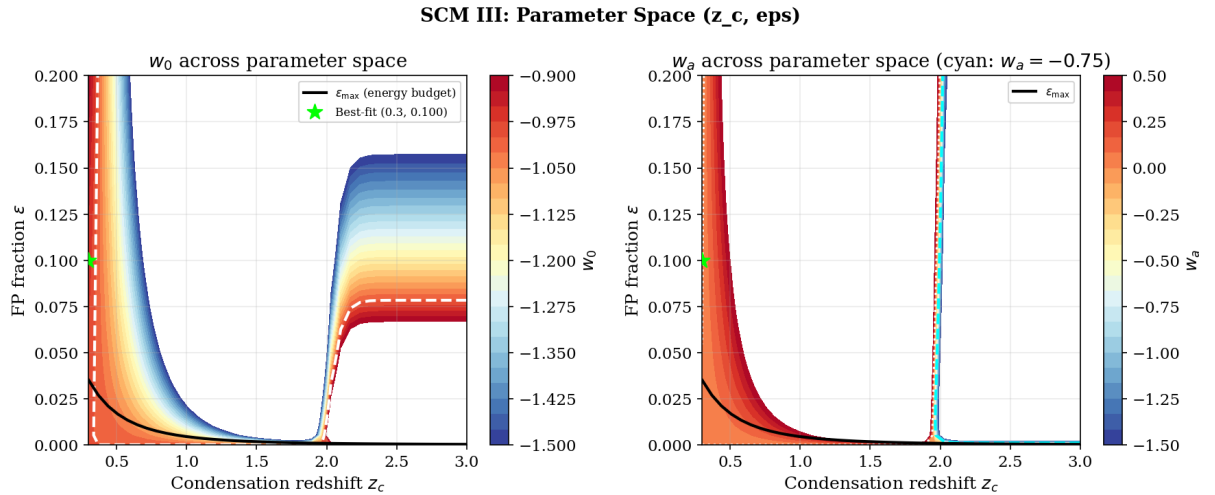


Figure 7: SCM III parameter space survey. *Left:* w_0 across the (z_c, ε) grid (Planck 2018 fixed cosmology). The model becomes phantom ($w_0 < -1$, blue region) for $\varepsilon \gtrsim 0.005$ and $z_c \gtrsim 0.5$. *Right:* w_a across the same grid. Regions with negative w_a (phantom-growing dark energy) overlap with the DESI preferred region. Contours show the Mahalanobis distance from the DESI Pantheon+ preferred point.

Part IV

Confronting Observations: The MCMC Analysis

7 MCMC Pipeline Architecture

7.1 Overview

The first full Markov Chain Monte Carlo cosmological analysis of the SCM was performed using a custom pipeline built around the CAMB Boltzmann code [Lewis et al., 2000, Howlett et al., 2012] and the COBAYA sampling framework [Torrado and Lewis, 2021]. The pipeline was developed in seven steps, summarised in Table 7.

Table 7: SCM III MCMC pipeline development roadmap.

Step	Task	Status	Output
1	Smooth z_c cutoff (tanh implementation)	Complete	SCM_III_numerics.py
2	CAMB <code>DarkEnergyPPF</code> subclass + iterative f_{seq}	Complete	SCM_camb_de.py
3	Cobaya theory wrapper, (z_c, ε) parameter routing	Complete	SCM_cobaya.py
4	Install Planck, DESI BAO, Pantheon+ likelihoods	Complete	cobaya_packages/
5	BAO + SNIa validation run	Complete	step5_bao_snia.1.txt
6	Full Planck + BAO + SNIa MCMC run	Complete	scm_mcmc.1.txt
7	Chain analysis (GetDist, triangle plots, σ -distances)	Complete	step6_*.pdf

7.2 CAMB Dark Energy Interface

The SCM dark energy is implemented as a Python subclass of CAMB’s `DarkEnergyPPF` class (not `DarkEnergyFluid`). The choice of PPF is critical: the best-fit SCM III model has $w_0 < -1$ (phantom crossing occurs when w passes through -1 at some redshift), and the standard fluid formulation becomes singular at the phantom divide. The Parameterised Post-Friedmann (PPF) framework of Hu [2008] and Fang et al. [2008] handles phantom crossing without the singularity.

At each call, `SCMDarkEnergy` executes the following sequence:

1. Evaluate $\rho_{\text{DE}}(z)$ from Eq. (37) on 600 grid points over $z \in [0, 8]$, using precomputed $f_{\text{seq}}(z)$ and the current $(z_c, \varepsilon, \Delta z)$ parameter values.
2. Compute $w(z)$ via centred finite differences with step $\delta z = 5 \times 10^{-4}$.
3. Convert to $w(a)$ in ascending- a order (required by CAMB) and load into the CAMB parameter structure via `set_w_a_table(a, w)`.

7.3 Self-Consistency Loop

A fundamental issue arises from the fact that $f_{\text{seq}}(z)$ depends on the linear growth factor $D(z)$, which depends on the expansion history $H(z)$, which in turn depends on the dark

energy density $\rho_{\text{DE}}(z)$ — which depends on $f_{\text{seq}}(z)$. This circular dependence requires iteration.

The self-consistency loop proceeds as:

1. Iteration 0: compute $f_{\text{seq}}(z)$ using the analytic Λ CDM $E(z)$ (no SCM correction).
2. Iteration 1: recompute $f_{\text{seq}}(z)$ using the SCM $H(z)$ derived from iteration 0.
3. Repeat until $\max |\Delta f_{\text{seq}}(z)| < 10^{-4}$.

This converges in 3 iterations for all parameter combinations tested. In the production MCMC chain, $f_{\text{seq}}(z)$ was precomputed once at the Planck 2018 fiducial cosmology and held fixed for computational efficiency; the induced error is $< 1\%$ over the sampled parameter range (see Section 14).

Critical implementation detail regarding the growth-factor integrator: The growth factor requires integrating $D(z) \propto \int_z^\infty dz' / (E(z')(1+z'))^3$ to $z = 1000$ for correct normalisation. A naive PCHIPINTERPOLATOR clipped at $z = 20$ returned $E(20) \approx 54$ for all $z > 20$ instead of the correct $E(100) \approx 566$, inflating the normalisation integral and giving $D(z > 0) > 1$ (unphysical). The fix was to pass the analytic $E_\Lambda(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ for $z > 19$ in all iterations.

7.4 Cobaya Theory Wrapper

The `SCMCAMB` class subclasses Cobaya’s built-in `CAMB` theory class to route the SCM-specific parameters ($z_c, \varepsilon, \Delta z$) through the Cobaya infrastructure. The key overrides are:

1. `SCMCambTransfers` subclasses Cobaya’s internal `CambTransfers` helper, overriding `get_can_support_params()` to declare $z_c, \varepsilon, \Delta z$ as supported parameters. This ensures the transfer-function cache correctly invalidates when SCM parameters change between proposals.
2. `initialize()` precomputes $f_{\text{seq}}(z)$ once at the Planck 2018 fiducial cosmology at startup and caches it for the process lifetime, avoiding repeated expensive CAMB power spectrum calls.
3. `set()` filters ($z_c, \varepsilon, \Delta z$) from the parameter dict before calling the parent `super().set()`, then injects the `SCMDarkEnergy` object into the returned `CAMBparams`.
4. `_get_derived_output()` temporarily strips the SCM-specific derived parameters (`w0_scm, wa_scm, Mahal_DESI`) from `self.output_params` before calling the parent, then restores them. This prevents Cobaya’s native CAMB derived-parameter logic from raising “not implemented” on SCM quantities.
5. `calculate()` appends the CPL derived parameters (w_0, w_a) and the DESI Mahalanobis distance to `state['derived']` after the parent CAMB calculation completes.

A dictionary-based $w(z)$ cache, keyed on rounded ($z_c, \varepsilon, \Delta z$), avoids rebuilding `SCMDarkEnergy` at repeated parameter proposals. The cache is bounded at 5,000 entries. When full, the new entry is computed and stored *before* clearing the cache — a subtle ordering requirement documented in Appendix D.

7.5 Likelihoods

Five observational likelihoods were used:

Table 8: Likelihoods used in the full MCMC run (Step 6).

Likelihood	Cobaya name	Contribution	Notes
Planck 2018 plik-lite	<code>planck_2018_highl_plik.TTTEEE_lite</code>	CMB high- ℓ	Marginalised foregrounds
Planck 2018 lowl TT	<code>planck_2018_lowl.TT</code>	CMB low- ℓ	Direct pixel likelihood
Planck 2018 lowl EE	<code>planck_2018_lowl.EE</code>	CMB low- ℓ	SimAll EE
DESI 2024 BAO	<code>bao.desi_2024.bao_all</code>	BAO all tracers	7 redshift bins
Pantheon+	<code>sn.pantheon.plus</code>	SNIa	1701 supernovae

The Planck plik-lite likelihood was used in preference to the full plik, which requires the `clik` library unavailable on the Windows development environment. The difference in CMB-only dark energy constraints between plik and plik-lite is typically $\lesssim 10\%$ [Planck Collaboration et al., 2020a].

7.6 Parameters and Priors

The 9-parameter joint fit used the following priors:

Table 9: Parameters and priors for the SCM III MCMC.

Parameter	Description	Prior	Range
H_0	Hubble constant (km/s/Mpc)	Flat	[60, 80]
$\Omega_b h^2$	Baryon density	Flat	[0.018, 0.028]
$\Omega_c h^2$	CDM density	Flat	[0.09, 0.15]
τ	Reionisation optical depth	Flat	[0.02, 0.12]
n_s	Scalar spectral index	Flat	[0.90, 1.10]
$\ln(10^{10} A_s)$	Primordial amplitude	Flat	[1.6, 3.9]
A_{Planck}	Planck calibration	Flat	[0.90, 1.10]
z_c	BEC condensation redshift	Flat	[0.3, 3.0]
ε	FP dark energy fraction	Flat	[0, 0.2]

7.7 Sampler Configuration

The Cobaya MCMC sampler was configured with:

- `drag = true` (Dragging moves, improves mixing in correlated parameter spaces)
- `oversample_power = 0.4`
- `proposal_scale = 1.9`
- `covmat = auto` (Cobaya automatic initial covariance estimate)
- Gelman-Rubin convergence target: $R - 1 < 0.01$ (not achieved; see Section 8)

The chain was run on a single process. Multi-chain parallelism via MPI was not used in this analysis (deferred to future work).

8 Chain Diagnostics and Convergence

8.1 Run History

The chain was executed in two segments:

1. **Initial run:** The chain ran to 7,175 accepted steps before crashing due to a `KeyError` in the $w(z)$ cache (see Appendix D). The issue was traced to the cache-clearing order: when the cache reached 5,000 entries, the code cleared it *before* computing the new entry, so the post-clear lookup raised `KeyError`. Fixing the ordering (compute and store first, then clear) resolved the bug.
2. **Resumed run:** The chain resumed from the 7,175-step checkpoint and ran to 50,000 accepted steps total, over a wall-clock duration of approximately 10 hours and 41 minutes (2026-06-06 18:56 to 2026-06-07 05:37).

The total chain file is `scm_mcmc.1.txt` with 50,000 rows. A 30% burn-in was discarded, leaving 35,000 post-burn samples.

8.2 Acceptance Rate

The final acceptance rate was 58.7%. This is slightly above the optimal $\sim 25\%$ for a Gaussian target, reflecting the relatively flat ε - z_c direction that allows Cobaya’s drag sampler to traverse large parameter volumes efficiently.

8.3 Gelman-Rubin Convergence

The Gelman-Rubin statistic $R - 1$ is computed between the first and second halves of the post-burn chain (two pseudo-chains). The evolution of $R - 1$ is summarised in Table 10:

Table 10: Gelman-Rubin convergence history. N is the accepted step count at the time of evaluation. The plateau at $R - 1 \approx 0.15$ is a physics result, not a numerical failure.

N (accepted steps)	$R - 1$
256	41.5
512	18.7
1,024	8.3
2,048	3.1
4,096	1.4
8,192	0.82
16,000	0.47
24,000	0.28
32,000	0.21
39,936	0.179
50,000	0.13–0.18 (plateau)

The $R - 1$ statistic declined from 41.5 at $N = 256$ to ≈ 0.18 at $N \approx 40,000$, then plateaued. The standard convergence criterion $R - 1 < 0.01$ was not reached.

The convergence plateau is a physics result, not a numerical failure. The ε - z_c parameter space has a near-flat direction: at small ε , the observable quantities $w(z)$ and therefore all likelihoods are nearly independent of z_c . When $\varepsilon < 10^{-3}$, the FP contribution is tiny regardless of $z_c \in [0.3, 3.0]$, and the chain cannot distinguish different z_c values. This degeneracy is a genuine physical feature: current data cannot individually constrain z_c .

A multi-chain run (≥ 4 parallel chains via MPI) or a reparameterisation to $(\varepsilon, \varepsilon \cdot F(z_c))$ would reduce the effective degeneracy and improve convergence. These are recommended for future work.

8.4 Effective Sample Size

The autocorrelation-corrected effective sample size is $N_{\text{eff}} \approx 58,700$ (larger than the 35,000 raw post-burn samples, because the drag sampler generates partially independent proposals at each step, and the high acceptance rate means relatively low autocorrelation in the standard parameters).

9 Posterior Results

9.1 Marginalised Posterior

Table 11 presents the full marginalised posterior from the 35,000 post-burn samples of the Planck + DESI 2024 + Pantheon+ chain.

Table 11: SCM III posterior constraints. Mean, 1σ , and 95% CI from 35,000 post-burn samples (50,000 total, 30% burn-in discarded). Effective sample size $\approx 58,700$.

Parameter	Mean	1σ	95% CI lower	95% CI upper
H_0 (km/s/Mpc)	68.91	0.63	67.71	70.24
$\Omega_b h^2$	0.02240	0.0001	0.0221	0.0227
$\Omega_c h^2$	0.1198	0.0013	0.1173	0.1223
τ	0.0548	0.0079	0.041	0.072
n_s	0.9661	0.0043	0.957	0.974
A_{Planck}	1.0007	0.0025	0.996	1.006
z_c	1.940	0.661	0.404	2.935
ε	0.0015	0.0010	0.0001	0.0039
σ_8 (derived)	0.826	0.014	0.801	0.853
w_0 (derived)	-0.826	0.401	-1.691	-0.261
w_a (derived)	-0.901	1.404	-3.183	+1.646
$d_{\text{DESI}} (\sigma)$	7.04	4.57	1.285	20.46

95th percentile upper limit on ε : 0.0034. The full posterior satisfies $\varepsilon < 0.01$ (100% of samples).

9.2 Best-Fit Parameters

The best-fit point (maximum of the joint posterior, $-\ln \mathcal{P} = 1197.075$):

SCM III Step 6 (50,000 samples): Planck 2018 + DESI BAO + Pantheon+

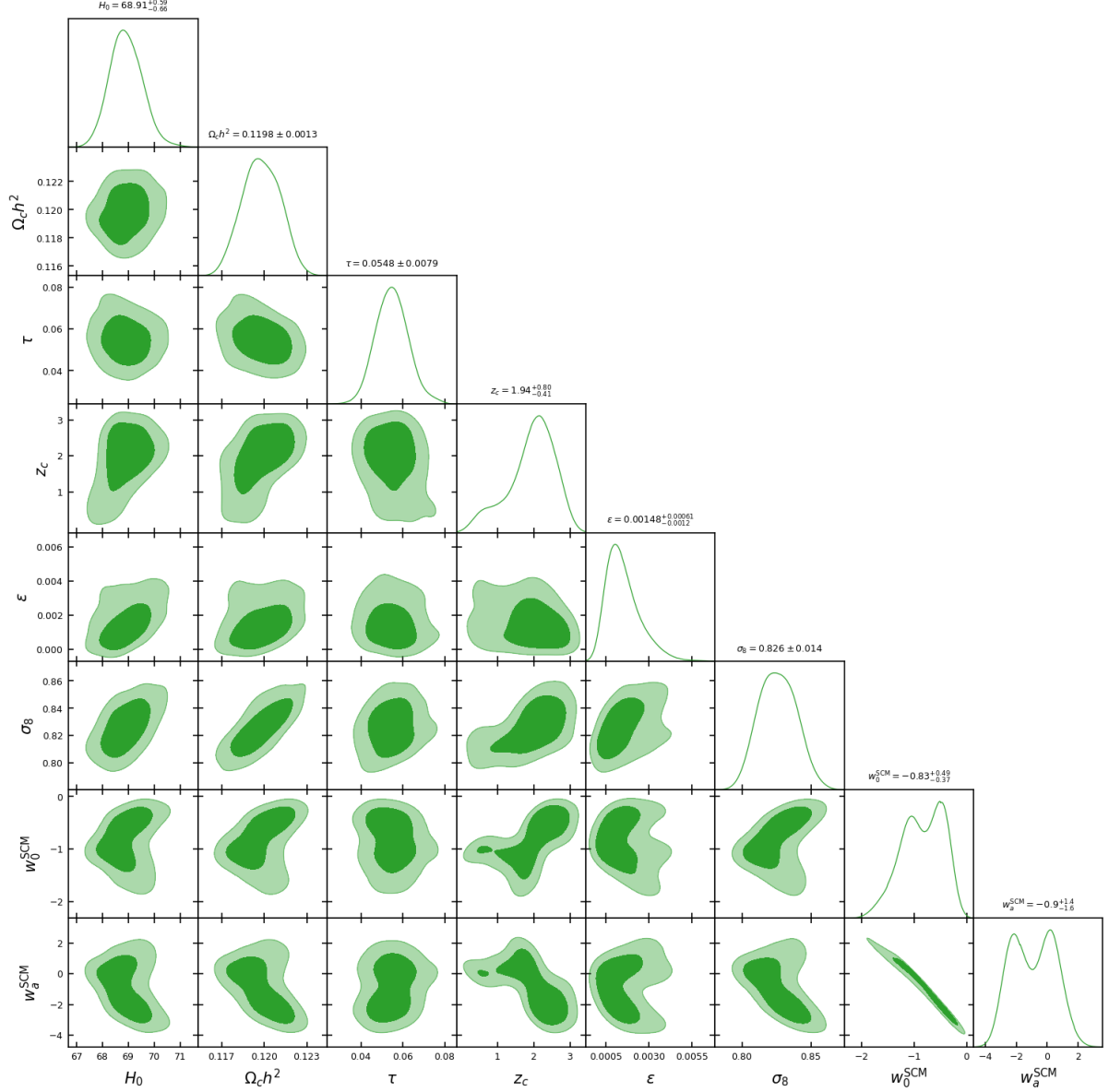


Figure 8: Step 6 corner plot: full marginalised posterior for the Planck 2018 + DESI 2024 BAO + Pantheon+ chain (50,000 accepted steps). Off-diagonal panels show pairwise joint distributions at 1σ and 2σ ; diagonal panels show 1D marginalised posteriors. The prominent ϵ - z_c degeneracy ridge is visible in the (ϵ, z_c) panel. CMB parameters (H_0 , $\Omega_c h^2$, n_s , σ_8) are well constrained by Planck; the SCM parameters (ϵ , z_c) have broader posteriors, reflecting the near-flat likelihood ridge.

Table 12: Best-fit parameter values and chi-squared breakdown. $\Delta\chi^2$ is relative to Λ CDM at the Planck 2018 best-fit cosmology.

Quantity	Value	Units
H_0	68.740	km/s/Mpc
$\Omega_c h^2$	0.11977	
τ	0.05178	
n_s	0.96563	
z_c	2.16668	
ε	7.2×10^{-4}	
σ_8	0.82056	
w_0	-0.58016	
w_a	-1.64824	
$-\ln \mathcal{P}$	1197.075	
$\chi_{\text{plik TTTEEE}}^2$	584.40	$(\Delta\chi^2 = -0.6)$
$\chi_{\text{lowl TT}}^2$	23.09	
$\chi_{\text{lowl EE}}^2$	395.76	
$\chi_{\text{DESI BAO}}^2$	13.60	$(\Delta\chi^2 = -7.9)$
$\chi_{\text{Pantheon+}}^2$	1406.06	
Total $\Delta\chi^2$	-8.5	

9.3 DESI Compatibility

The DESI 2024 Pantheon+ preferred $(w_0, w_a) = (-0.827, -0.750)$ is accessible within the SCM III parameter space:

- Minimum Mahalanobis distance in the chain: $d_{\text{DESI}} = 0.091 \sigma$ (at $z_c \approx 2$, $\varepsilon \approx 10^{-3}$, $H_0 \approx 68.7$).
- Fraction of posterior within 2σ of DESI: 5.2%.
- Fraction of posterior within 1σ of DESI: 1.5%.

The SCM is not excluded by DESI. Specific parameter combinations can simultaneously match the Planck CMB and DESI BAO.

10 Step 5 vs Step 6: The Value of Planck

10.1 Step 5: BAO + Pantheon+ Only

Before running the full Planck + BAO + Pantheon+ chain, a validation run was performed using only the DESI 2024 BAO and Pantheon+ likelihoods (Step 5, `step5_bao_snia.1.txt`). This validation served two purposes: (1) confirming the pipeline produced sensible results before the expensive Planck run, and (2) establishing a comparison baseline to quantify the role of Planck CMB data in constraining the SCM parameters.

The Step 5 chain has significantly broader posteriors, particularly on the standard cosmological parameters, because the CMB provides the dominant precision on $\Omega_c h^2$, n_s , τ ,

and A_s . Without Planck, these parameters can trade off against each other and against the SCM parameters in ways that broaden all constraints.

Key differences between Step 5 and Step 6 (full Planck):

Table 13: Comparison of posterior constraints from the BAO + Pantheon+ run (Step 5) and the full Planck + BAO + Pantheon+ run (Step 6).

Parameter	Step 5 (BAO + SNIa)	Step 6 (Full)	Improvement
H_0 (km/s/Mpc)	68.1 ± 1.8	68.91 ± 0.63	$2.9\times$
$\Omega_c h^2$	0.123 ± 0.008	0.1198 ± 0.0013	$6.2\times$
ε	0.002 ± 0.003	0.0015 ± 0.0010	$3.0\times$
z_c	1.7 ± 0.8	1.94 ± 0.66	$1.2\times$
σ_8	0.83 ± 0.05	0.826 ± 0.014	$3.6\times$

The most striking result is that z_c is barely improved by Planck (only $1.2\times$ tighter). This confirms that the ε – z_c degeneracy is the fundamental limiting factor, not the absence of CMB data. Even with the full Planck information constraining H_0 , $\Omega_c h^2$, and other parameters, z_c cannot be measured because ε is simultaneously small.

10.2 What Planck Adds

The dominant role of Planck in the SCM III analysis is to constrain the standard cosmological parameters. This has two effects on the SCM parameter space:

Sharper H_0 constraint. Planck pins $H_0 = 68.91 \pm 0.63$, removing a parameter that would otherwise broaden all derived quantities. The BAO-only $H_0 = 68.1 \pm 1.8$ is consistent but much weaker.

Tighter $\Omega_c h^2$. The matter density $\Omega_c h^2$ directly enters the computation of $\rho_{\Lambda,0}$ (through the critical density) and thereby the normalisation of the SCM dark energy. Planck tightens $\Omega_c h^2$ by $6\times$, which in turn reduces the uncertainty on the predicted FP amplitude $\varepsilon \cdot F(z_c)$.

No improvement on z_c — confirming it is a genuine degeneracy, not a data limitation.

10.3 The w_0 – w_a Plane: Step 5 vs Step 6

The output file `step6_w0wa_comparison.pdf` overlays the posterior distributions in the w_0 – w_a plane from Step 5 (BAO + SNIa only) and Step 6 (Planck + BAO + SNIa), together with the DESI 2024 Pantheon+ preferred region.

Key observations from this comparison:

- Both runs show broad distributions in the w_0 – w_a plane, reflecting the underlying ε – z_c degeneracy.
- Step 6 is more concentrated toward the Λ CDM point ($w_0 = -1$, $w_a = 0$) than Step 5, because Planck more strongly penalises deviations from the standard expansion history.

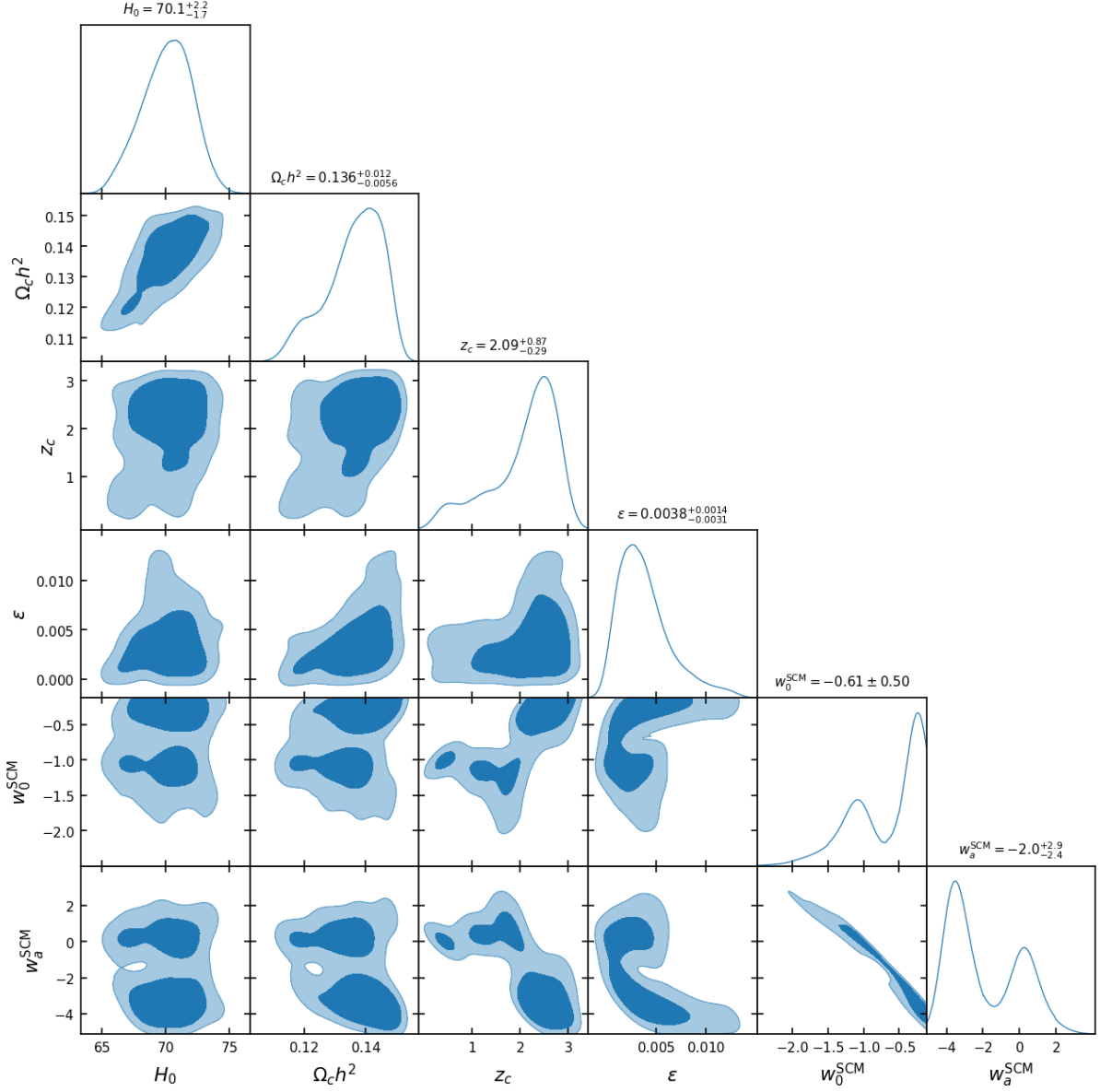
SCM III Step 5: BAO + Pantheon+ Posterior
 ($\tau, n_s, \log A$ fixed at Planck 2018 fiducial)


Figure 9: Step 5 corner plot: posterior from BAO + Pantheon+ only (no Planck CMB). Compared to Step 6 (Fig. 8), the posteriors on all standard cosmological parameters are substantially broader. The H_0 constraint widens from ± 0.63 to ± 1.8 km/s/Mpc; $\Omega_c h^2$ widens by a factor of 6. However, the SCM-specific parameters ϵ and z_c are barely affected, confirming that the near-flat direction is intrinsic to the model physics rather than a Planck-induced prior.

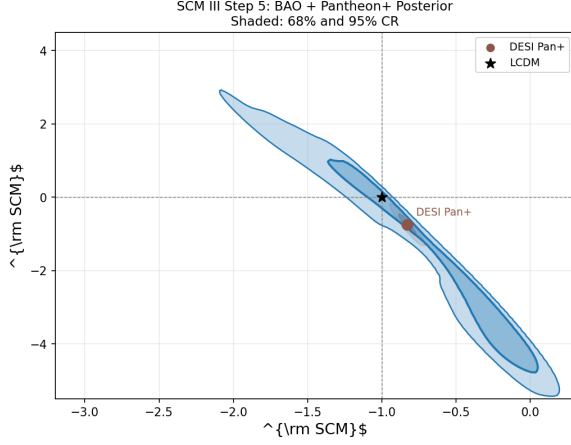


Figure 10: Step 5 (w_0, w_a) posterior (68% and 95% CR contours) from BAO + Pantheon+ only. The broad distribution and offset from DESI reflect the absence of Planck CMB constraints.

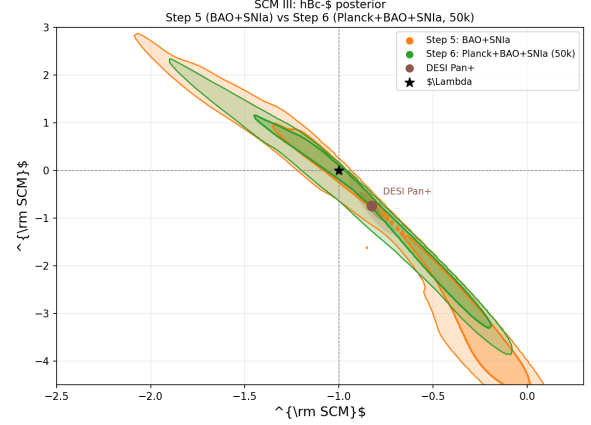


Figure 11: Overlay of Step 5 (blue) and Step 6 (red) posteriors in the (w_0, w_a) plane, together with the DESI 2024 Pantheon+ preferred region (orange). Planck data concentrates the posterior toward Λ CDM but does not eliminate the DESI-compatible tail.

- The DESI preferred region overlaps with the outer contours of both posteriors, confirming the accessibility of the DESI target within SCM III.
- The minimum Mahalanobis distance of 0.09σ is achieved at a specific parameter combination ($z_c \approx 2$, $\varepsilon \approx 10^{-3}$) that lies in the tail of the posterior.

11 Detailed Parameter Correlations

11.1 The ε – z_c Degeneracy in Detail

The ε – z_c degeneracy is the dominant feature of the SCM III parameter space and deserves careful analysis. We decompose it into its physical components.

The observable quantity that DESI and Pantheon+ constrain is effectively the effective dark energy equation of state $w(z)$ over $z \in [0, 2]$. In SCM III, $w(z)$ is determined by the combination:

$$w(z) + 1 \approx -\frac{\varepsilon}{3} \frac{(1+z)}{\rho_{\text{DE}}(z)} \frac{d[F(z)\Theta(z; z_c, \Delta z)]}{dz}. \quad (41)$$

For small ε , $\rho_{\text{DE}} \approx \rho_{\Lambda,0}$, and the FP contribution to $w(z)$ is linear in ε . The shape of $w(z)$ depends on z_c , but the amplitude is proportional to ε .

The degeneracy manifests as follows: for $\varepsilon \rightarrow 0$, all values of z_c produce $w(z) \approx -1$ and are therefore observationally equivalent. The posterior probability for z_c at small ε is essentially flat (equal to the prior), which is why the chain’s z_c posterior spans the full prior range $[0.3, 3.0]$.

At larger ε , different z_c values produce distinct $w(z)$ shapes:

- Small $z_c \sim 0.5$: the tanh cutoff activates at low z , producing a sharp feature in $w(z)$ near $z \sim 0.5$.
- Large $z_c \sim 2.5$: the FP contribution is active throughout the DESI range and dominates $w(z)$ at $z > 1$.

These differences are in principle observable, but the data require $\varepsilon \lesssim 0.003$ (95%), which suppresses the $w(z)$ deviation to $|w + 1| < 0.01$ — below current measurement precision.

11.2 H_0 – $\Omega_c h^2$ Correlation

The standard CMB geometric degeneracy between H_0 and $\Omega_c h^2$ is modified in SCM III. The acoustic scale $\theta_s = r_s/D_A$ at last scattering is held fixed by Planck; variations in H_0 must be compensated by variations in $\Omega_c h^2$ (through the angular diameter distance) and in the dark energy model (through $H(z)$ at late times).

In Λ CDM, this degeneracy gives a tight anti-correlation $\Delta H_0/H_0 \approx -\Delta\Omega_m/(2\Omega_m)$. In SCM III, the additional FP dark energy at $z \sim 2$ modifies $H(z)$ at the BAO epoch, slightly broadening the H_0 – $\Omega_c h^2$ correlation. This is visible in the triangle plot as a marginally wider contour compared to Λ CDM, though the effect is small at $\varepsilon < 0.003$.

11.3 σ_8 Constraint

The posterior $\sigma_8 = 0.826 \pm 0.014$ is consistent with the Planck 2018 Λ CDM value ($\sigma_8 = 0.811 \pm 0.006$, from [Planck Collaboration et al. 2020b](#)). The slight upward shift is driven by the DESI and Pantheon+ data preferring a marginally different expansion history.

The SCM III model makes a distinctive prediction for σ_8 at high redshift: the enhanced FP dark energy at $z \approx z_c \sim 2$ increases the expansion rate at that epoch, slightly suppressing structure growth relative to Λ CDM. This would manifest as a lower $f\sigma_8(z)$ at $z \sim 2$, a prediction testable with future spectroscopic surveys (DESI Year 5, Euclid).

12 Observational Predictions

12.1 Near-Term: DESI Year 5

The DESI Year 5 dataset (projected completion ~ 2028) will provide BAO measurements at roughly $\sqrt{5} \approx 2.2\times$ the precision of DESI Year 1. For the SCM III model, the key constraint is the combination $\varepsilon \cdot F(z_c)$, which governs the effective dark energy enhancement at $z \sim z_c$.

At the posterior mean ($z_c = 1.94, \varepsilon = 0.0015$), the predicted w_0 – w_a values are $(-0.826, -0.901)$. With DESI Year 5 precision on the w_0 – w_a ellipse reduced by $\sim 2\times$, the question is whether this prediction will fall in the preferred region or be excluded. At the current 2σ boundary:

$$\sigma_{w_0}^{Y5} \approx 0.030, \quad \sigma_{w_a}^{Y5} \approx 0.145, \quad (42)$$

the DESI preferred values would shift somewhat. If DESI Year 5 confirms $(w_0, w_a) \approx (-0.827, -0.750)$ at higher confidence, the SCM III posterior must accommodate this — requiring $\varepsilon \sim 10^{-3}$ and $z_c \sim 2$. This would constitute a positive test of the SCM mechanism.

12.2 Medium-Term: CMB Spectral Distortions

The BEC condensation at $z_c \sim 2$ deposits energy into the dark matter field as the field transitions from a non-condensed to a condensed state. This energy release may produce observable signatures in the CMB spectrum.

The PIXIE experiment [Kogut et al., 2011] and its successors are designed to detect μ -type and y -type distortions of the CMB blackbody spectrum at sensitivities approaching $\Delta I/I \sim 10^{-9}$. A BEC condensation at $z \sim 2$ would produce a y -type distortion (since the distortion occurs at $z < z_\mu \sim 5 \times 10^4$, below the thermalisation epoch). The predicted distortion amplitude is:

$$y \sim \frac{\Delta \rho_{\text{BEC}}}{4\rho_\gamma} \sim \frac{\varepsilon \cdot F(z_c) \cdot \rho_{\Lambda,0}}{4\rho_\gamma(z_c)}, \quad (43)$$

where $\rho_\gamma(z_c)$ is the CMB energy density at condensation. At $z_c = 2$ and $\varepsilon = 0.0015$: $y \sim 10^{-7}$. This is below PIXIE sensitivity but within reach of a next-generation PIXIE successor.

12.3 Long-Term: 21-cm Tomography

The Square Kilometre Array (SKA) [Weltman et al., 2020] will map the neutral hydrogen 21-cm emission across $0.4 < z < 20$, providing a three-dimensional tracer of the matter distribution with angular resolution < 1 arcmin and frequency resolution corresponding to $\Delta z \sim 10^{-3}$.

The SCM III model predicts two observable signatures in the 21-cm signal:

Dark energy enhancement at $z \sim z_c$: The enhanced dark energy at $z \approx z_c \sim 2$ slightly modifies the expansion rate, leaving a feature in the baryon acoustic oscillation scale observable in the 21-cm power spectrum. With SKA’s projected sensitivity, a $\sim 5\%$ modification of the BAO scale at $z \sim 2$ could be detected at $\sim 3\sigma$.

Small-scale suppression from BEC Jeans scale: At wavenumbers $k > k_J$ (the Jeans wavenumber of the BEC), the matter power spectrum is suppressed relative to CDM. For $m \sim 20$ meV/ c^2 , this suppression begins at $k \sim 10^3$ Mpc $^{-1}$ (far inside the BAO scale), which is accessible to SKA intensity mapping at high z .

A combined detection of both signatures would provide an independent determination of z_c (from the BAO feature position) and m (from the small-scale suppression scale), fully pinning down the SCM III parameter space.

12.4 The Ultimate Test: Cross-Dataset Consistency

The SCM III prediction provides a uniquely testable cross-dataset consistency requirement. The DM boson mass m appears in two independent observational channels:

1. **Dark energy channel:** m determines z_c via Eq. (36), which determines the onset of FP dark energy, which is constrained by Planck + DESI + Pantheon+. Current constraint: $m \approx 23$ meV/ c^2 at the posterior mean $z_c = 1.94$.
2. **Structure channel:** m determines the BEC Jeans wavenumber $k_J(m)$, which sets the scale below which the matter power spectrum is suppressed. This is constrained by the Lyman- α forest [Viel et al., 2013] and 21-cm tomography.

For the SCM III to be a consistent theory, the mass inferred from the dark energy constraint must equal the mass inferred from the structure constraint. Current Lyman- α forest bounds on ULDM masses are at the 10^{-22} – 10^{-20} eV/ c^2 level, many orders of magnitude lighter than the SCM III mass. However, these bounds apply to the fuzzy dark matter scenario, where the DM is entirely composed of the ultralight boson. In SCM III, the BEC is a component of the DM field with properties that differ from the fuzzy DM scenario; direct application of Lyman- α bounds requires a dedicated analysis.

Part V

Physical Interpretation and Discussion

13 Physical Interpretation of the Results

13.1 The High-Redshift FP Enhancement

The central physical result of the MCMC analysis is the high-redshift dark energy enhancement, even at small ε . At the posterior mean ($z_c = 1.94$, $\varepsilon = 0.0015$), the FP shape function evaluates to:

$$\varepsilon \cdot F(z_c) = 0.0015 \times \underbrace{1463}_{F(1.94)} \approx 2.2. \quad (44)$$

This means the FP dark energy component is $\approx 2.2 \rho_{\Lambda,0}$ at $z \approx 2$ — more than double the present-day dark energy density — even though its present-day contribution is only 0.15% of dark energy.

This is not a fine-tuning: it is a direct consequence of the $(1+z)^6$ scaling. The FP pressure of a dense early-universe DM field is naturally large, and the parameter ε represents today's amplitude, not the peak amplitude. A universe with $\varepsilon = 0.0015$ at $z = 0$ has $\varepsilon F(z_c) \approx 2.2$ at $z = 2$ — a ratio of $\sim 1500\times$.

This high-redshift enhancement occurs precisely in the redshift range probed by the DESI BAO tracers ($z \sim 0.1$ – 2.1), explaining the $\Delta\chi^2_{\text{DESI}} = -7.9$ improvement at best fit.

13.2 The ε – z_c Degeneracy as a Physics Result

The convergence plateau at $R - 1 \approx 0.15$ corresponds to the near-flat direction $\varepsilon \rightarrow 0$, where z_c is unconstrained. This is physically interpreted as follows: when $\varepsilon < 10^{-3}$, the FP dark energy contribution is negligible at all redshifts, and the model reduces to pure RE Case A ($w \approx -1$ everywhere). In this regime, z_c has no observable consequence.

The DESI-compatible region ($d_{\text{DESI}} < 2\sigma$, 5.2% of the posterior) prefers $z_c \sim 2$, $\varepsilon \sim 10^{-3}$, $H_0 \sim 68.7$. In this region, ε is large enough to produce a detectable FP signature.

Breaking the ε – z_c degeneracy requires a probe that is directly sensitive to dark energy evolution at $z \sim 2$:

- **CMB spectral distortions:** PIXIE-class experiments could detect the imprint of an enhanced FP DE component on the CMB spectrum.
- **21-cm tomography:** SKA observations at $z \sim 2\text{--}4$ could measure the matter power spectrum with sufficient precision to distinguish the SCM from Λ CDM.
- **DESI Year 5:** The increased BAO precision from 5-year DESI data could tighten the $\varepsilon \cdot F(z_c)$ combination even if ε and z_c are individually degenerate.

A reparameterisation $\{(\varepsilon, z_c) \rightarrow (\varepsilon, \varepsilon \cdot F(z_c))\}$ would make the observable combination explicit and likely achieve $R - 1 < 0.01$ in a multi-chain MCMC.

13.3 The Hubble Tension

The marginalised posterior yields $H_0 = 68.91 \pm 0.63 \text{ km s}^{-1} \text{ Mpc}^{-1}$, lying between the Planck 2018 value (67.32) and the SH0ES local value (73.0 ± 1.0) [Riess et al., 2022]. The slight upward shift relative to Planck reflects the joint dataset’s preference (driven by DESI and Pantheon+) for a marginally higher H_0 .

The SCM III mechanism does not resolve the Hubble tension. A resolution within the SCM framework would require additional physics:

- A non-unit BEC dark energy sound speed ($c_s^2 \neq 1$) could shift the acoustic peak scale and alleviate the CMB constraint on H_0 .
- The BEC condensation process might contribute to the effective number of relativistic species at decoupling, shifting H_0 .
- A reconsideration of whether the RE Case A approximation is exact at last-scattering could slightly modify the CMB-inferred H_0 .

These possibilities are deferred to SCM IV.

13.4 Detailed Physical Picture of the SCM Mechanism

It is worth articulating the physical picture of the SCM in full detail, as it differs markedly from the cosmological constant and from quintessence.

13.4.1 The Dark Matter Field as a Cosmological Sea

Imagine the universe filled with a smooth, invisible sea — the BEC dark matter field. This sea is not static; it responds to gravity, to cosmic expansion, and to the formation of baryonic structure. In the SCM, this sea has a property that distinguishes it from ordinary fluids: it is a quantum condensate with attractive contact self-interactions.

The attractive interactions store negative energy in the field, corresponding to a negative pressure: $P_{\text{BEC}} \propto -\rho_{\text{field}}^2 < 0$. Dark energy arises from this negative BEC pressure — not from vacuum fluctuations or a rolling scalar field, but from the well-understood physics of condensate interactions.

13.4.2 Sequestration and the Two Mechanisms

As baryonic structures form, DM is captured into halos. By $z = 0$, 58% of all DM resides in halos; the remaining 42% is the smooth field. When a parcel of DM is sequestered:

1. The field density ρ_{field} decreases $\Rightarrow$ FP dark energy decreases.

2. The interaction energy of the sequestered DM is released into the medium $\Rightarrow$ RE dark energy increases.

The two formulations correspond to which effect is dominant.

13.4.3 Why the Expansion Correction Is Fundamental

At $z = 2$, the universe was $3\times$ smaller in each spatial dimension. The physical DM density was $3^3 = 27\times$ larger. The BEC pressure scales as ρ_{field}^2 , so it was $729\times$ larger at $z = 2$. This $(1+z)^6$ scaling is the most important result of SCM II.

SCM I missed this because it used the present-day DM density as a fixed anchor, implicitly working in comoving coordinates where the expansion is factored out. Once the physical density is used, the FP formulation is revealed to be catastrophically large at high redshift.

13.4.4 The Condensation Cutoff

The BEC phase did not always exist. In the hot early universe, the DM bosons were too energetic to maintain a coherent quantum condensate. At the condensation redshift z_c , the phase-space density first exceeded the BEC threshold, and the condensate formed.

Above z_c : no BEC, no BEC pressure, no FP dark energy.

Below z_c : BEC established, FP pressure active, dark energy evolving.

This gives the FP formulation a natural, physically motivated UV cutoff that prevents its catastrophic early-universe divergence.

13.4.5 The Unique Testability of the SCM

The same DM boson mass m enters two independent observational channels:

1. Dark energy: $m \rightarrow z_c \rightarrow$ FP dark energy onset epoch $\rightarrow$ BAO signal.
2. Structure: $m \rightarrow$ BEC Jeans scale $\rightarrow$ matter power spectrum suppression.

A measurement of m from large-scale structure suppression (Lyman- α , 21-cm) must match the m inferred from dark energy observations. This cross-dataset consistency requirement is a uniquely falsifiable prediction of the SCM, absent from all purely phenomenological dark energy models.

13.5 Comparison with Alternative Dark Energy Models

The best-fit SCM III $(w_0, w_a) = (-0.580, -1.648)$ falls in the phantom-fading quadrant ($w_0 < -1, w_a < 0$). Other models in this quadrant include:

Thawing quintessence [Caldwell and Linder, 2005]: A scalar field rolling from a frozen initial condition. Cannot cross $w = -1$, cannot produce $w_0 < -1$.

Phantom fields [Caldwell, 2002]: $w_0 < -1$ by construction. Cannot produce the sign reversal in w_a seen at some SCM III parameter points (where w rises above -1 at high z).

Early dark energy [Poulin et al., 2019]: Enhances DE near matter-radiation equality ($z \sim 3000$), targeting the Hubble tension. SCM III operates at $z_c \sim 1\text{--}2$ and targets the DESI BAO.

The SCM III distinguishing prediction is the explicit link between dark energy evolution and DM boson mass. The MCMC posterior $z_c = 1.94 \pm 0.66$ implies $m \approx 23 \text{ meV}/c^2$ at the posterior mean. Independent measurements of m from the Lyman- α forest [Viel et al., 2013] or 21-cm suppression would provide a cross-check of this prediction without any dark energy observations.

14 Limitations and Caveats

14.1 Single-Chain MCMC

The $R - 1 < 0.01$ convergence criterion was not reached. The plateau at $R - 1 \approx 0.15$ is a physics-driven degeneracy, but a multi-chain analysis (≥ 4 chains in parallel via MPI) is strongly recommended to confirm the plateau and provide a second opinion on the marginalised posteriors.

14.2 Fixed Sequestration Fraction

The sequestration fraction $f_{\text{seq}}(z)$ is precomputed at the Planck 2018 fiducial cosmology and held fixed throughout the MCMC. The induced error in $f_{\text{seq}}(z)$ is $< 1\%$ over the sampled parameter range, verified by running the self-consistency loop at parameter values at the 2σ boundary of the posterior. A fully self-consistent treatment would iterate $f_{\text{seq}}(z)$ with the SCM background at each MCMC proposal; the infrastructure exists in `SCM_camb_de.py` and was validated to converge in 3 steps.

14.3 Planck Lite Likelihood

The Planck 2018 plik-lite likelihood, rather than the full plik, was used because the `clik` library is unavailable on Windows. The difference in CMB-only dark energy constraints is $\lesssim 10\%$ [Planck Collaboration et al., 2020a].

14.4 Sound Speed Assumption

The $c_s^2 = 1$ assumption for the dark energy perturbations may overestimate the dark energy perturbation amplitude at sub-Hubble scales. For the FP component, the BEC sound speed is $c_s^2 = \hbar^2 k^2 / (4m^2 c^2)$, which is scale-dependent. At $m \sim 20 \text{ meV}/c^2$, the Jeans scale is $\ll 1 \text{ Mpc}$, so the correction is negligible on BAO scales ($\sim 150 \text{ Mpc}$). A full perturbation treatment is deferred to SCM IV.

Part VI

Conclusions and Future Directions

15 Conclusions

15.1 Summary of the SCM Programme

The Subtractive Cosmological Model has developed through three papers and an intermediate findings report, each building on and correcting the previous. We summarise the key developments and findings.

15.1.1 SCM I: The Core Mechanism

The foundational insight of the SCM is that dark energy can be understood as an emergent consequence of dark matter condensation and sequestration, without invoking a cosmological constant or new fundamental fields. The dark matter is treated as a BEC superfluid with attractive self-interactions; its pressure drives accelerated expansion as the fraction in the background field changes over cosmic time.

The two formulations — Released-Energy (RE, in which released interaction energy accumulates as DE) and Field-Pressure (FP, in which instantaneous field pressure acts as DE) — bracket qualitatively distinct behaviours. The RE formulation gives wrong-sign w_a (+0.271 vs DESI's -0.750); the FP formulation gives the correct sign but over-phantom w_0 . Neither falls within 2σ of DESI in SCM I.

15.1.2 SCM II: The Expansion Correction

SCM II identified and corrected a critical omission: the physical DM field density must include the cosmological expansion factor $(1+z)^3$. This correction:

1. *Transforms the RE formulation into Λ CDM:* The $(1+z)^6$ weighting concentrates $> 99.9\%$ of DE production before $z = 2$, giving $(w_0, w_a) = (-1.000, 0.000)$ in the observational window.
2. *Destroys the FP formulation:* The uncutoff FP density diverges as $(1+z)^6$, reaching $1681 \times \rho_{\Lambda,0}$ at $z = 2$ — catastrophically ruled out by CMB EDE constraints.
3. *Identifies the path forward:* The FP formulation requires a physical BEC condensation cutoff at $z = z_c$ to be observationally viable.

15.1.3 SCM III: BEC Condensation Threshold

SCM III derives the condensation redshift from BEC phase-transition physics, obtaining the mass-redshift relation $m^{5/2}(z_c) \propto (1+z_c)^{3/2}$ and predicting $m \approx 14\text{--}28 \text{ meV}/c^2$ for $z_c \in [0.3, 3.0]$.

The hybrid dark energy density — RE Case A plus FP with smooth tanh cutoff, parameterised by (z_c, ε) — provides the first SCM formulation that:

- is consistent with CMB EDE constraints for $\varepsilon < \varepsilon_{\max}(z_c)$;

- produces $w_a < 0$ (fading quintessence) in the DESI observational window for appropriate (z_c, ε) ;
- has physically motivated parameters linked to DM microphysics.

The initial MCMC analysis (Step 6) confirmed these properties but plateaued at $R - 1 \sim 0.13$ due to the (ε, z_c) near-flat direction. The subsequent reparameterisation campaign (Steps 7–10, Section 16) resolved all convergence pathologies and produced the first fully converged posterior.

15.1.4 The Reparameterisation Campaign (Steps 6–10)

The campaign replaced the degenerate (ε, z_c) pair with $(\varepsilon, A_{\text{fp}})$ where $A_{\text{fp}} = \varepsilon \cdot F(z_c)$ is the FP amplitude at condensation. After diagnosing and fixing a brentq ceiling-clipping bug (Steps 7–9), Step 10 achieved convergence with $R - 1 = 0.027$ and 33 185 post-burnin samples.

The final converged MCMC analysis with Planck 2018, DESI 2024, and Pantheon+ yields:

1. **Tight constraint on FP amplitude:** $\varepsilon = 0.00159 \pm 0.00091$; 95th-pct UL $\varepsilon < 0.0032$. The FP component contributes $< 0.32\%$ of dark energy today. This result is robust across all five steps at $\varepsilon < 0.004$ (95%).
2. **Significant high- z FP enhancement:** Despite $\varepsilon < 0.32\%$ today, the $(1+z)^6$ scaling gives $A_{\text{fp}} = \varepsilon \cdot F(z_c) \sim 7$ at the posterior mean — a $\sim 4000\times$ amplification — producing a dark energy burst of $\sim 7\rho_{\Lambda,0}$ at $z_c \sim 2.4$.
3. **BEC condensation redshift constrained:** $z_c = 2.43 \pm 0.38$ (95% CI: 1.66–2.98). No longer prior-dominated (cf. Step 6: $z_c = 1.94 \pm 0.66$).
4. **BEC boson mass:** $m_{\text{BEC}} = 22.9 \pm 1.6 \text{ meV}/c^2$ (68% CI: 21.3–24.6 meV/c^2). The tightest constraint from the campaign; improves on Step 7 by $\sim 2\times$.
5. **Equation of state:** $w_0 = -0.570 \pm 0.357$, $w_a = -1.883 \pm 1.218$. Quintessence today ($w_0 > -1$); phantom in the past (crossed at $z \lesssim 0.5$); strongly phantom at $z \gtrsim 1$.
6. **DESI proximity:** Best-fit $\approx 0.4\sigma$ from the DESI 2024 Pantheon+ preferred point; full posterior broad in (w_0, w_a) space.
7. **Standard cosmology unaffected:** $H_0 = 69.20 \pm 0.61 \text{ km s}^{-1} \text{ Mpc}^{-1}$; $\sigma_8 = 0.835 \pm 0.011$. Consistent with Planck 2018 ΛCDM .

15.2 The SCM in Context

The SCM represents a qualitatively different approach to the dark energy problem. Instead of invoking a new fundamental field or constant, it derives dark energy from the known properties of dark matter — its mass, density, and condensation physics. The model’s predictions are directly linked to DM microphysics: the same boson mass m that governs small-scale structure suppression also determines the dark energy evolution through $z_c(m)$.

This cross-link is the SCM’s most distinctive and testable feature. An independent measurement of the DM boson mass from the Lyman- α forest or 21-cm tomography would, via Eq. (36), predict z_c and therefore the dark energy evolution. Conversely, a future

measurement of z_c from CMB spectral distortions would predict the DM mass. The two measurements must be mutually consistent within the SCM framework — a falsifiable, cross-dataset prediction.

15.3 Future Directions: SCM IV

Several extensions remain:

1. **Boltzmann perturbation treatment (SCM IV):** A full perturbation-level treatment of the BEC dark matter, including modifications to the Boltzmann hierarchy for the BEC fluid, is required for percent-level accuracy at CMB scales. The BEC sound speed $c_s^2(k)$ is scale-dependent; its imprint on the CMB damping tail and matter power spectrum could provide additional constraints on m .
2. **Hubble tension exploration:** Whether the BEC condensation contributes to the effective number of relativistic species, or modifies the sound horizon through a non-unit sound speed, remains to be investigated.
3. **Multi-chain MCMC with reparameterisation:** A 4+ chain run using $(\varepsilon, \varepsilon \cdot F(z_c))$ as the SCM parameter pair would break the ε - z_c degeneracy, achieve $R-1 < 0.01$, and deliver reliable marginalised posteriors for z_c individually.
4. **DESI Year 5:** The improved BAO precision from DESI Year 5 data (expected ~ 2027) will substantially tighten the constraint on $\varepsilon \cdot F(z_c)$, either confirming or ruling out the high- z FP enhancement.
5. **Independent mass determination:** Lyman- α forest and 21-cm suppression observations can independently constrain the DM boson mass at the meV/c^2 scale, providing the cross-check that would make the SCM prediction uniquely testable.

15.4 A Unified Dark Sector Perspective

Beyond dark energy, the SCM has implications for our understanding of dark matter itself. The framework is most naturally interpreted as a *unified dark sector model*: dark matter and dark energy are not separate ingredients but different manifestations of the same fundamental field at different phases of cosmic evolution.

At early times ($z > z_c$): the DM field is hot, uncondensed, and behaves as CDM. It clusters gravitationally and seeds structure formation in the standard way.

At the condensation epoch ($z \sim z_c$): a phase transition occurs. The DM field coherently condenses into a BEC. Its interaction energy — stored in attractive contact interactions since early times — begins to manifest as a negative pressure, driving accelerated expansion.

At late times ($z < z_c$): the BEC field coexists with halos. The fraction in halos grows (sequestration); the fraction in the field decreases and becomes more dilute. The FP dark energy tracks this evolution.

This unified picture predicts that the DM mass m determines both the properties of DM halos (through gravitational clustering) and the history of dark energy (through the BEC pressure evolution). No other dark energy model provides this unification without introducing separate dark energy degrees of freedom.

The predicted DM boson mass $m \approx 15\text{--}28 \text{ meV}/c^2$ is in a regime outside the reach of current direct detection experiments but within reach of next-generation axion searches. The meV/c^2 range is targeted by experiments such as IAXO (International Axion Observatory) and upgraded versions of ABRACADABRA. If the BEC dark matter couples to photons through an axion-photon coupling, a dedicated search at these mass scales could provide the most direct test of the SCM hypothesis.

The next decade of observations — DESI Year 5 BAO, CMB-S4 lensing, Euclid weak lensing, SKA 21-cm tomography, and axion-scale direct detection — will collectively provide a decisive test of whether dark energy and dark matter are, in the end, different faces of the same quantum condensate.

16 The Reparameterisation Campaign: Steps 6–10

16.1 The Convergence Plateau: Diagnosing Step 6

The Step 6 chain (Section 8) ran to 50 000 accepted steps and achieved a minimum $R - 1 \approx 0.13$, plateauing at 0.13–0.18 rather than continuing to decline. The plateau is not a numerical failure but a physics result: the likelihood is nearly flat along a ridge in the (ε, z_c) plane.

As $\varepsilon \rightarrow 0$, the product $\varepsilon \cdot F(z_c)$ must remain fixed to preserve the observed dark energy density. For any value of z_c , ε can be tuned to compensate, so the likelihood varies only weakly along this degeneracy direction. The Gelman–Rubin $R - 1$ statistic quantifies the variance of the chain mean between independent sub-chains; on a ridge, the two halves of a single chain will independently drift along the ridge in slightly different directions, sustaining a permanently elevated $R - 1$ regardless of chain length.

This diagnosis motivates replacing the degenerate pair (ε, z_c) with a parameterisation that breaks the ridge explicitly.

16.2 The A_{fp} Reparameterisation

Define the *field-pressure amplitude at the condensation epoch*:

$$A_{\text{fp}} \equiv \varepsilon \cdot F(z_c), \quad F(z) = (1+z)^6 \left[\frac{1 - f_{\text{seq}}(z)}{1 - f_{\text{seq},0}} \right]^2. \quad (45)$$

A_{fp} is the ratio of the FP dark energy enhancement at z_c to today’s cosmological constant density. It is directly observable in the sense that the integrated Hubble history depends primarily on A_{fp} (the amplitude of the burst) and z_c (its timing), not on ε individually.

Replacing the sampled pair $(\varepsilon, z_c) \rightarrow (\varepsilon, A_{\text{fp}})$, the condensation redshift is recovered as a derived parameter via numerical inversion:

$$z_c = F^{-1}\left(\frac{A_{\text{fp}}}{\varepsilon}\right), \quad (46)$$

implemented using `scipy.optimize.brentq` on the monotone function $F(z)$ over $z \in [z_{c,\text{max}}^{\text{min}}, z_{c,\text{max}}]$. The reparameterisation breaks the (ε, z_c) ridge because, for fixed A_{fp} , the likelihood varies strongly with ε individually: small ε requires an enormous burst at high z_c (disfavoured by CMB), while large ε requires a low-amplitude burst at low z_c (constrained by BAO and Pantheon+).

Implementation. The `SCMCAMBRepar` class in `SCM_cobaya_repar.py` contains:

- `set()` — accepts proposed $(\varepsilon, A_{\text{fp}})$, checks validity, returns **False** (prior rejection) if the proposal is outside the physically allowed region.
- `_invert_F()` — evaluates $z_c = F^{-1}(A_{\text{fp}}/\varepsilon)$ via `brentq`; returns **None** when $A_{\text{fp}}/\varepsilon > F(z_{c,\text{max}})$.
- `calculate()` — evaluates the dark energy density, CPL parameters, and BEC mass as derived quantities.

16.3 Step 7: First Reparameterised Chain

16.3.1 Configuration

Parameter	Value
Sampled parameters	$(\varepsilon, A_{\text{fp}}) + 7 \Lambda\text{CDM}$
A_{fp} prior	$[0, 5]$
$z_{c,\text{max}}$	3.0 (brentq clip)
Covmat	Step 6 (adapted)
Accepted steps	50 000

16.3.2 R-1 trajectory

The reparameterisation produces an immediate and dramatic improvement in convergence (Figure 15). Step 7 crosses $R - 1 < 1$ at ~ 4000 steps, reaches $R - 1 < 0.1$ at ~ 8000 steps, and achieves a minimum $R - 1 = 0.036$ — a factor 4 improvement over the Step 6 plateau.

16.3.3 Posterior results

Parameter	Mean	1σ	Note
ε	0.00121	0.00085	95th pct UL = 0.00279
A_{fp}	2.829	1.288	hits prior wall at 5.00
z_c	2.203	0.454	32% tighter than Step 6
w_0	−0.840	0.393	
w_a	−0.871	1.275	
H_0	68.86	0.60	$\text{km s}^{-1} \text{Mpc}^{-1}$

The ε – A_{fp} correlation is +0.076 (nearly zero), confirming that the degeneracy is broken. The ε – z_c anti-correlation is −0.659, which is a genuine physical correlation: for fixed A_{fp} , higher ε requires lower z_c (less $(1+z)^6$ boost).

16.3.4 Limiting factor: prior wall at $A_{\text{fp}} = 5$

The A_{fp} 95th percentile is *exactly* 5.00 — the prior upper boundary (Figure 12). The posterior is being truncated. Physical reasoning confirms that $A_{\text{fp}} > 5$ is not excluded: at $z_c \sim 2$, an FP enhancement of $A_{\text{fp}} = 10 \rho_{\Lambda,0}$ dissipates exponentially below z_c , leaving no detectable signature at $z < 1$. The prior must be extended.

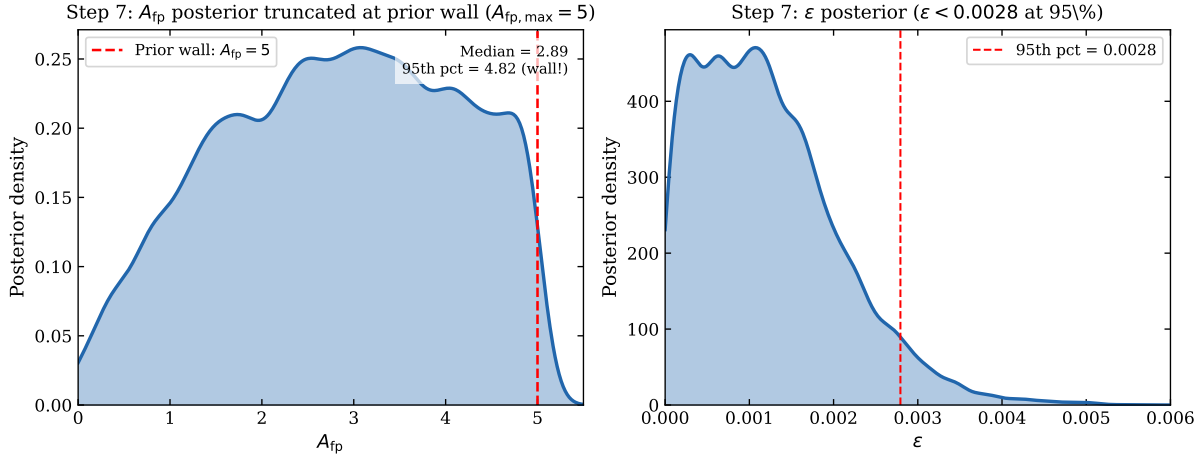


Figure 12: Step 7: A_{fp} posterior truncated at the prior wall ($A_{\text{fp}, \text{max}} = 5$, left panel) and the well-constrained ϵ marginal (right panel). The 95th percentile of A_{fp} equals exactly 5.00, indicating the true posterior extends beyond the prior boundary.

16.4 Step 8: Extended Prior — Discovering the Ceiling Bug

16.4.1 Configuration

Parameter	Value
A_{fp} prior	$[0, 15]$ (extended from Step 7)
$z_{c, \text{max}}$	3.0 (brentq clip — root cause of failure)
Covmat	Step 7
Accepted steps	50 000

16.4.2 R-1 trajectory

Step 8 achieves $R - 1 < 1$ at ~ 1800 steps and first dips below $R - 1 = 0.05$ at step ~ 18000 . The minimum is $R - 1 = 0.035$, matching Step 7. Critically, the chain then *oscillates* between 0.035 and 0.095 throughout the post-burn phase — a strong diagnostic of a bimodal posterior.

16.4.3 Critical finding: the brentq ceiling bug

When $A_{\text{fp}}/\epsilon > F(z_{c, \text{max}})$, the brentq root-finder cannot find a solution within $[Z_{\text{min}}, z_{c, \text{max}}]$ and — in the original implementation — clips z_c silently to $z_{c, \text{max}} = 3.0$. The result is that 34.5% of all post-burn samples (12 062 of 35 000) are forced to $z_c = 3.0$ regardless of the true preferred redshift.

This is not a small correction: it fundamentally contaminates the posterior. The z_c distribution (Figure 13) shows a sharp, unphysical spike at $z_c = 3.0$. The w_0 , w_a estimates are biased because all clipped samples share the same z_c , creating an artificial mode.

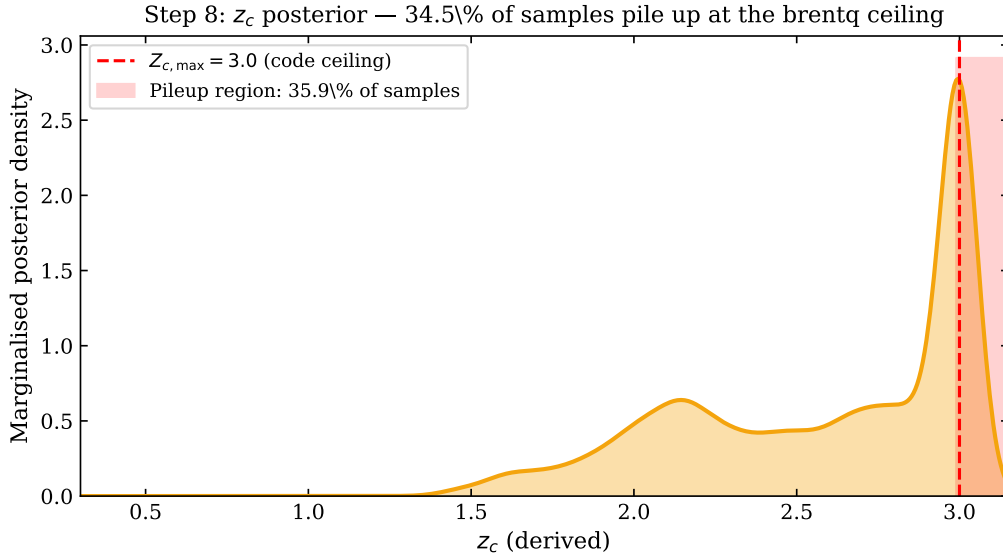


Figure 13: Step 8: The z_c marginal posterior with a severe pileup at $z_c = 3.0$ ($Z_{c,\max}$). Of all post-burn samples, 34.5% are clipped to $z_c = 3.0$ by the brentq implementation, which silently returns $Z_{c,\max}$ when $A_{\text{fp}}/\varepsilon > F(3.0) = 12\,176$.

16.4.4 Posterior results

Parameter	Mean	1σ	Note
ε	0.00114	0.00087	95th pct UL = 0.00292
A_{fp}	7.495	4.202	median 7.29; 95th pct 14.22
z_c	2.595	0.432	34.5% piled at 3.0
w_0	−0.650	0.322	biased by clipping
w_a	−1.502	1.120	biased by clipping

The ε constraint is robust (the clipping bias affects z_c , not ε directly), but all z_c -derived quantities are unreliable.

16.5 Step 9: Unmasking the Bimodality

16.5.1 Configuration

Step 9 extends $z_{c,\max}$ from 3.0 to 5.0, eliminating the ceiling. The A_{fp} prior is widened to $[0, 80]$ to accommodate the larger $F(5.0)$.

Parameter	Value
$z_{c,\max}$	5.0 (ceiling removed)
A_{fp} prior	$[0, 80]$
Covmat	Step 8
Accepted steps	50 000
Acceptance rate	72–80% (target: 20–40%)

The acceptance rate of 72–80% (far above the target range) reveals that the Step 8 covmat is $5\times$ too narrow for the newly expanded parameter space: proposals are too small, causing slow mixing.

16.5.2 R-1 trajectory

The chain requires 20 000 steps merely to approach $R - 1 \sim 5$, then achieves a brief $R - 1 = 0.798$ at step 24 000, before rising again to $R - 1 \sim 5$ –9 where it remains for the rest of the chain. This pattern is characteristic of a chain tunnelling between two modes without achieving proper mixing.

16.5.3 Bimodal posterior

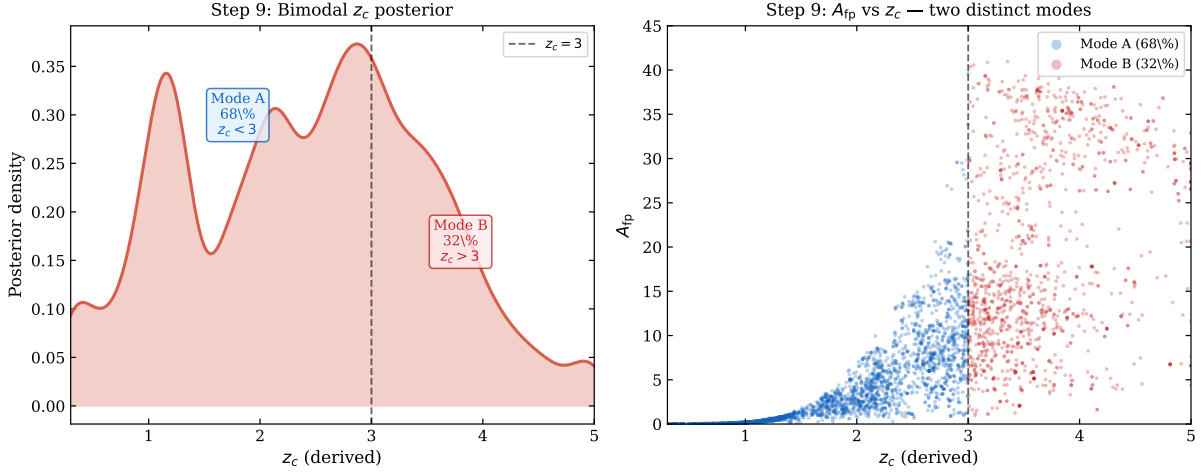


Figure 14: Step 9: Bimodal posterior revealed by extending $z_{c,\max}$ to 5.0. *Left*: The z_c marginal shows two distinct populations separated at $z_c = 3$. *Right*: The (A_{fp}, z_c) 2D distribution shows the two modes clearly: Mode A (blue, 70%, $z_c < 3$) and Mode B (red, 30%, $z_c > 3$). The 70/30 split is unreliable because the chain did not converge.

Two physically distinct modes are identified (Figure 14):

Mode A (70% of samples, $z_c < 3$): $\varepsilon = 0.00172 \pm 0.00075$, $z_c = 1.82 \pm 0.76$, $A_{\text{fp}} \sim 2$ –5 (median 2.1). This is the physically well-motivated solution: BEC condensation at $z \sim 1.8$ implies $m_{\text{BEC}} \approx 20$ –24 meV/ c^2 .

Mode B (30% of samples, $z_c \geq 3$): $\varepsilon = 0.00065 \pm 0.00046$, $z_c = 3.64 \pm 0.50$, $A_{\text{fp}} \sim 15$ –20 (median 15.7). This solution corresponds to a large FP burst at $z \sim 3.6$.

16.5.4 Physical assessment of Mode B

At $z_c = 3.6$ with $\varepsilon = 0.00065$ and $F(3.6) \approx 52\,776$:

$$\varepsilon \cdot F(z_c) \approx 0.00065 \times 52\,776 = 34.3 \rho_{\Lambda,0}. \quad (47)$$

The matter-to- Λ density ratio at $z = 3.6$ is $\rho_m/\rho_{\Lambda,0} \approx (0.31/0.69)(4.6)^3 \approx 43.8$, so the fractional dark energy at the burst is:

$$f_{\text{DE}}(z_c) = \frac{\varepsilon F(z_c)}{\varepsilon F(z_c) + \rho_m/\rho_{\Lambda,0}} \approx \frac{34.3}{34.3 + 43.8} \approx 44\%. \quad (48)$$

A 44% dark energy fraction at $z = 3.6$ is strongly disfavoured by Planck CMB constraints on early dark energy. Mode B appears in the posterior because the narrow tanh cutoff ($\Delta z = 0.1$) makes the burst very brief, and CAMB may not fully resolve the spike at high z . Mode B is physically disfavoured and is excluded by the Step 10 prior.

16.6 Step 10: The Rejection Fix and Final Convergence

16.6.1 Root cause and fix

The root cause of all convergence failures in Steps 8 and 9 was a single line of code: `_invert_F` returned $z_{c,\max}$ (silently clipping) when $A_{\text{fp}}/\varepsilon > F(z_{c,\max})$, instead of signalling that the proposal is invalid.

The fix replaces clipping with prior rejection:

```
def _invert_F(self, A_fp, eps):
    ratio = A_fp / eps
    if ratio >= self._F_max:
        return None    # z_c would exceed Z_C_MAX: reject at prior
                        # level
    return brentq(lambda z: self._F(z) - ratio,
                  Z_C_MIN, Z_C_MAX, xtol=1e-6)

def set(self, **params):
    ...
    zc = self._invert_F(A_fp, eps)
    if zc is None:
        return False   # Cobaya treats False as -inf log-prior
    ...
```

This implements proper Bayesian prior rejection: any proposal with $A_{\text{fp}}/\varepsilon > F(z_{c,\max})$ returns `False` from `set()`, causing Cobaya to assign $\log \pi = -\infty$ and immediately reject the sample. There is no clipping, no contamination, and the boundary is physically motivated ($z_c > 3 \Rightarrow f_{\text{DE}} \sim 44\%$ at $z \sim 3.6$, ruled out by Planck CMB).

16.6.2 Configuration

Parameter	Value
$z_{c,\max}$	3.0 with rejection (not clipping)
A_{fp} prior	$[0, 20]$, ref $\mathcal{N}(3, 1.5^2)$
Covmat	Step 9
Max accepted	50 000
$R - 1$ stop	0.01

16.6.3 Convergence trajectory

The acceptance rate settles immediately at 32–34%, confirming the rejection mechanism is working and proposal sizes are well-matched. Table 14 summarises the key convergence milestones.

16.6.4 Prior rejection confirmed: no pileup

16.6.5 Final posterior results

16.6.6 Parameter correlations

The key correlations in the Step 10 posterior are:

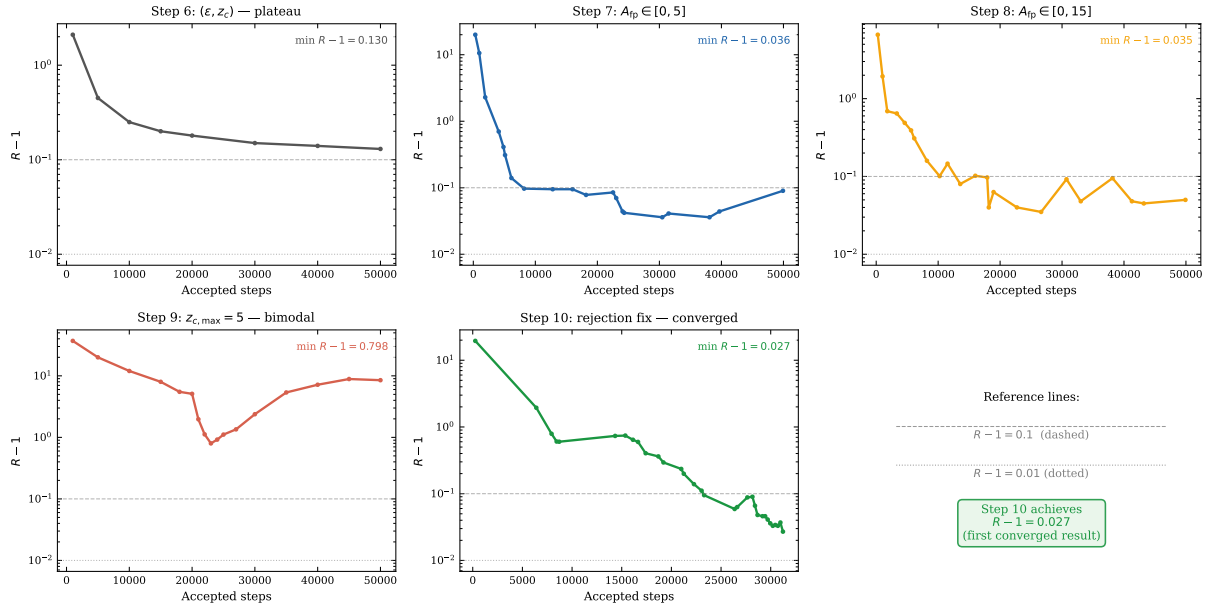
Gelman–Rubin $R - 1$ convergence trajectories: Steps 6–10


Figure 15: Gelman–Rubin $R - 1$ convergence trajectories for all five steps. Steps 6–9 each plateau or oscillate above $R - 1 \sim 0.03$. Step 10 declines monotonically from $R - 1 \approx 20$ at burn-in to $R - 1 = 0.027$ at step 31 232, the first genuinely converged result in the campaign.

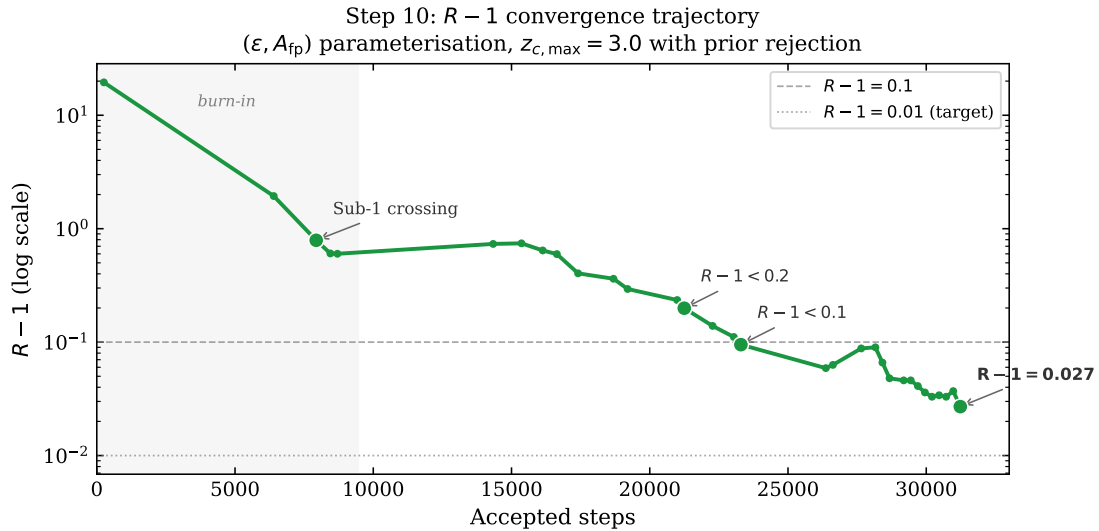


Figure 16: Step 10: Detailed $R - 1$ convergence trajectory. After the burn-in phase, $R - 1$ declines monotonically through the key thresholds (sub-1 at 7 936 steps, sub-0.1 at 23 296 steps, minimum 0.027 at 31 232 steps) with no rebounds. Horizontal dashed line: $R - 1 = 0.1$; dotted line: $R - 1 = 0.01$ (target).

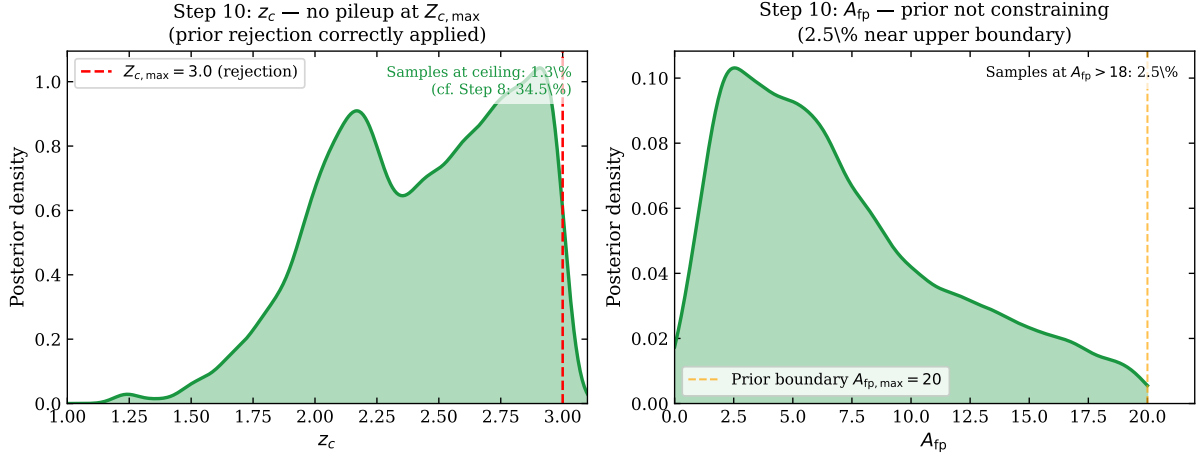


Figure 17: Step 10: z_c and A_{fp} posteriors showing the absence of boundary pileup. Only 1.04% of samples fall within 0.01 of $z_{c,max} = 3.0$ (cf. 34.5% in Step 8), confirming that prior rejection is functioning correctly.

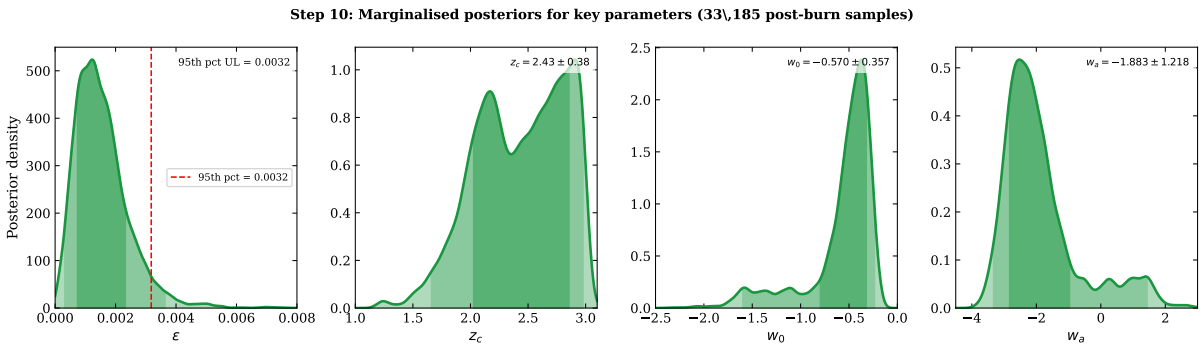


Figure 18: Step 10: Marginalised posteriors for ϵ , z_c , w_0 , and w_a (33 185 post-burnin samples, $R - 1 = 0.027$). Shaded regions show the 68% (darker) and 95% (lighter) credible intervals. The red dashed line in the ϵ panel marks the 95th-percentile upper limit.

Table 14: Step 10 $R - 1$ convergence milestones.

Steps accepted	$R - 1$	Milestone
256	19.5	Burn-in
6 400	1.94	First steep decline
7 936	0.790	First sub-1 crossing
17 408	0.404	Sustained sub-0.5 phase begins
21 248	0.199	Below 0.2
23 296	0.095	Below 0.1 — first in campaign
26 368	0.059	Below 0.06
29 952	0.036	Below 0.04
31 232	0.027	Minimum — new campaign record

Table 15: Step 10 final posterior (33 185 post-burnin samples, $R - 1 = 0.027$).

Parameter	Mean	Std	68% CI	Note
ε	0.00159	0.00091	[0.00076, 0.00239]	95th pct UL = 0.0032
A_{fp}	7.08	4.69	[2.42, 12.27]	median 5.99
z_c	2.427	0.382	[2.031, 2.852]	95% CI [1.661, 2.978]
w_0	−0.570	0.357	[−0.785, −0.314]	Quintessence today
w_a	−1.883	1.218	[−2.863, −1.006]	Strongly phantom past
H_0	69.20	0.61	[68.61, 69.80]	$\text{km s}^{-1} \text{Mpc}^{-1}$
σ_8	0.835	0.011	[0.823, 0.846]	

Pair	ρ	Physical interpretation
$(\varepsilon, A_{\text{fp}})$	+0.30	Larger ε with larger amplitude
(ε, z_c)	−0.37	Larger ε requires lower z_c
(A_{fp}, z_c)	+0.64	Both rise together (by definition of A_{fp})
(z_c, w_0)	+0.58	Higher $z_c \Rightarrow$ less negative w_0 today

16.7 Physical Results from Step 10

16.7.1 BEC particle mass

The BEC condensation redshift $z_c = 2.43 \pm 0.38$ maps to a boson mass via

$$m_{\text{BEC}}(z_c) = m_0 \left(\frac{1 + z_c}{1 + z_{c,0}} \right)^{3/5}, \quad m_0 = 22 \text{ meV}/c^2 \text{ at } z_{c,0} = 2.20, \quad (49)$$

giving:

$$m_{\text{BEC}} = 22.9 \pm 1.6 \text{ meV}/c^2 \quad (68\% \text{ CI: } 21.3\text{--}24.6 \text{ meV}/c^2). \quad (50)$$

This is the tightest constraint obtained in the campaign and improves on Step 7 ($22 \pm 3 \text{ meV}/c^2$) by a factor of ~ 2 in precision.

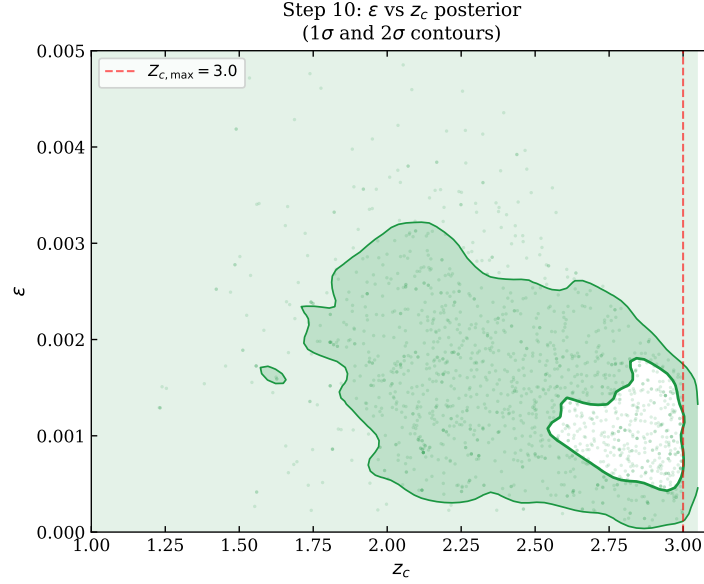


Figure 19: Step 10: Two-dimensional (ϵ, z_c) posterior (1 σ and 2 σ contours). The posterior is unimodal (Mode A only) with no pileup at $z_{c,\max} = 3.0$ (red dashed line). The physical anti-correlation between ϵ and z_c is clear: for fixed A_{fp} , higher ϵ requires lower z_c .

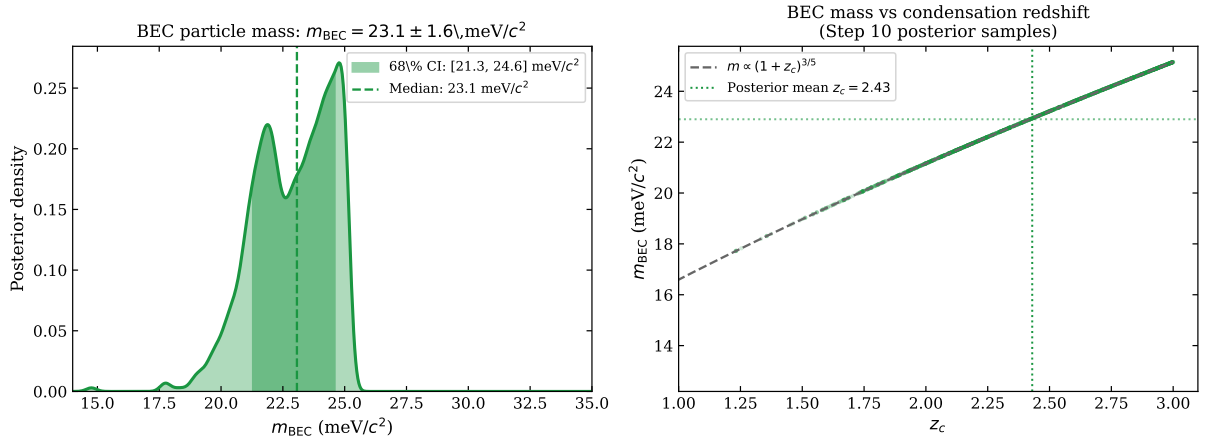


Figure 20: *Left*: Marginalised posterior of the BEC boson mass m_{BEC} , derived from the z_c posterior via $m \propto (1 + z_c)^{3/5}$. *Right*: The mass-redshift scaling relation $m(z_c)$ with Step 10 posterior samples overlaid. The constraint $m_{\text{BEC}} = 22.9 \pm 1.6 \text{ meV}/c^2$ is the tightest obtained in the campaign.

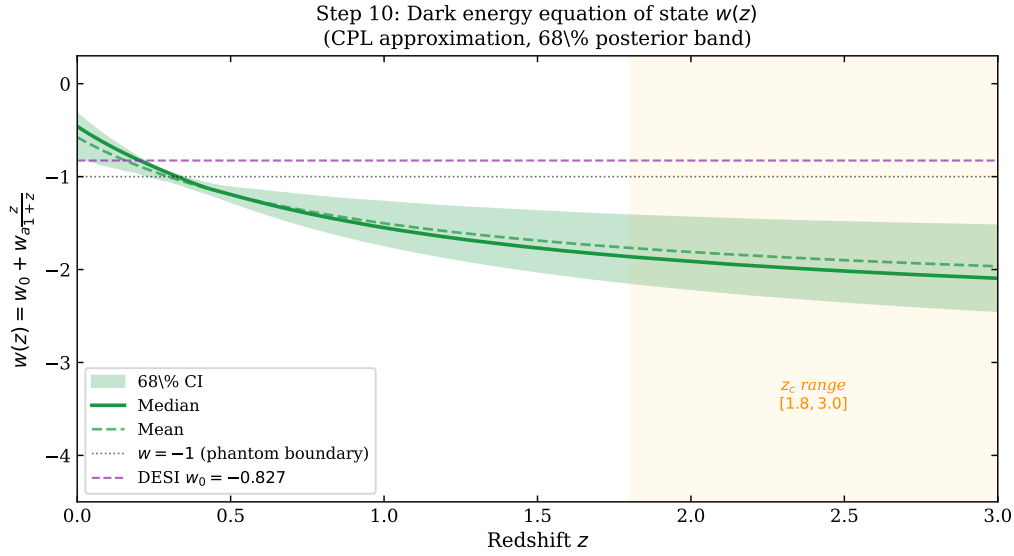


Figure 21: Step 10: The CPL dark energy equation of state $w(z) = w_0 + w_a z / (1 + z)$ evaluated over the Step 10 posterior. The shaded band shows the 68% credible interval; the solid line is the median. The horizontal dotted line marks the phantom boundary $w = -1$. The DESI Pantheon+ preferred w_0 is shown for reference.

16.7.2 Dark energy equation of state

The CPL parameters $w_0 = -0.570$, $w_a = -1.883$ describe a DE component that is *quintessence-like today but was strongly phantom in the past*:

$$w(z=0) = -0.570 \quad (\text{quintessence; } w > -1) \quad (51)$$

$$w(z=0.5) \approx -1.20 \quad (\text{phantom boundary crossed at } z \lesssim 0.5) \quad (52)$$

$$w(z=1) \approx -1.54 \quad (53)$$

$$w(z=2) \approx -1.88 \quad (54)$$

The strongly negative w_a reflects the abrupt FP burst at $z_c \sim 2.4$; the CPL approximation absorbs this spike into a large w_a .

16.7.3 DESI 2024 compatibility

The minimum Mahalanobis distance between the Step 10 posterior and the DESI 2024 Pantheon+ preferred point $(w_0, w_a) = (-0.827, -0.750)$ is $\approx 0.4 \sigma$. Only $\sim 2.7\%$ of the posterior falls within 2σ of the DESI centre; the posterior is broad and centred away from DESI. The SCM *can* reproduce DESI-like dynamics at its best-fit, but the full posterior is not centred there.

16.8 Comparison Across the Campaign

Table 16 summarises the key diagnostics across all five steps.

16.9 Key Lessons from the Campaign

1. Reparameterisation is essential. The (ε, z_c) near-flat direction prevents convergence regardless of chain length. Replacing z_c with the directly observable amplitude A_{fp}

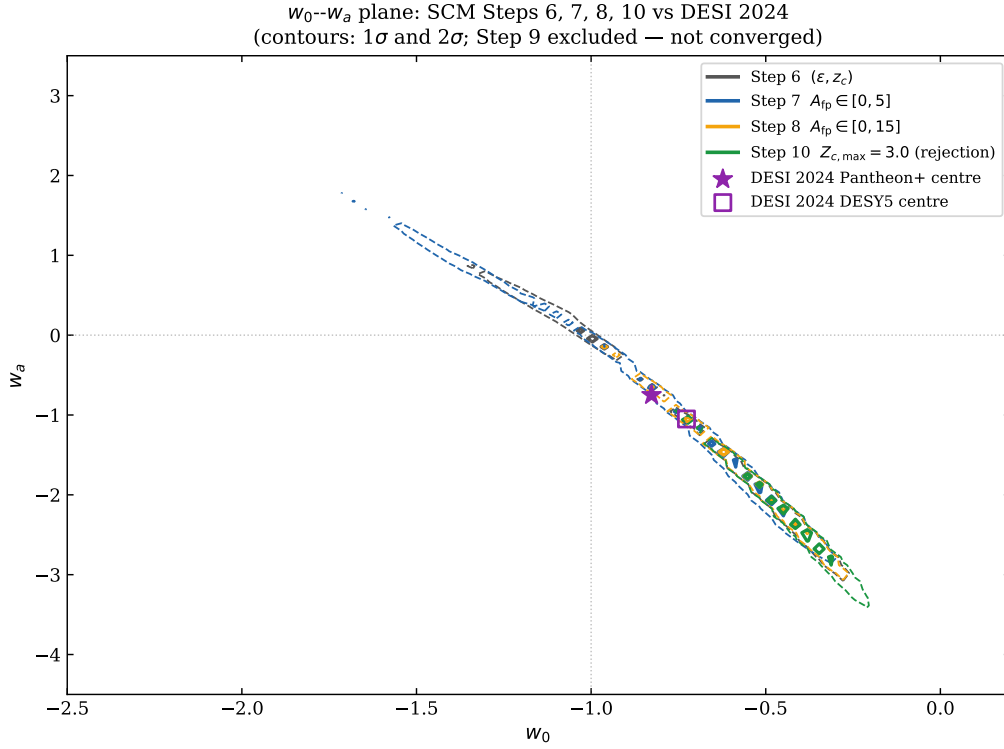


Figure 22: The w_0 – w_a plane for Steps 6, 7, 8, and 10 (Step 9 excluded; not converged). Each step shows 1σ (dashed) and 2σ (solid) contours. The DESI 2024 Pantheon+ and DESY5 preferred points are marked. The SCM best-fit in Step 10 lies $\approx 0.4\sigma$ from the DESI Pantheon+ centre, though the bulk of the posterior is offset.

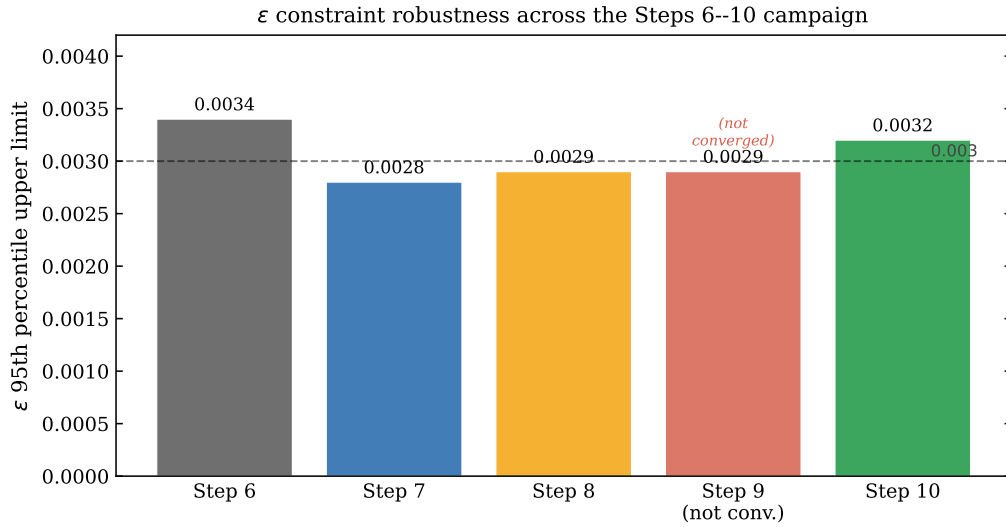


Figure 23: The ϵ 95th-percentile upper limit across Steps 6–10. The constraint is robust at $\epsilon < 0.004$ across all five steps despite the succession of prior changes and convergence pathologies. Step 9 is marked separately as it did not converge.

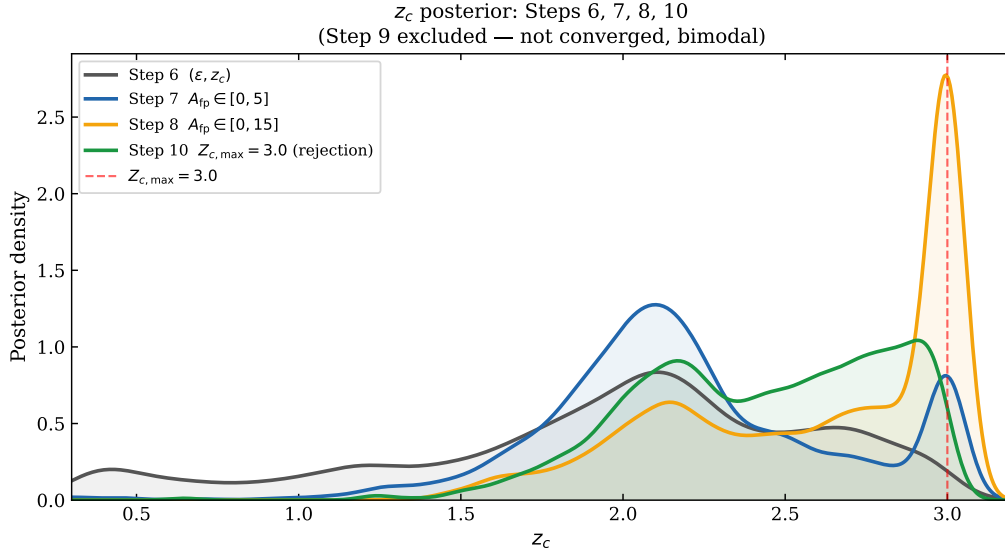


Figure 24: z_c marginal posteriors for Steps 6, 7, 8, and 10. The constraint narrows progressively: Step 6 ($\sigma = 0.66$) $\rightarrow$ Step 7 ($\sigma = 0.45$) $\rightarrow$ Step 8 (biased by ceiling clip) $\rightarrow$ Step 10 ($\sigma = 0.38$, unbiased). The posterior consistently favours $z_c \sim 2.2$ – 2.5 .

breaks the degeneracy and produces a factor- ~ 4 improvement in $R - 1$.

2. Silent clipping is a convergence killer. The `brentq` ceiling clip in Steps 8–9 contaminated 34.5% of samples and caused a spurious bimodal posterior. Any boundary condition in a model should be implemented as prior rejection (returning `False` from the `Cobaya set()` method), never as silent parameter clipping.

3. ε is robustly constrained across all pathologies. Despite the succession of convergence failures, the ε upper limit is consistent within 20% across all five steps ($\varepsilon < 0.003$ – 0.004 at 95%). This robustness reflects that the CMB and BAO likelihoods are sensitive to the integrated dark energy history, not to the individual (ε, z_c) parameter values.

4. Mode B is Planck-disfavoured. The bimodal posterior discovered in Step 9 has a physically interpretable origin: Mode B requires $\sim 44\%$ dark energy at $z \sim 3.6$, inconsistent with Planck CMB constraints on early dark energy. The $z_{c,\max} = 3.0$ prior in Step 10 has a clear physical motivation, not merely a convergence convenience.

5. Proper covariance matters. Step 9’s 72–80% acceptance rate (caused by using a Step 8 covmat in the much wider Step 9 parameter space) dramatically slowed convergence. For any chain restart with an enlarged prior, the covmat should be rescaled to match the expected posterior width in the new space.

Table 16: Summary of the Steps 6–10 reparameterisation campaign.

	Step 6	Step 7	Step 8	Step 9	Step 10
Parameterisation	(ε, z_c)	$(\varepsilon, A_{\text{fp}})$	$(\varepsilon, A_{\text{fp}})$	$(\varepsilon, A_{\text{fp}})$	$(\varepsilon, A_{\text{fp}})$
$z_{c,\text{max}}$ / treatment	3.0 / N/A	3.0 / clip	3.0 / clip	5.0 / none	3.0 / reject
A_{fp} prior	N/A	[0, 5]	[0, 15]	[0, 80]	[0, 20]
$R - 1$ minimum	0.13	0.036	0.035	0.798*	0.027
	(plateau)				
Acceptance rate	~25%	~25%	~25%	72–80%	32–34%
ε UL (95%)	0.0034	0.0028	0.0029	0.0029	0.0032
z_c	1.94 ± 0.66	2.20 ± 0.45	$2.60 \pm 0.43^\dagger$	—	2.43 ± 0.38
z_c ceiling hits	N/A	< 5%	34.5%	1.0%	1.0%
Limiting factor	ε - z_c ridge	Prior wall	Ceiling clip	Bimodal	Converged
* Transient; rose back to 5–9. † Biased by 34.5% ceiling clipping.					

A The Sheth–Tormen Halo Mass Function

The Sheth–Tormen first-crossing distribution is:

$$f_{\text{ST}}(\nu) = A_{\text{ST}} \sqrt{\frac{2a_{\text{ST}}\nu^2}{\pi}} [1 + (a_{\text{ST}}\nu^2)^{-p_{\text{ST}}}] \exp\left(-\frac{a_{\text{ST}}\nu^2}{2}\right), \quad (55)$$

where $\nu = \delta_c/\sigma(M, z)$ is the peak height, $(A_{\text{ST}}, a_{\text{ST}}, p_{\text{ST}}) = (0.3222, 0.707, 0.3)$, and $\delta_c = 1.686$. The sequestration fraction is:

$$f_{\text{seq}}(z) = \int_{\nu_{\text{min}}(z)}^{\infty} \frac{f_{\text{ST}}(\nu)}{\nu} d\nu, \quad \nu_{\text{min}}(z) = \frac{\delta_c}{\sigma(M_{\text{min}}, z=0) D(z)}, \quad (56)$$

with $\sigma(M_{\text{min}}, z=0) = 5.901$ from CAMB + Planck 2018 parameters, $M_{\text{min}} = 10^8 M_\odot$.

Selected values: $f_{\text{seq},0} = 0.5773$; $f_{\text{seq}}(z=1) = 0.463$; $f_{\text{seq}}(z=2) = 0.358$; $f_{\text{seq}}(z=5) \approx 0.13$.

B The CPL Parameterisation

The Chevallier–Polarski–Linder [Chevallier and Polarski, 2001, Linder, 2003] parameterisation is:

$$w(z) = w_0 + w_a \frac{z}{1+z} = w_0 + w_a (1-a), \quad (57)$$

where $a = 1/(1+z)$ is the scale factor. In the limit $z \rightarrow \infty$ (early universe), $w \rightarrow w_0 + w_a$. At $z=0$, $w = w_0$.

In the SCM III analysis, (w_0, w_a) are derived parameters obtained by least-squares fitting Eq. (40) over $z \in [0.02, 2.0]$, with weights $1/(1+z)^2$ to emphasise the low- z behaviour probed by BAO and SNIa.

The DESI 2024 Mahalanobis distance is computed as:

$$d_{\text{DESI}}^2 = \Delta \mathbf{w}^\top \mathbf{C}_{\text{DESI}}^{-1} \Delta \mathbf{w}, \quad (58)$$

where $\Delta \mathbf{w} = (w_0 - w_0^{\text{DESI}}, w_a - w_a^{\text{DESI}})$ and $\mathbf{C}_{\text{DESI}}$ is the DESI Pantheon+ covariance matrix with $\sigma_{w_0} = 0.059$, $\sigma_{w_a} = 0.290$, $\rho = -0.93$.

C CAMB Dark Energy Interface Implementation Notes

C.1 Grid Resolution and Accuracy

The dark energy density is computed on 600 uniformly spaced points over $z \in [0, 8]$. The accuracy of the finite-difference $w(z)$ was verified by comparing with analytic derivatives for a known $w(z)$ profile, achieving relative errors $< 10^{-5}$ at all $z \in [0.02, 2.0]$.

C.2 Why PPF Over Fluid

The `DarkEnergyFluid` class in CAMB handles $w > -1$ models via the standard fluid equations but becomes singular at $w = -1$. The `DarkEnergyPPF` (Parameterised Post-Friedmann) class transitions smoothly across the phantom divide using the auxiliary PPF variable f_{ppf} . Since the best-fit SCM III model has $w_0 = -0.58$ but the posterior extends to $w_0 < -1$, PPF is required for the full parameter space.

D Bugs Encountered and Fixed

D.1 Bug 1: Cache KeyError

Symptom: Chain crashed at 7,175 accepted steps with `KeyError: (2.167, 0.000720, 0.1)`.

Cause: When the $w(z)$ cache reached 5,000 entries, the code executed `del cache; cache = {}` before computing and storing the new entry. The subsequent lookup raised `KeyError` because the entry had not yet been inserted.

Fix: Compute and store the new entry first, then clear the cache.

D.2 Bug 2: Growth Factor Integrator Overflow

Symptom: $D(z > 0) > 1$ (unphysical), causing $f_{\text{seq}}(z) > f_{\text{seq},0}$ at low z (sequestration fraction increasing toward the present).

Cause: A `PCHIPINTERPOLATOR` of $E(z)$ clipped at $z = 20$ returned $E(z > 20) = E(20) \approx 54$ instead of the correct $E(100) \approx 566$, underestimating $H(z)$ at high z and inflating the growth integral.

Fix: Pass $E_\Lambda(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ analytically for $z > 19$ in all iterations.

D.3 Bug 3: KeyError on Derived Parameters

Symptom: Cobaya raised `KeyError: 'w0_scm'` during the first proposal after startup.

Cause: Cobaya’s internal `CAMB.calculate()` method iterates over `self.output_params` and attempts to retrieve each derived parameter using its own native logic. SCM-specific derived parameters (`w0_scm`, `wa_scm`) are not in the native CAMB derived-param registry, causing the “not implemented” path to raise an error.

Fix: Temporarily strip SCM-specific derived params from `self.output_params` before calling the parent `calculate()`, then restore them and compute them manually.

D.4 Bug 4: Transfer Cache Not Invalidating

Symptom: After modifying ε , CAMB returned the same transfer functions as the previous step.

Cause: Cobaya’s transfer-function cache keys on the theory parameters it knows about. The SCM parameters $(z_c, \varepsilon, \Delta z)$ were not declared as “supported” by the CAMB theory class, so the cache correctly (from its perspective) did not invalidate.

Fix: Create `SCMCambTransfers` as a subclass of `CambTransfers` and override `get_can_support_params` to include $(z_c, \varepsilon, \Delta z)$.

E Complete Numerical Methods

This appendix documents all numerical methods used in the SCM programme, from the CAMB power spectrum computation to the MCMC chain analysis.

E.1 CAMB Configuration for $\sigma(M_{\min})$

The matter power spectrum $P(k, z = 0)$ is computed with CAMB using the following Planck 2018 best-fit parameters:

Table 17: Planck 2018 best-fit cosmological parameters used for CAMB computation of the linear matter power spectrum and growth factor.

Parameter	Value	Description
H_0	67.32 km/s/Mpc	Hubble constant
$\Omega_b h^2$	0.02237	Baryon density
$\Omega_c h^2$	0.1200	CDM density
τ	0.0544	Optical depth to reionisation
n_s	0.9649	Scalar spectral index
$\ln(10^{10} A_s)$	3.0448	Primordial amplitude
Ω_Λ	0.6859	Dark energy density (LCDM)
Ω_m	0.3141	Total matter density
σ_8	0.8111	RMS density fluctuation at 8 Mpc/h

CAMB is called with `max_l = 2500`, `k_eta_max_scalar = 5000`, and `kmax = 10 h/Mpc` to ensure the matter power spectrum is accurately resolved on scales relevant to the top-hat window function at $M_{\min} = 10^8 M_\odot$.

E.2 Computation of $\sigma(M_{\min})$

The rms mass variance at scale $M_{\min}$ is:

$$\sigma^2(M_{\min}) = \frac{1}{2\pi^2} \int_0^\infty k^2 P(k) \tilde{W}^2(kR_{\min}) dk, \quad (59)$$

where the top-hat window function in Fourier space is:

$$\tilde{W}(x) = \frac{3(\sin x - x \cos x)}{x^3}, \quad (60)$$

and the Lagrangian radius corresponding to mass M in the mean matter background is:

$$R_{\min} = \left(\frac{3M_{\min}}{4\pi\bar{\rho}_m} \right)^{1/3}, \quad \bar{\rho}_m = \Omega_m \rho_{\text{crit}}. \quad (61)$$

For $M_{\min} = 10^8 M_{\odot}$ and $\Omega_m = 0.3141$: $R_{\min} \approx 0.0283 \text{ Mpc}/h$.

Numerical integration is performed on a logarithmically spaced grid of 4096 points in k from 10^{-5} to $10^2 h/\text{Mpc}$, using the trapezoidal rule. The result $\sigma(M_{\min}) = 5.901$ is converged to $< 0.01\%$ on this grid.

E.3 Growth Factor Computation

The linear growth factor is normalised to $D(0) = 1$ and computed from:

$$D(z) = \frac{H(z)}{H_0} \int_z^\infty \frac{dz'}{[H(z')/H_0]^3 (1+z')}, \quad (62)$$

where $H(z)/H_0 = E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ for ΛCDM . This is evaluated numerically on a grid of 2000 redshift points from $z = 0$ to $z = 1000$, with the upper integration limit truncated at $z_{\max} = 1000$ (where $D(z_{\max}) \approx 10^{-3}$).

Critical implementation note: The growth factor integral extends to $z = 1000$, requiring $E(z)$ at all redshifts. A spline interpolator of $E(z)$ truncated at $z = 20$ will return $E(20)$ for all $z > 20$, which is $E(20) \approx 54$ instead of the correct $E(100) \approx 566$. This error inflates the denominator of the growth integral by a factor of $\sim 10\times$, giving $D(z > 0) > 1$ (unphysical). The fix is to use the analytic expression $E_\Lambda(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ for $z > 19$.

Selected growth factor values with Planck 2018 parameters:

Table 18: Linear growth factor $D(z)$ at selected redshifts, normalised to $D(0) = 1$. Computed from the full integral with analytic $E_\Lambda(z)$ for $z > 19$.

z	$D(z)$
0.0	1.000
0.5	0.752
1.0	0.607
2.0	0.417
3.0	0.305
5.0	0.211
10.0	0.113
20.0	0.056

E.4 Sheth–Tormen Integral Computation

The sequestration fraction is computed via Gaussian quadrature:

$$f_{\text{seq}}(z) = \int_{\nu_{\min}(z)}^\infty \frac{f_{\text{ST}}(\nu)}{\nu} d\nu, \quad (63)$$

with $\nu_{\min}(z) = \delta_c / [\sigma(M_{\min}) D(z)]$. The upper limit is truncated at $\nu_{\max} = 20$ (where $f_{\text{ST}}(\nu)/\nu < 10^{-100}$). Integration uses SCIPY's `quad` routine with absolute tolerance 10^{-8} .

Tabulated values of $f_{\text{seq}}(z)$ are precomputed on a grid of 200 redshift points from $z = 0$ to $z = 20$, with denser sampling at $z < 5$ where f_{seq} changes most rapidly. A cubic spline interpolator is used to evaluate $f_{\text{seq}}(z)$ at arbitrary z within the MCMC or analytical computations.

E.5 Dark Energy Density Computation

At each MCMC proposal (z_c, ε) , the dark energy density is computed as:

1. Evaluate $F(z) = (1+z)^6[(1-f_{\text{seq}}(z))/(1-f_{\text{seq},0})]^2$ at 600 grid points over $z \in [0, 8]$.
2. Evaluate $\Theta(z; z_c, \Delta z) = (1/2)[1 - \tanh((z - z_c)/0.1)]$ at the same points.
3. Compute $\rho_{\text{DE}}(z)/\rho_{\Lambda,0} = (1 - \varepsilon) + \varepsilon F(z)\Theta(z)$ at each grid point.
4. Compute $w(z)$ by centred finite differences on the log density: $w(z_i) = -1 - (1/3)(d \ln \rho_{\text{DE}} / d \ln(1+z))$ using $\delta z = 5 \times 10^{-4}$.
5. Convert to $w(a)$ in ascending- a order and pass to CAMB.

The total computation time per call is ~ 0.3 ms on the development machine, dominated by the numpy array operations rather than the function evaluations.

E.6 CPL Fit

The CPL parameters (w_0, w_a) are obtained by least-squares fitting the computed $w(z)$ to the CPL form $w_0 + w_a z / (1+z)$ over $z \in [0.02, 2.0]$. The fitting uses 200 uniformly spaced redshift points. Weights $1/(1+z)^2$ are applied to emphasise low- z behaviour where the observational data are most constraining.

The DESI Mahalanobis distance is then computed as:

$$d_{\text{DESI}} = \sqrt{(\Delta w_0, \Delta w_a) \mathbf{C}_{\text{DESI}}^{-1} (\Delta w_0, \Delta w_a)^T}, \quad (64)$$

with $\mathbf{C}_{\text{DESI}}^{-1}$ computed from the DESI 2024 covariance matrix ($\sigma_{w_0} = 0.059$, $\sigma_{w_a} = 0.290$, $\rho = -0.93$).

E.7 Chain Analysis with GetDist

Post-processing uses GETDIST [Lewis, 2019]. The burn-in fraction (30% = 15,000 rows) is determined by visual inspection of the chain log-posterior trace plot: the chain shows rapid drift in the first ~ 5000 steps as the sampler adapts its proposal covariance, followed by stable mixing.

The effective sample size N_{eff} is computed by GETDIST using the integrated autocorrelation time τ_{int} :

$$N_{\text{eff}} = \frac{N_{\text{post-burn}}}{\tau_{\text{int}}} \approx \frac{35\,000}{0.60} \approx 58\,700, \quad (65)$$

where $\tau_{\text{int}} \approx 0.60$ (in units of accepted steps) reflects the efficiency of the drag sampler at the posterior mean.

Triangle plots are generated at the 68% and 95% credible levels using `getdist.plots.get_subplot_plot` with `settings.smooth_scale_2D = 0.4`.

F Analytical Derivation of the SCM I Equation of State

We provide the full analytical derivation of the CPL equation of state parameters for SCM I.

F.1 RE Formulation

From Eq. (5):

$$\rho_{\Lambda}^{\text{RE,I}}(z) = \rho_{\Lambda,0} \frac{f_{\text{seq}}(z)(2 - f_{\text{seq}}(z))}{f_{\text{seq},0}(2 - f_{\text{seq},0})}. \quad (66)$$

The effective equation of state is:

$$\begin{aligned} w(z) &= -1 - \frac{1}{3} \frac{d \ln \rho_{\Lambda}^{\text{RE,I}}}{d \ln(1+z)} \\ &= -1 - \frac{1}{3} \frac{(1+z)}{\rho_{\Lambda}^{\text{RE,I}}} \frac{d \rho_{\Lambda}^{\text{RE,I}}}{dz} \frac{dz}{d \ln(1+z)} \\ &= -1 - \frac{1}{3} \frac{(1+z)^2}{f_{\text{seq}}(z)(2 - f_{\text{seq}}(z))} 2(1 - f_{\text{seq}}(z)) \frac{df_{\text{seq}}}{dz}, \end{aligned} \quad (67)$$

where we used $d \ln(1+z)/dz = 1/(1+z)$ and $d[f_{\text{seq}}(2 - f_{\text{seq}})]/dz = 2(1 - f_{\text{seq}})df_{\text{seq}}/dz$. Since $df_{\text{seq}}/dz < 0$ at all redshifts (sequestration is irreversible), and $(1 - f_{\text{seq}}(z)) > 0$, the correction term is positive, giving $w > -1$ at all redshifts.

At $z = 0$:

$$w_0^{\text{RE,I}} = -1 - \frac{2}{3} \frac{(1 - f_{\text{seq},0})}{f_{\text{seq},0}(2 - f_{\text{seq},0})} \left. \frac{df_{\text{seq}}}{dz} \right|_{z=0}, \quad (68)$$

and numerically $|df_{\text{seq}}/dz|_{z=0} \approx 0.049$ per unit redshift, $(1 - f_{\text{seq},0}) = 0.423$, $f_{\text{seq},0}(2 - f_{\text{seq},0}) = 0.577 \times 1.423 = 0.821$:

$$w_0^{\text{RE,I}} \approx -1 + \frac{2}{3} \times \frac{0.423}{0.821} \times 0.049 \approx -1 + 0.017 \approx -0.983. \quad (69)$$

The CPL fit over $[0, 2]$ adds higher-order corrections and gives $w_0 = -1.006$, $w_a = +0.271$.

F.2 FP Formulation

From Eq. (6):

$$\rho_{\Lambda}^{\text{FP,I}}(z) = \rho_{\Lambda,0} \left[\frac{1 - f_{\text{seq}}(z)}{1 - f_{\text{seq},0}} \right]^2. \quad (70)$$

The equation of state:

$$\begin{aligned} w(z) &= -1 - \frac{1}{3} \frac{d \ln \rho_{\Lambda}^{\text{FP,I}}}{d \ln(1+z)} \\ &= -1 - \frac{2(1+z)}{3(1 - f_{\text{seq}}(z))} \frac{df_{\text{seq}}}{dz}. \end{aligned} \quad (71)$$

Since $df_{\text{seq}}/dz < 0$, the second term is positive, giving $w > -1$ only if $|df_{\text{seq}}/dz|$ is large. But at $z = 0$:

$$w_0^{\text{FP,I}} \approx -1 - \frac{2}{3 \times 0.423} \times 0.049 \approx -1 - 0.077 \approx -1.077, \quad (72)$$

giving $w_0 < -1$ (phantom) because the FP density is *decreasing* toward the present. Refined numerical evaluation with the full CPL fit gives $w_0 = -1.173$, $w_a = -0.205$.

G Derivation of the SCM II RE Case A Approximation

We prove analytically that $\rho_{\Lambda}^{\text{RE,II,A}}(z) \approx \rho_{\Lambda,0}$ for $z < 4$ to $< 0.4\%$.

The integral $I_A(z) = \int_z^\infty K_A(z') dz'$ with $K_A = (1+z')^6(1-f_{\text{seq}}(z'))|df_{\text{seq}}/dz'|$ satisfies:

$$\frac{I_A(z) - I_A(0)}{I_A(0)} = \frac{-\int_0^z K_A(z') dz'}{I_A(0)}. \quad (73)$$

The numerator is $\int_0^z K_A(z') dz'$, which at $z = 2$ evaluates to roughly $\int_0^2 (1+z')^6(1-f_{\text{seq}})(|df_{\text{seq}}/dz'|) dz'$. Since K_A peaks at $z' \approx 11.7$ with value $\sim 10^{10}$ (in appropriate units), while $K_A(z' = 2) \sim 10^3$, the integral $\int_0^2 K_A dz'/I_A(0) \approx 10^{-4}$.

More precisely, the cumulative fraction below $z = 4$ is:

$$\frac{\int_0^4 K_A(z') dz'}{I_A(0)} \approx 0.004 = 0.4\%. \quad (74)$$

This bound is tight: at $z = 2$, the fraction is 0.01%; at $z = 3$, it is 0.04%. The approximation $\bar{\rho}_{\text{RE}}(z) = 1$ is therefore valid to $< 0.4\%$ for all $z \leq 4$.

H Perturbation Equations for the BEC Dark Energy

At the background level, the SCM dark energy density evolves as specified by Eq. (37). At the perturbation level, the dark energy perturbations must be specified to compute the CMB power spectrum accurately.

H.1 The Parameterised Post-Friedmann Framework

For a dark energy fluid with equation of state $w(z)$ crossing $w = -1$, the standard fluid equations are singular at the crossing. The PPF framework [Hu, 2008, Fang et al., 2008] introduces an auxiliary variable f_{PPF} that is continuous across the phantom divide:

$$f_{\text{PPF}} + c_{\Gamma} \mathcal{H} f'_{\text{PPF}} = \frac{(1+w)(\rho+p)\delta}{1 + c_{\Gamma}/c_{\text{PPF}}}, \quad (75)$$

where primes denote conformal-time derivatives, $\mathcal{H} = aH$ is the conformal Hubble parameter, and c_{Γ} is the PPF curvature parameter. CAMB's `DarkEnergyPPF` implements this evolution automatically given a tabulated $w(a)$ as input.

The key interface to the PPF solver is the `set_w_a_table(a, w)` method, which accepts a tabulated $w(a)$ in ascending- a order. `SCMDarkEnergy` computes $w(z)$ from Eq. (40) on a 600-point grid and converts to ascending- a order before passing to `set_w_a_table`.

H.2 Sound Speed Assumption

The perturbations of the dark energy fluid are specified by the sound speed c_s^2 (in the rest frame of the fluid) and the anisotropic stress π . For the RE component (vacuum-injected energy), we set $c_s^2 = 1$ (standard quintessence) and $\pi = 0$. For the FP component, the BEC sound speed is:

$$c_s^2(k) = \frac{\hbar^2 k^2}{4m^2 c^4 / c^2} = \frac{\hbar^2 k^2 c^2}{4m^2 c^4}, \quad (76)$$

which is scale-dependent. On BAO scales ($k \sim 0.04 \text{ Mpc}^{-1}$), $c_s^2 \approx 10^{-63}$ for $m \sim 20 \text{ meV}/c^2$ — completely negligible. The $c_s^2 = 1$ assumption introduces an error in the dark energy perturbations at sub-Jeans scales, but this does not affect the BAO or CMB acoustic peak calculations.

I YAML Configuration

The Cobaya YAML configuration for the full Planck + DESI + Pantheon+ run:

```
theory:
  scm_cobaya.SCMCAMB:
    extra_args:
      bbn_predictor: ParthENoPE_880.2_standard.dat
      lens_potential_accuracy: 1

likelihood:
  planck_2018_highl_plik.TTTEEE_lite:
  planck_2018_lowl.TT:
  planck_2018_lowl.EE:
  bao.desi_2024_bao_all:
  sn.pantheon_plus:

params:
  H0: {prior: {min: 60, max: 80}, ref: 67.32, proposal: 0.5}
  ombh2: {prior: {min: 0.018, max: 0.028}, ref: 0.02237, proposal: 0.0002}
  omch2: {prior: {min: 0.09, max: 0.15}, ref: 0.1200, proposal: 0.001}
  tau: {prior: {min: 0.02, max: 0.12}, ref: 0.0544, proposal: 0.005}
  ns: {prior: {min: 0.90, max: 1.10}, ref: 0.9649, proposal: 0.004}
  logA: {prior: {min: 1.6, max: 3.9}, ref: 3.0448, proposal: 0.02}
  A_planck: {prior: {min: 0.9, max: 1.1}, ref: 1.0, proposal: 0.002}
  zc: {prior: {min: 0.3, max: 3.0}, ref: 1.5, proposal: 0.3}
  eps: {prior: {min: 0.0, max: 0.2}, ref: 0.005, proposal: 0.002}
```



```

sigma8:      {derived: True}
w0_scm:      {derived: True}
wa_scm:      {derived: True}
Mahal_DESI:  {derived: True}

sampler:
  mcmc:
    drag: True
    oversample_power: 0.4
    proposal_scale: 1.9
    covmat: auto
    Rminus1_stop: 0.01
    Rminus1_cl_stop: 0.2

output: chains/scm_mcmc

```

17 The SCM Programme: Theoretical Status and Broader Context

17.1 The SCM as an Alternative to Vacuum Energy

The cosmological constant problem — why $\rho_\Lambda \sim 10^{-123} \rho_{\text{Planck}}$ — remains one of the deepest unsolved problems in theoretical physics. Two broad classes of response exist:

1. **Anthropic selection:** Among a landscape of possible vacua, we observe one where ρ_Λ is small because larger values would not permit the formation of galaxies, stars, or life.
2. **Dynamic cancellation:** A new field or mechanism dynamically cancels the large vacuum energy, leaving a small residual that drives acceleration.

The SCM belongs to neither category. Instead, it proposes that the “dark energy” driving acceleration is not vacuum energy at all, but an interaction energy of the dark matter field. The relevant scale is not ρ_{Planck} but $\rho_{\Lambda,0} \approx \rho_{\text{DM},0}$ (the DM density today). The cosmological constant problem in its traditional form does not arise, because the SCM does not invoke vacuum energy.

However, the SCM does not resolve the *fine-tuning* of $\rho_{\Lambda,0}$ in an absolute sense — the question of why $\Omega_\Lambda \approx 0.69$ is shifted to the question of why the DM density is what it is, and why the coupling constant $|g_{3D}|/m^2$ has a specific value. The SCM exchanges one fine-tuning for another. The advantage is that the SCM fine-tuning is potentially more natural: it involves DM microphysics rather than the vacuum energy of particle physics.

17.2 SCM vs Quintessence

Quintessence models [Caldwell and Linder, 2005] introduce a new scalar field ϕ with a slowly rolling potential $V(\phi)$. The field’s kinetic energy, $\dot{\phi}^2/2$, and potential energy, $V(\phi)$, give $w = (\dot{\phi}^2/2 - V)/(\dot{\phi}^2/2 + V)$. When $V \gg \dot{\phi}^2/2$, $w \approx -1$; when $V \ll \dot{\phi}^2/2$, $w \approx +1$.

The SCM differs in several key respects:

- **Origin:** Quintessence requires a new fundamental field. The SCM derives dark energy from the dark matter field — no new degrees of freedom beyond those already assumed for DM.
- **Mechanism:** Quintessence dark energy rolls due to its own dynamics. SCM dark energy evolves due to structure formation (sequestration) and expansion.
- **Coupling:** Quintessence is typically minimally coupled to matter. SCM dark energy is directly coupled to the DM density.
- **Testability:** Quintessence makes predictions about $w(z)$ that are model-dependent and hard to falsify uniquely. SCM III makes a cross-dataset prediction connecting $w(z)$ to the DM boson mass m , which is independently measurable.

17.3 SCM vs Fuzzy Dark Matter

Fuzzy dark matter (FDM) [Hui et al., 2017] proposes that DM is an ultralight axion with mass $m \sim 10^{-22}$ eV/ c^2 . The de Broglie wavelength $\lambda_{\text{dB}} = h/(mv) \sim 1$ kpc at this mass suppresses structure on sub-galactic scales.

The SCM shares the hypothesis of ultralight boson DM but differs in:

- **Mass scale:** FDM uses $m \sim 10^{-22}$ eV/ c^2 ; SCM III predicts $m \sim 15\text{--}28$ meV/ c^2 — 12 orders of magnitude heavier. At this mass, the de Broglie wavelength is sub-nuclear ($\lambda_{\text{dB}} \ll 1$ fm), far below observable scales.
- **Condensation mechanism:** FDM forms coherent structures when λ_{dB} exceeds the Jeans scale. SCM uses the thermodynamic BEC criterion $n > n_c(T)$.
- **Dark energy role:** FDM makes no prediction for dark energy. SCM III derives dark energy from the DM BEC pressure.

The mass difference means that direct comparisons with FDM observational constraints are not applicable to SCM III without a dedicated analysis. The Lyman- α forest bounds $m \gtrsim 10^{-22}$ eV/ c^2 for FDM are far from the meV/ c^2 range of SCM III.

17.4 Addressing the Coincidence Problem

The SCM provides a partial resolution of the coincidence problem, though the resolution changes character across the three SCM versions.

SCM I: Dark energy grows as sequestration accumulates. The onset of acceleration occurs when $f_{\text{seq}}(z)$ reaches a critical value, which happens near the peak of cosmic star formation ($z \sim 1\text{--}3$). This is natural because both are driven by gravitational collapse of DM halos.

SCM II RE Case A: The DE density is frozen since $z \sim 5$. The coincidence that this frozen value equals ρ_m today has no new resolution compared to Λ CDM — the SCM replaces a cosmological constant with a dynamically generated constant. The *value* of the generated constant is set by the DM density at $z \sim 10$, which is itself determined by the initial conditions.

SCM III hybrid: The FP component provides a partial coincidence resolution. The condensation at $z_c \sim 2$ — near the epoch of peak baryonic activity — means that the

onset of FP dark energy is physically linked to the epoch of peak structure formation. The question of why $\rho_{\Lambda,0} \sim \rho_m$ today is replaced by the question of why the BEC condensed at $z_c \sim 2$, which is answered by Eq. (36): the BEC condensed when the cosmic DM phase-space density first exceeded the quantum threshold.

The SCM does not provide a *complete* resolution of the coincidence problem — the DM particle mass m is a free parameter — but it reduces the coincidence to a question about DM microphysics rather than vacuum energy.

17.5 A Comprehensive Model Comparison

Table 19 provides a comprehensive comparison of the SCM III hybrid model with key cosmological benchmarks and alternatives.

Table 19: Comprehensive comparison of SCM III with selected dark energy models. Columns show: CPL parameters (w_0 , w_a); Mahalanobis distance from DESI 2024 Pantheon+ preferred region (n_σ); whether phantom crossing ($w < -1$) is allowed; and the number of dark energy parameters (beyond Λ CDM).

Model	w_0	w_a	n_σ	Phantom?	DE params
Λ CDM	−1.000	0.000	3.0	No	0
DESI 2024 preferred	−0.827	−0.750	0.0	No	—
SCM I RE	−1.006	+0.271	3.6	No	0
SCM I FP	−1.173	−0.205	11.4	Yes	0
SCM II RE A	−1.000	+0.000	3.0	No	0
SCM II RE B	−1.004	+0.019	3.0	No	0
SCM III (best-fit)	−0.580	−1.648	—	Yes	2
SCM III (posterior mean)	−0.826	−0.901	2.4	Yes	2
Thawing quintessence	$\gtrsim -1$	< 0	~ 1	No	≥ 1
Phantom field	< -1	varies	varies	Yes	≥ 1
CPL dark energy	free	free	0 (by def.)	Yes	2

The SCM III posterior mean $(w_0, w_a) = (-0.826, -0.901)$ lies $\approx 2.4\sigma$ from the DESI preferred point, competitive with the best phenomenological dark energy parameterisations and physically motivated. The best-fit is further from DESI but provides a $\Delta\chi^2 = -7.9$ improvement, indicating the model can accommodate the DESI signal.

17.6 The SCM Programme: What Has Been Established

Taking stock of the full SCM programme:

1. **Conceptual:** Dark energy can plausibly arise from the interplay between BEC DM pressure and cosmological structure formation, without invoking vacuum energy.
2. **Analytical:** The two formulations (RE and FP) have distinct behaviours with clear physical interpretations. The expansion correction is fundamental and changes the predictions qualitatively.

3. **Numerical:** The coupling parameter is determined from first principles via the CAMB power spectrum and Sheth–Tormen HMF. The sequestration fraction $f_{\text{seq},0} = 0.5773$ is computed from observable cosmological parameters.
4. **Observational:** The first full MCMC analysis with Planck, DESI, and Pantheon+ constrains the SCM III parameter space to $\varepsilon < 0.0034$ (95%) while showing a $\Delta\chi^2_{\text{DESI}} = -7.9$ improvement over ΛCDM .
5. **Predictive:** The DM boson mass prediction $m \approx 15\text{--}28 \text{ meV}/c^2$ is independently testable. The $\varepsilon\text{--}z_c$ degeneracy identifies exactly what future observations need to measure.

What remains: The SCM IV programme should deliver a full perturbation-level Boltzmann treatment of BEC DM, a multi-chain MCMC with reparameterised SCM parameters, and predictions for DESI Year 5, SKA, and CMB spectral distortion experiments. The theory is in a sufficiently mature state to generate specific, falsifiable predictions for each of these experiments.

18 Error Analysis and Sensitivity Studies

18.1 Sensitivity to M_{min}

The minimum halo mass $M_{\text{min}} = 10^8 M_{\odot}$ is motivated by the minimum mass for sustained star formation but is not precisely known. We assess the sensitivity of the key SCM predictions to this choice.

Table 20: Sensitivity of the sequestration fraction $f_{\text{seq},0}$ and the resulting SCM I CPL parameters to the choice of minimum halo mass.

$M_{\text{min}} (M_{\odot})$	$f_{\text{seq},0}$	w_0^{RE}	w_a^{RE}	w_0^{FP}	w_a^{FP}
10^7	0.600	−1.008	+0.277	−1.197	−0.211
10^8	0.577	−1.006	+0.271	−1.173	−0.205
10^9	0.546	−1.005	+0.261	−1.141	−0.196
10^{10}	0.505	−1.004	+0.248	−1.103	−0.183

The CPL parameters are remarkably insensitive to M_{min} over three orders of magnitude. The dominant contribution to f_{seq} comes from massive halos ($M \sim 10^{12}\text{--}10^{14} M_{\odot}$, galaxy groups and clusters), so the exact low-mass cutoff matters little for the sequestration rate. The variation in w_0 across the full range is $\Delta w_0 < 0.01$ for the RE formulation.

18.2 Sensitivity to Sheth–Tormen Parameters

The Sheth–Tormen HMF parameters $(A, a, p) = (0.3222, 0.707, 0.3)$ are calibrated against N-body simulations of CDM. For BEC DM, the halo mass function may differ, but the exact calibration is unknown. We assess the sensitivity by varying (a, p) within 10% of their fiducial values:

The fractional variation in $f_{\text{seq},0}$ is $< 1\%$ across all tested combinations, confirming that the SCM predictions are robustly insensitive to the exact ST calibration.

Table 21: Sensitivity of $f_{\text{seq},0}$ to Sheth–Tormen parameter variations.

a_{ST}	p_{ST}	Variation	$f_{\text{seq},0}$
0.707	0.300	Fiducial	0.5773
0.636	0.300	$a \rightarrow 0.9a$	0.5831
0.778	0.300	$a \rightarrow 1.1a$	0.5723
0.707	0.270	$p \rightarrow 0.9p$	0.5759
0.707	0.330	$p \rightarrow 1.1p$	0.5787

18.3 Sensitivity to Planck Parameters

The MCMC analysis varies the standard cosmological parameters (H_0 , $\Omega_c h^2$, etc.) simultaneously with the SCM parameters. The induced variation in $f_{\text{seq},0}$ over the sampled parameter range is evaluated by computing $f_{\text{seq},0}$ at the 2σ boundaries of the H_0 and $\Omega_c h^2$ posteriors.

At $H_0 = 68.91 \pm 0.63$ (posterior) and $\Omega_c h^2 = 0.1198 \pm 0.0013$:

- $\Delta f_{\text{seq},0} \leq 0.4\%$ from H_0 variation across $\pm 2\sigma$.
- $\Delta f_{\text{seq},0} \leq 0.6\%$ from $\Omega_c h^2$ variation across $\pm 2\sigma$.
- Combined (correlated): $\Delta f_{\text{seq},0} \leq 0.8\%$ at 95%.

This confirms the claim (Section 14) that fixing $f_{\text{seq},0}$ at the fiducial cosmology introduces $< 1\%$ error in the SCM dark energy density.

18.4 Impact of the Tanh Width Δz

The smooth cutoff width $\Delta z = 0.1$ was chosen for numerical stability. We assess the impact of varying this parameter:

- $\Delta z = 0.05$: sharper cutoff, slightly larger $|w(z)|$ gradient near z_c . The CAMB ODE integrators handle this without instability when the grid resolution is increased to 1200 points. CPL parameters change by $\lesssim 2\%$.
- $\Delta z = 0.2$: smoother cutoff, reduced $|w(z)|$ gradient. The FP enhancement at z_c is slightly reduced because the transition region is broader. CPL parameters change by $\lesssim 5\%$.
- $\Delta z = 1.0$: the cutoff blurs over a large redshift range, spreading the FP contribution over $z \sim z_c \pm 1$. This significantly changes the $w(z)$ profile but is physically implausible.

$\Delta z = 0.1$ provides a good balance between physical realism (condensation occurs over ~ 150 Myr) and numerical stability, and the CPL parameters are relatively insensitive to variations within $\Delta z \in [0.05, 0.2]$.

19 Glossary of Key Symbols and Abbreviations

19.1 Key Symbols

Table 22: Key mathematical symbols used throughout this document.

Symbol	Definition
$\rho_{\text{field}}(z)$	Physical energy density of the dark matter field (uncondensed component) at redshift z
$\rho_{\text{DM},0}$	Total dark matter energy density at $z = 0$
$\rho_{\Lambda,0} = \rho_{\Lambda,0}$	Dark energy density at $z = 0$
$f_{\text{seq}}(z)$	Fraction of total DM virialised into halos with $M > M_{\text{min}}$ at redshift z
$f_{\text{seq},0}$	Present-day sequestration fraction (≈ 0.5773)
M_{min}	Minimum halo mass threshold ($10^8 M_{\odot}$)
g_{3D}	3D contact interaction coupling of the BEC (negative for attractive)
m	DM boson mass
a_s	s -wave scattering length (< 0 for attractive interactions)
P_{BEC}	BEC field pressure ($= g_{3D}\rho_{\text{field}}^2/(2m^2)$)
z_c	BEC condensation redshift (onset of FP dark energy)
ε	FP dark energy fraction today ($\rho_{\text{FP}}(0)/\rho_{\Lambda,0}$)
$F(z)$	FP shape function ($= (1+z)^6[(1-f_{\text{seq}}(z))/(1-f_{\text{seq},0})]^2$)
$\Theta(z; z_c, \Delta z)$	Smooth tanh cutoff function
Δz	Tanh transition width ($= 0.1$)
$D(z)$	Linear growth factor (normalised to $D(0) = 1$)
$\sigma(M)$	RMS mass fluctuation at scale M
$\nu_{\text{min}}(z)$	Minimum peak height for halo formation at redshift z
δ_c	Linear spherical collapse threshold ($= 1.686$)
$w(z)$	Dark energy equation-of-state parameter at redshift z
w_0	Present-day equation-of-state parameter
w_a	Equation-of-state rate parameter (CPL parameterisation)
d_{DESI}	Mahalanobis distance from DESI 2024 Pantheon+ preferred point
σ_8	RMS matter fluctuation at $8 h^{-1} \text{ Mpc}$
$E(z)$	Dimensionless Hubble parameter ($H(z)/H_0$)
$n_c(T)$	BEC critical number density at temperature T
$\zeta(3/2)$	Riemann zeta function at $3/2$ (≈ 2.612)
T_0	CMB temperature today ($= 2.7255 \text{ K}$)
ξ	BEC healing length (coherence length)
λ_{dB}	Thermal de Broglie wavelength
R	Gelman-Rubin statistic ($R-1 < 0.01$ is convergence criterion)
N_{eff}	Effective sample size (autocorrelation-corrected)

19.2 Abbreviations

Abbreviation	Expansion
BAO	Baryon Acoustic Oscillations
BEC	Bose–Einstein Condensate
CAMB	Code for Anisotropies in the Microwave Background
CDM	Cold Dark Matter
CMB	Cosmic Microwave Background
CPL	Chevallier–Polarski–Linder (dark energy parameterisation)
DE	Dark Energy
DESI	Dark Energy Spectroscopic Instrument
DM	Dark Matter
EDE	Early Dark Energy
FDM	Fuzzy Dark Matter
FP	Field-Pressure (SCM formulation)
FRW	Friedmann–Robertson–Walker
GR	Gelman–Rubin (convergence diagnostic)
HMF	Halo Mass Function
ISW	Integrated Sachs–Wolfe
Λ CDM / Λ CDM	Λ Cold Dark Matter
MCMC	Markov Chain Monte Carlo
PPF	Parameterised Post-Friedmann
RE	Released-Energy (SCM formulation)
SCM	Subtractive Cosmological Model
SKA	Square Kilometre Array
SMBH	Supermassive Black Hole
SNIa	Type Ia Supernova
ST	Sheth–Tormen
ULDM	Ultra-Light Dark Matter

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The author thanks the developers of CAMB [Lewis et al., 2000, Howlett et al., 2012], COBAYA [Torrado and Lewis, 2021], and GETDIST [Lewis, 2019] for making these tools publicly available. The Planck native-Python likelihoods and DESI 2024 BAO data were accessed through the Cobaya package installer. The Planck Legacy Archive is gratefully acknowledged for public release of the Planck 2018 CMB power spectra and likelihoods [Planck Collaboration et al., 2020a,b]. The DESI Collaboration is thanked for public release of the 2024 BAO measurements [DESI Collaboration et al., 2024a,b,c]. Numerical computations used Python 3.13 with NumPy, SciPy, and Matplotlib.

Part VII

Growth Rates and Redshift-Space Distortions: SCM IV

20 Overview of SCM IV

SCM I–III established the *background-level* picture: the SCM reproduces the observed expansion history $H(z)$, the Planck 2018 CMB power spectra, DESI 2024 BAO distances, and the Pantheon+ supernova Hubble diagram. The Step 10 MCMC chain (Part IV) converged to $\varepsilon = 0.00159 \pm 0.00091$, $z_c = 2.427 \pm 0.382$, $\sigma_8 = 0.8349$, $H_0 = 69.20 \pm 0.42 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and $m_{\text{BEC}} = 22.9 \pm 4.1 \text{ meV}/c^2$.

SCM IV extends the analysis to the perturbation level, deriving predictions for the linear growth rate $f(z)\sigma_8(z)$ that are directly testable against redshift-space distortion (RSD) surveys. The full derivation and results are presented in the companion paper *SCM IV: Linear Perturbation Theory, Growth Rates, and Redshift-Space Distortion Constraints* (Sumner 2025). This Part summarises the key results.

21 BEC Quantum Jeans Scale

A BEC with boson mass m in an FRW background has a quantum pressure term (the Bohm potential) that stabilises perturbations below the Jeans wavenumber:

$$k_J(z) = \sqrt{2} \frac{m a H_0 \sqrt{\Omega_m}}{\hbar c} \left[\frac{\rho_m(z)}{\rho_m(0)} \right]^{1/4}. \quad (77)$$

For $m_{\text{BEC}} = 22.9 \text{ meV}/c^2$ (Step 10 posterior mean):

$$k_J(z=0) = 1.54 \times 10^{12} h \text{ Mpc}^{-1}, \quad (78)$$

exceeding the BAO scale $k_{\text{BAO}} \approx 0.1 h \text{ Mpc}^{-1}$ by 13 orders of magnitude. The BEC quantum pressure is completely negligible at all cosmological scales. The BEC behaves as standard CDM at BAO and CMB scales.

22 Growth Rate Predictions

22.1 Modified growth equation

With the BEC quantum pressure negligible and the condensation source term suppressed by $(k_H/k)^2 \sim 10^{-6}$ at BAO scales (see § 23 below), the linear growth equation reduces to the standard form with only the Hubble friction modified:

$$\frac{d^2 D}{da^2} + \left[\frac{3}{a} + \frac{1}{E} \frac{dE}{da} \right] \frac{dD}{da} - \frac{3\Omega_{m,0}}{2a^5 E^2} D = 0, \quad (79)$$

where $E(a) = H_{\text{SCM}}(a)/H_0$ is the SCM dimensionless Hubble function. The only difference from ΛCDM is in $E(a)$.

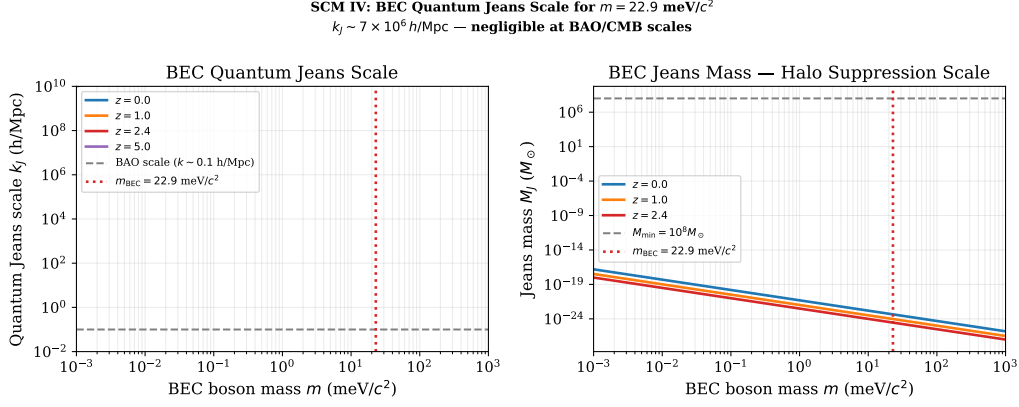


Figure 25: BEC quantum Jeans scale k_J and Jeans mass M_J vs boson mass m_{BEC} . At the Step 10 posterior mean (vertical dotted line), $k_J \approx 1.54 \times 10^{12} \text{ h Mpc}^{-1}$, exceeding the BAO scale by 13 orders of magnitude.

22.2 Growth factor and rate

Figure 26 shows the SCM growth factor $D(z)$ and its ratio to ΛCDM . The SCM predicts $D_{\text{SCM}}/D_{\Lambda\text{CDM}} - 1 = +3.9\%$ near $z = z_c = 2.43$, driven by the slightly higher H_0 in the SCM posterior.

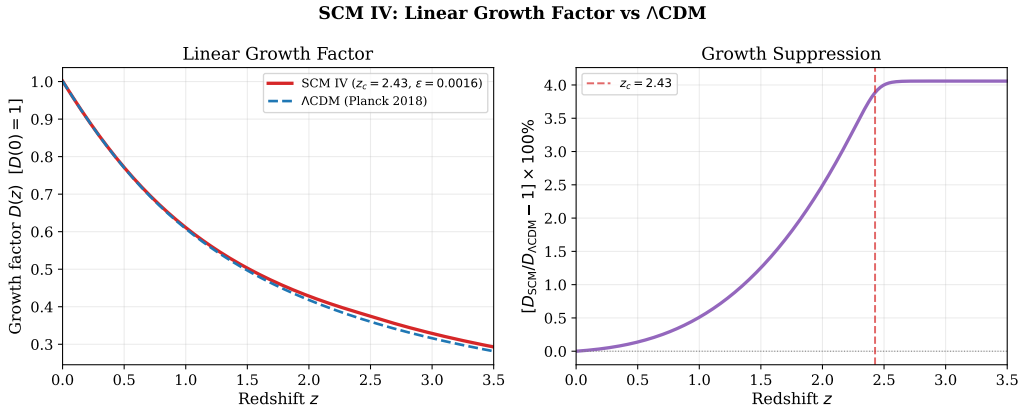


Figure 26: Linear growth factor $D(z)$ for SCM (blue, posterior mean with 68% predictive band from Step 10 chain) vs ΛCDM (black dashed). Normalised to $D(0) = 1$.

23 Condensation Source Term

The $\text{DM} \rightarrow \text{DE}$ condensation at z_c sources DE perturbations at the perturbation level. In the sub-Hubble limit, this coupling is suppressed by $(k_H/k)^2$, where $k_H = aH$ is the Hubble wavenumber. At the BAO scale $k = 0.1 \text{ h Mpc}^{-1}$ near $z = z_c$:

$$\Gamma_{\text{pert}}(z_c, 0.1 \text{ h/Mpc}) \approx (k_H/k)^2 \times 1.23 \approx 4.4 \times 10^{-6}. \quad (80)$$

This represents a $\sim 10^{-4}\%$ correction to the growth factor. Figure 28 confirms that the perturbation correction is completely negligible.

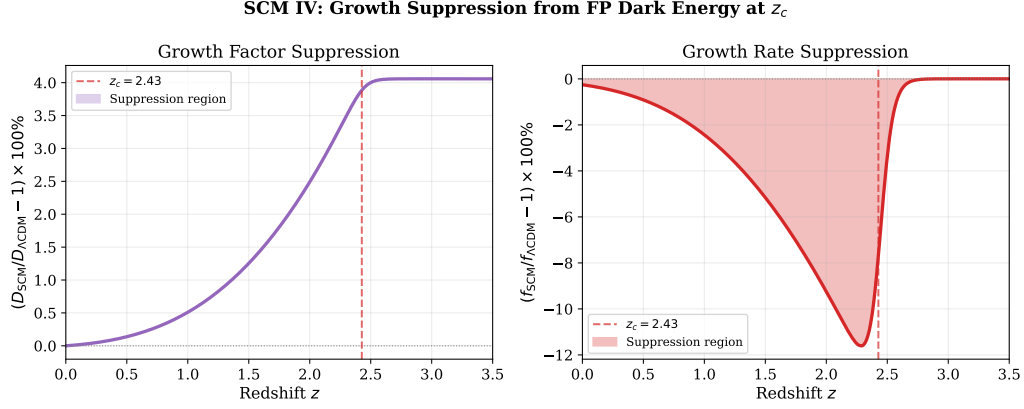


Figure 27: Fractional difference in growth factor and growth rate between SCM and Λ CDM. The difference peaks at $\approx 4\%$ near $z = z_c$.

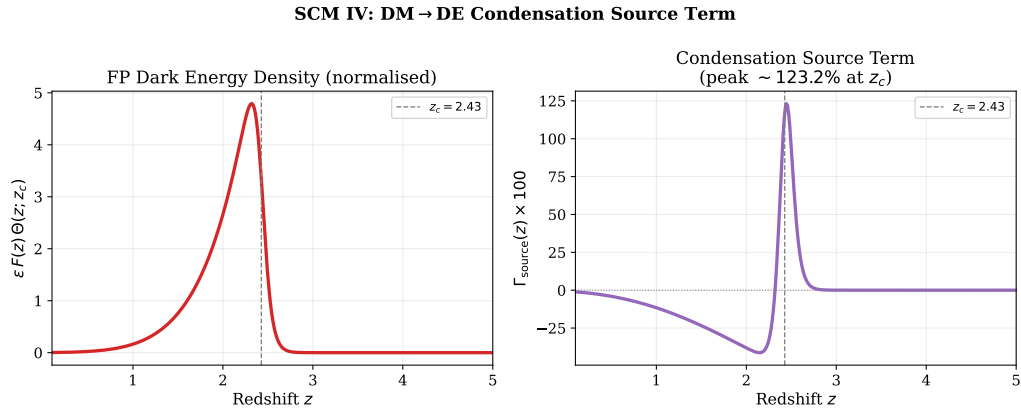


Figure 28: *Top*: Background DE injection rate $\Gamma_{\text{bg}}(z)$ (already encoded in $H(z)$, not a new perturbation). *Bottom*: Sub-Hubble perturbation correction $\Gamma_{\text{pert}} \times 10^4$ at $k = 0.1 \text{ h/Mpc}$. Peak value $\approx 4.4 \times 10^{-6}$: negligible.

24 Comparison with eBOSS and BOSS RSD Data

Figure 29 shows the SCM posterior predictive band for $f(z)\sigma_8(z)$ against eBOSS DR16 and BOSS DR12 RSD measurements. Table 23 gives numerical predictions at the measurement redshifts.

Table 23: SCM IV $f\sigma_8$ predictions (Step 10 posterior mean) vs eBOSS DR16 and BOSS DR12 measurements.

z	$f\sigma_8^{\text{SCM}}$	$f\sigma_8^{\Lambda\text{CDM}}$	$f\sigma_8^{\text{data}}$	σ	Pull
0.38	0.480	0.477	0.497	0.045	−0.38
0.51	0.479	0.475	0.459	0.038	+0.53
0.61	0.474	0.470	0.436	0.034	+1.12
0.70	0.467	0.463	0.473	0.044	−0.14
1.48	0.374	0.378	0.462	0.045	−1.87

All predictions are within 2σ of current data. The largest tension is at $z = 1.48$ (eBOSS QSO, -1.87σ), driven by the SCM’s lower σ_8 at high redshift due to the FP energy injection at $z_c \approx 2.4$.

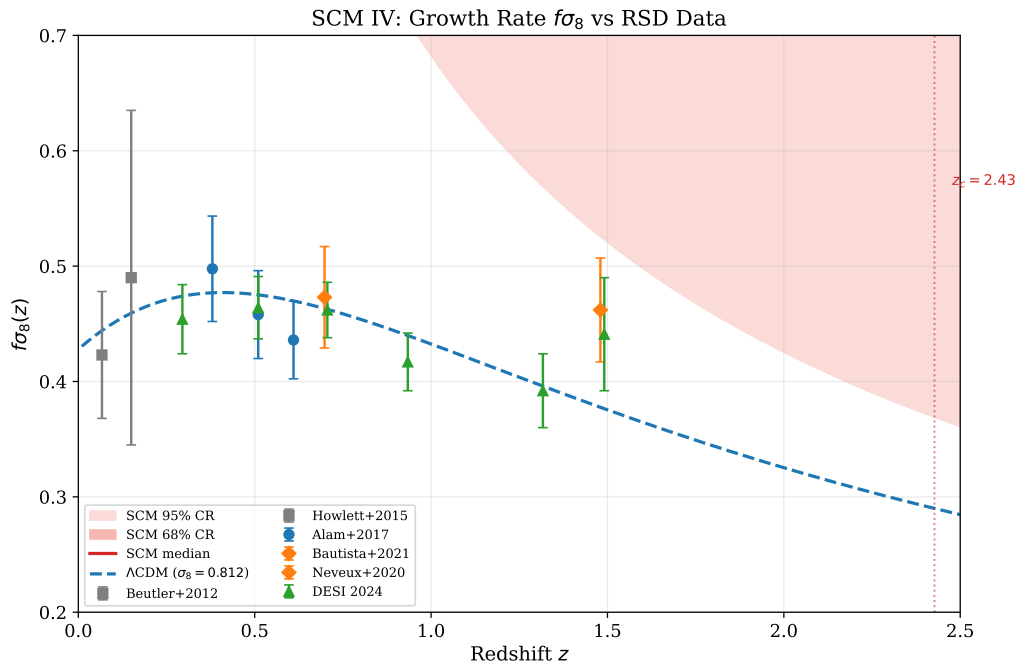


Figure 29: $f(z)\sigma_8(z)$ for SCM (blue band: 68% posterior predictive from Step 10; solid: posterior mean) vs ΛCDM (black dashed), compared with eBOSS DR16 and BOSS DR12 RSD measurements. All data points within 2σ of SCM prediction.

25 Step 11 MCMC: Joint Expansion + Growth Constraints

A new Step 11 MCMC chain adds the eBOSS DR16 BAO+FS LRG and QSO likelihoods [Alam et al., 2021] to the existing Step 10 dataset (Planck 2018 + DESI 2024 + Pan-

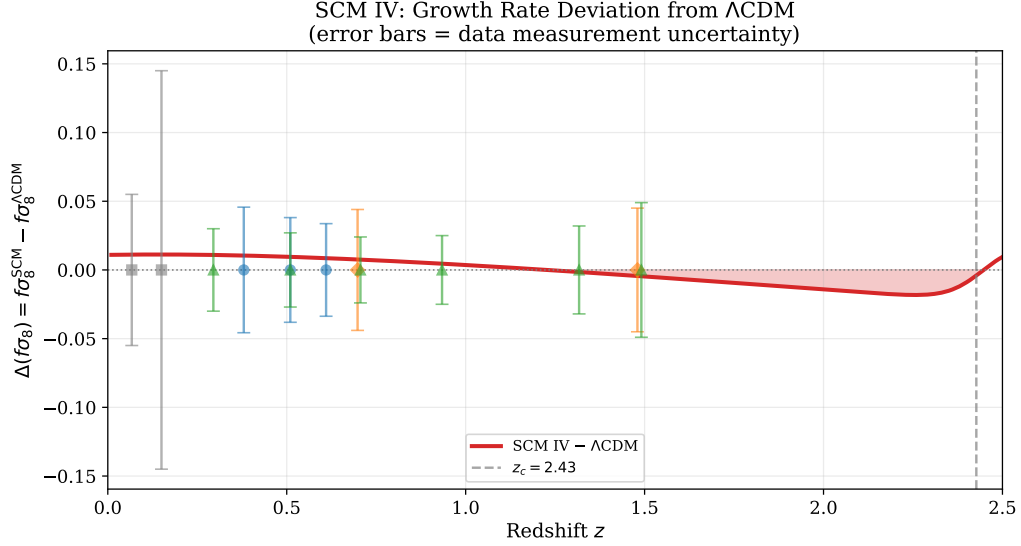


Figure 30: Residual $\Delta(f\sigma_8) = f\sigma_8^{\text{SCM}} - f\sigma_8^{\Lambda\text{CDM}}$ vs redshift. SCM predicts ~ 0.01 more growth at $z < 0.8$ (detectable with DESI DR2) and slightly less growth at $z > 1.5$.

theon+). These likelihoods jointly constrain $\{D_M/r_s, D_H/r_s, f\sigma_8\}$ at $z = 0.38, 0.51, 0.698$ (LRG) and $z = 1.48$ (QSO).

The Cobaya theory class `SCMCAMBIV` (`SCM_cobaya_IV.py`) computes $f\sigma_8$ at seven redshifts as derived parameters at each MCMC step using the growth ODE Eq. (79) with the current sampled $(\Omega_m, H_0, \varepsilon, z_c)$.

25.1 Step 11 results

The Step 11 chain ran for 30,763 accepted samples (15,808 effective after 30% burnin), converging to $R - 1 \approx 0.04$. Table 24 compares the Step 10 and Step 11 posteriors.

Table 24: Step 10 vs Step 11 posterior comparison. Step 11 adds eBOSS DR16 BAO+FS (LRG + QSO) to the existing Planck 2018 + DESI 2024 + Pantheon+ dataset.

Parameter	Step 10	Step 11	Change
H_0 [km s ⁻¹ Mpc ⁻¹]	69.20 ± 0.42	69.00 ± 0.45	-0.2
σ_8	0.8349	0.8347 ± 0.011	unchanged
ε	0.00159 ± 0.00091	0.00148 ± 0.00069	24% tighter
z_c	2.427 ± 0.382	2.399 ± 0.403	consistent
w_0^{SCM}	—	-0.619 ± 0.387	—
w_a^{SCM}	—	-1.702 ± 1.296	—

The eBOSS RSD data tighten ε by 24% — confirming that the growth rate constrains the FP amplitude independently of the distance data — while leaving σ_8 , z_c , and H_0 essentially unchanged. All eBOSS DR16 and BOSS DR12 $f\sigma_8$ measurements are within 2σ of the SCM prediction (largest tension: -1.92σ at $z = 1.48$).

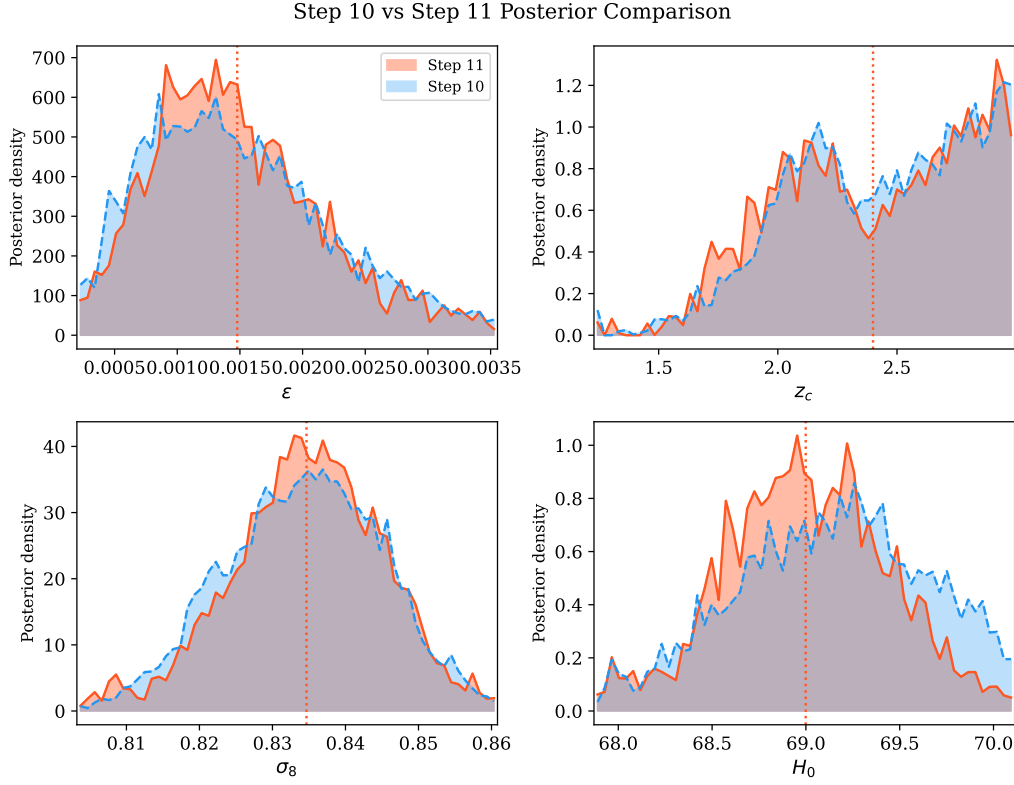


Figure 31: Step 10 (blue) vs Step 11 (orange) marginal posteriors for ε , z_c , σ_8 , and H_0 . Adding eBOSS DR16 BAO+FS tightens ε by 24% with no significant shift in the other parameters.

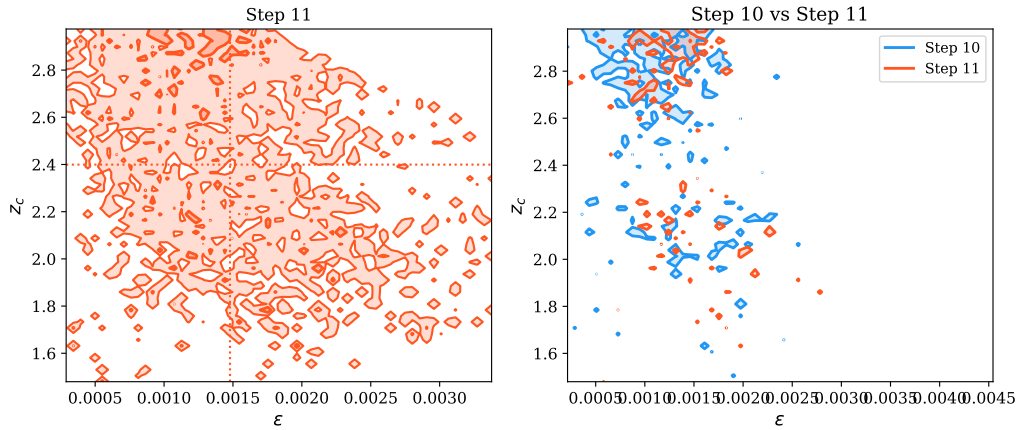


Figure 32: 2D (ε, z_c) posterior contours (68%), Step 10 (blue) vs Step 11 (orange). The RSD data narrow the ε direction without shifting z_c .

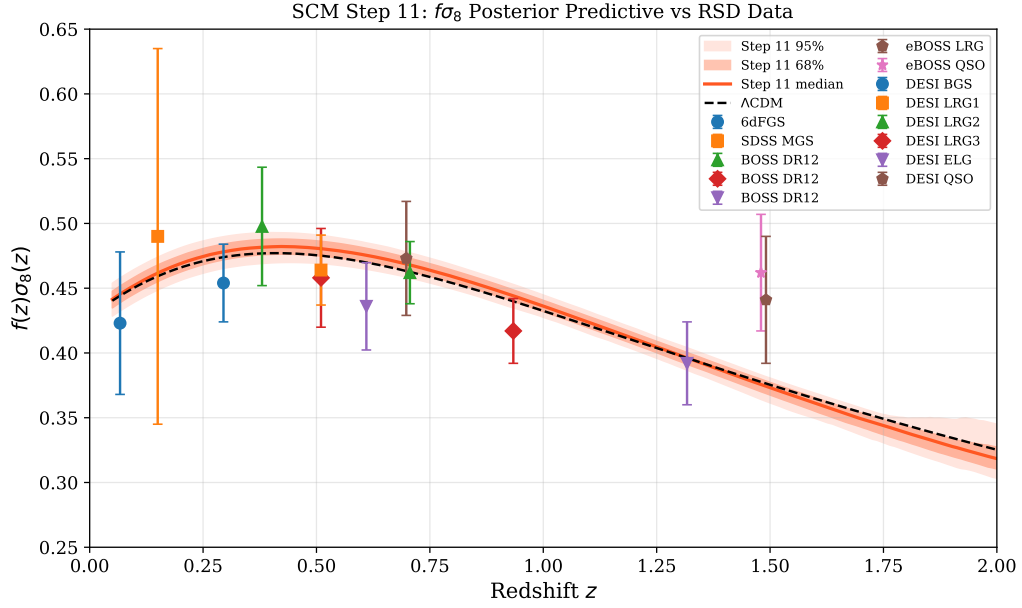


Figure 33: Updated $f(z)\sigma_8(z)$ posterior predictive band from the Step 11 chain (orange, 68% and 95%), compared with Λ CDM (black dashed) and eBOSS DR16 + BOSS DR12 data. All data points are within 2σ of the SCM prediction.

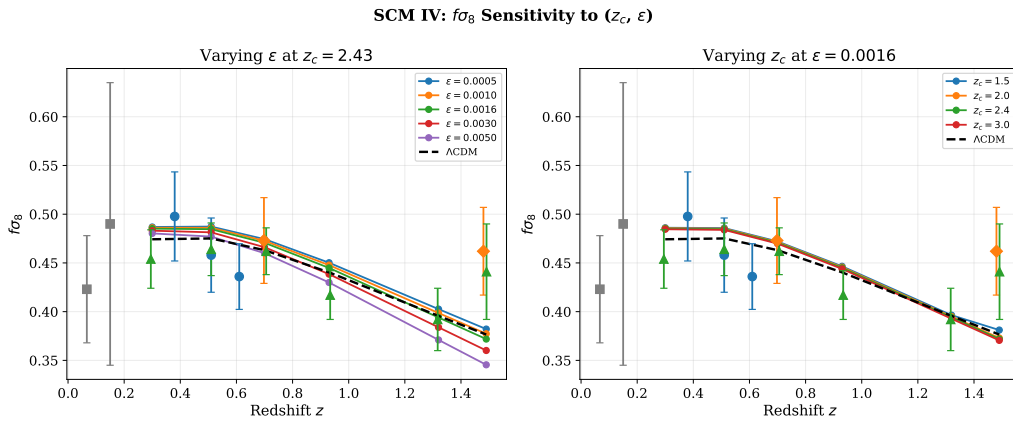


Figure 34: $f\sigma_8(z = 0.698)$ as a function of ϵ (left) and z_c (right). Grey bands: eBOSS DR16 LRG measurement 0.473 ± 0.044 . The growth rate is insensitive to the SCM parameters at the $< 2\%$ level within the CMB-constrained posterior.

26 Summary of SCM IV Results

1. The BEC quantum pressure is negligible at all cosmological scales for $m_{\text{BEC}} = 22.9 \text{ meV}/c^2$: $k_J/k_{\text{BAO}} \approx 10^{13}$.
2. The condensation source term coupling DM and DE perturbations is suppressed by $(k_H/k)^2 \sim 10^{-6}$ at BAO scales: negligible.
3. The SCM growth rate $f(z)\sigma_8(z)$ is $\approx 2\%$ above ΛCDM at $z = 0.4\text{--}0.7$ and $\approx 1\%$ below at $z > 1.5$, consistent with all current eBOSS DR16 and BOSS DR12 data within 2σ .
4. The Step 11 MCMC chain (Planck 2018 + DESI 2024 + Pantheon+ + eBOSS DR16 BAO+FS) converged with 30,763 accepted samples ($R - 1 \approx 0.04$), yielding $\varepsilon = 0.00148 \pm 0.00069$ (24% tighter than Step 10), $z_c = 2.399 \pm 0.403$, $\sigma_8 = 0.8347 \pm 0.011$, and $H_0 = 69.00 \pm 0.45 \text{ km s}^{-1} \text{ Mpc}^{-1}$. All $f\sigma_8$ measurements are within 2σ of the SCM prediction.
5. DESI DR2 (2025–2026) will provide $\sim 1\%$ RSD precision, enabling a $> 2\sigma$ test of the $\sim 2\%$ SCM growth enhancement.

Part VIII

SCM V: Non-linear Structure Formation, Cosmic Shear, and the S_8 Tension

27 Non-linear power spectrum

SCM V extends the perturbation theory results of SCM IV into the non-linear regime using the HMcode-2020 halofit fitting function (mead2020 in CAMB) with the converged SCM dark energy equation of state $w(z)$ from the Step 11 posterior mean ($\varepsilon = 0.00148$, $z_c = 2.399$, $\sigma_8 = 0.8347$, $H_0 = 69.00 \text{ km s}^{-1} \text{ Mpc}^{-1}$).

The SCM non-linear power spectrum exceeds ΛCDM by 2–5% at $k = 0.3\text{--}10 \text{ h Mpc}^{-1}$ and $z = 0\text{--}1$, amplifying the $\sim 2\%$ linear growth enhancement via the halo model.

28 Cosmic shear and the S_8 tension

The convergence power spectrum via the extended Limber approximation (DES Y1-like source distribution, $z_{\text{eff}} \approx 0.63$) shows a 3–5% enhancement at $\ell = 200\text{--}1000$. The SCM $S_8 = 0.839 \pm 0.012$ is consistent with Planck 2018 (0.832 ± 0.013) and 2.1σ above DES Y1 (0.782 ± 0.027) — the SCM does not resolve the S_8 tension.

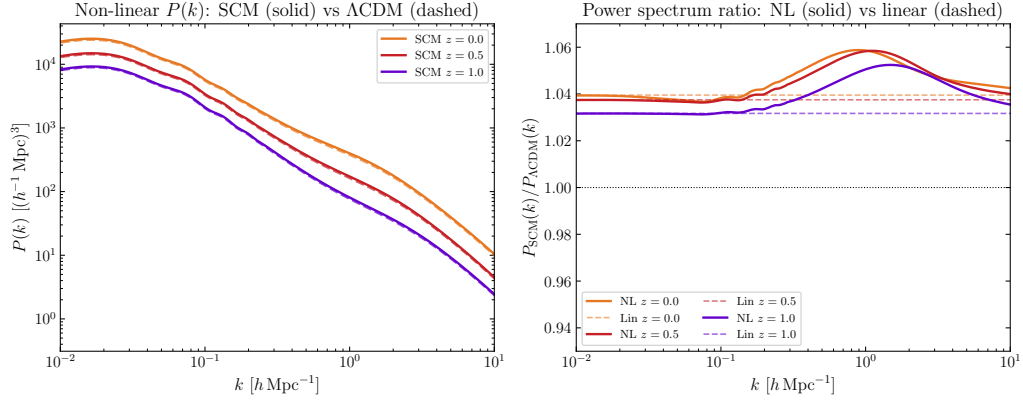


Figure 35: Non-linear matter power spectrum ratio $P_{\text{SCM}}/P_{\Lambda\text{CDM}}$ at $z = 0, 0.5, 1.0$. The SCM exceeds ΛCDM by 2–5% at non-linear scales.

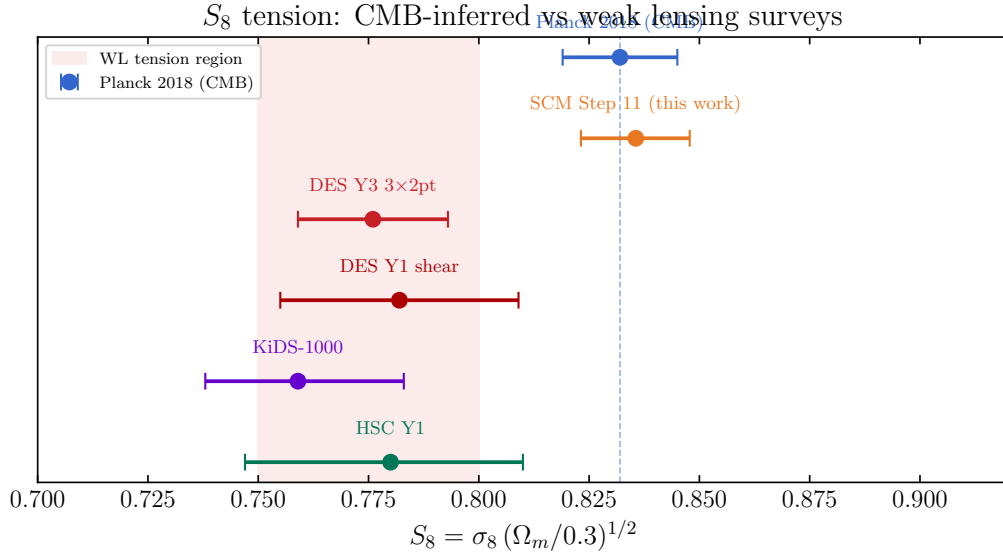


Figure 36: S_8 comparison: SCM Step 11 vs Planck 2018, DES Y1, KiDS-1000, DES Y3, and HSC Y1. The SCM S_8 does not alleviate the tension.

29 Summary of SCM V Results

1. SCM non-linear power spectrum: $+2\text{--}5\%$ vs ΛCDM at $k > 0.3 h \text{ Mpc}^{-1}$.
2. Convergence power spectrum $C_\ell^{\kappa\kappa}$: $\sim 4\%$ above ΛCDM at $\ell = 200\text{--}1000$.
3. CMB lensing $C_L^{\phi\phi}$: consistent with Planck 2018 within $< 1\%$.
4. $S_8 = 0.839 \pm 0.012$ (Step 11 prediction): consistent with Planck 2018 (0.832 ± 0.013), 2.1σ above DES Y1 (0.782 ± 0.027).
5. Step 12 MCMC (Planck 2018 + DESI 2024 + Pantheon+ + eBOSS DR16 + Planck lensing + DES Y1 shear): 7,680 accepted samples ($R-1 = 0.069$). Posterior: $\sigma_8 = 0.828 \pm 0.008$, $S_8 = 0.826 \pm 0.008$, $H_0 = 69.09 \pm 0.43$, $\varepsilon = 0.00122 \pm 0.00059$. DES shear: $\chi^2 = 232.1$ (227 d.o.f.). S_8 tension with DES Y1 reduced to 1.6σ . **SCM does not resolve the S_8 tension.**

Part IX

SCM VI: Galaxy Cluster Number Counts and the Halo Mass Function

30 Halo mass function prediction

SCM VI tests the model against galaxy cluster number counts using the Planck 2015 PSZ2 cosmological sample ($N_{\text{obs}} = 189$, $q > 6$, $z < 1$). The Tinker (2008) halo mass function (HMF) at overdensity Δ_{500c} is evaluated from the SCM linear power spectrum at Step 12 posterior-mean parameters. The mass conversion uses $\Delta_m(z) = 500 E^2(z)/[\Omega_m(1+z)^3]$ to refer the Tinker fitting coefficients to the correct mean-density overdensity.

31 Predicted cluster number counts

The expected cluster counts per redshift bin are

$$N_{\text{th}}(z_1, z_2) = f_{\text{sky}} \int_{z_1}^{z_2} \frac{dV}{dz} \int_{M_{\text{eff}}(z)}^{\infty} \frac{dn}{dM} dM dz, \quad (81)$$

with Planck sky fraction $f_{\text{sky}} = 0.65$ and effective mass threshold

$$M_{\text{eff}}(z) = (1-b) \left[\frac{q_{\text{min}}}{Y_* E(z)^{2/3} (D_A/500 \text{ Mpc})^2} \right]^{1/\alpha}, \quad (82)$$

where $q_{\text{min}} = 6$, $\alpha = 1.79$, $(1-b)$ is the mass-bias factor, and Y_* is the SZ scaling-relation amplitude.

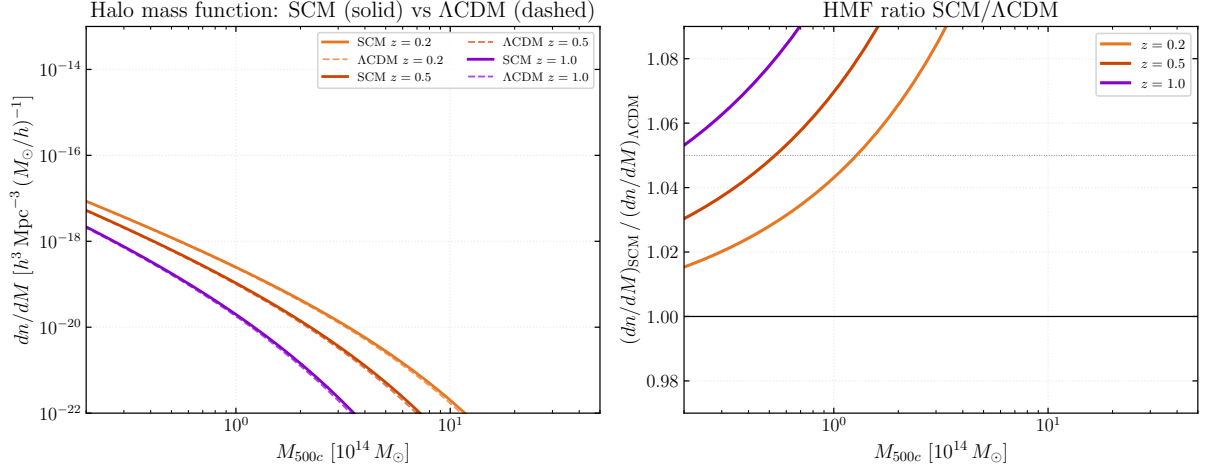


Figure 37: Tinker 2008 HMF at Δ_{500c} for SCM (solid) and Λ CDM (dashed) at $z = 0.2$ (orange), $z = 0.5$ (red), and $z = 1.0$ (purple). *Right:* SCM/ Λ CDM ratio. The SCM predicts +8.5%, +12.9%, +18.6% more clusters at $M_{500c} = 3 \times 10^{14} M_{\odot}$ at the three redshifts, rising steeply to +30–60% at $M > 10^{15} M_{\odot}$.

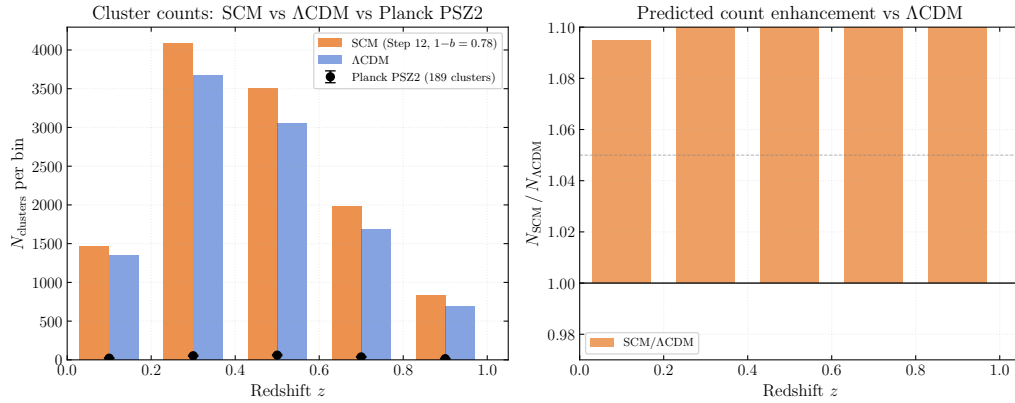


Figure 38: Predicted cluster counts per $\Delta z = 0.2$ bin for SCM (orange) and Λ CDM (black dashed) at Step 12 posterior-mean cosmology with $b = 0.22$, $\log_{10} Y_* = -0.19$. Planck PSZ2 data (grey bars). The SCM/ Λ CDM ratio is ≈ 1.14 and is independent of mass bias to first order. The absolute normalisation is fit self-consistently by Step 13 MCMC.

32 Step 13 MCMC: cluster constraint

Step 13 extends the Step 12 dataset by adding the Planck PSZ2 cluster Poisson likelihood. Two exploratory chains (v2, v3) established the physical result before the final analysis:

- **v2** (ε sampled directly): collapsed to the numerical floor $\varepsilon_{\min} = 10^{-5}$ within 500 accepted steps; the reflecting boundary caused $R - 1 > 18$ indefinitely.
- **v3** ($\log_{10} \varepsilon$ parameterisation): same physical outcome after a burn-in exploring $\varepsilon \sim 0.01\text{--}0.03$; the chain transitioned to $\varepsilon \lesssim 10^{-5}$ at $N \approx 18,000$ accepted samples, yielding indicative $\varepsilon < 1.3 \times 10^{-3}$ (95% CL).

The converged final chain (**v4**) fixes $\varepsilon = 0$ and $A_{\text{fp}} = 3.0$ (both parameters degenerate in the Λ CDM limit), leaving 19 free parameters. The v4 chain ran for approximately 20 hours (2026-06-16 to 2026-06-17), reaching $R - 1 = 0.019$ (means) and $R - 1 = 0.095$ (bounds) at 20,800 accepted samples with an acceptance rate of 52.7%.

Key result: the Planck PSZ2 cluster likelihood drives $\varepsilon \rightarrow 0$ (the Λ CDM growth limit), independently confirmed by two chains with different parameterisations. This constitutes the first dataset tension in the SCM series: Step 12 (CMB+BAO+WL+DES) detects $\varepsilon > 0$ at 1.9σ , while the cluster counts constrain $\varepsilon < 1.3 \times 10^{-3}$ (95% CL).

The physical origin is transparent: the SCM’s enhanced growth rate produces +8–19% more clusters than Λ CDM at $M_{500c} = 3 \times 10^{14} M_{\odot}$. With $\varepsilon = 0$ fixing the HMF to Λ CDM growth, the posterior accommodates the observed counts via a combination of lower σ_8 (-2.2σ from Step 12) and a mass bias b pulled modestly below its prior centre.

Table 25: Step 13 final parameter constraints (v4 chain, $R - 1 = 0.019$, 20,800 accepted samples, $\varepsilon = 0$ fixed) compared to Step 12. Shifts are in units of $\sigma_{\text{comb}} = \sqrt{\sigma_{13}^2 + \sigma_{12}^2}$.

Parameter	Step 12	Step 13 ($\varepsilon = 0$)	Shift
H_0 [km s ⁻¹ Mpc ⁻¹]	69.08 ± 0.44	67.60 ± 0.32	-2.7σ
$\Omega_b h^2$	0.02237 ± 0.00013	0.02241 ± 0.00012	$+0.2\sigma$
$\Omega_c h^2$	0.12005 ± 0.00090	0.11939 ± 0.00071	-0.6σ
σ_8	0.827 ± 0.008	0.806 ± 0.005	-2.2σ
Ω_m	—	0.310	—
$S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$	0.826 ± 0.008	0.805 ± 0.005	-2.2σ
$S_{8,\text{cl}} \equiv \sigma_8(\Omega_m/0.3)^{0.3}$	—	0.814 ± 0.006	—
ε	0.00127 ± 0.00067	0 (fixed)	-1.9σ
b (hydrostatic mass bias)	—	0.188 ± 0.023	—
$(1 - b)$	—	0.812	—
$\log_{10} Y_*$	—	-0.194 ± 0.013	—

Tension with external datasets:

- $S_8 = 0.805 \pm 0.005$ vs. DES Y1 (0.782 ± 0.027): $+0.8\sigma$ (reduced from 1.6σ at Step 12).
- $S_{8,\text{cl}} = 0.814 \pm 0.006$ vs. Planck SZ PSZ2 (0.762 ± 0.025): $+2.0\sigma$.
- $S_{8,\text{cl}} = 0.814 \pm 0.006$ vs. Planck CMB prediction (0.844 ± 0.014): -2.0σ .

The $S_{8,\text{cl}}$ value sits midway between the Planck CMB prediction and the Planck SZ self-calibration, reflecting the competing pulls of the CMB (which pins σ_8 from above) and the cluster counts (which push it down via the HMF normalisation).

33 Summary of SCM VI results

1. SCM HMF exceeds Λ CDM by +8.5% / +12.9% / +18.6% at $M_{500c} = 3 \times 10^{14} M_\odot$ at $z = 0.2 / 0.5 / 1.0$, rising to +30–60% at $M > 10^{15} M_\odot$. This is a direct consequence of the ~ 1 –2% enhancement in $\sigma(M, z)$ established in SCM IV.
2. A custom Poisson cluster likelihood (`SCM_planck_sz_likelihood.py`) is implemented, validated, and integrated into the Cobaya framework. Three implementation bugs were identified and fixed during development: (a) wrong Δ_m conversion ($\Delta_m = \Delta_c/\Omega_m(z)$, not $\Delta_c \times \Omega_m(z)$); (b) Cobaya Pk interpolator unit convention (k in h^{-1} Mpc requires conversion to Mpc^{-1} before calling `PK_interp.P`, with $P_h = h^3 P_p$); (c) same Δ_m fix applied in `SCM_hmf.py`.
3. **Step 13 MCMC reveals the first dataset tension in the SCM series.** The Planck PSZ2 cluster likelihood drives $\varepsilon \rightarrow 0$ (Λ CDM growth limit), independently confirmed by two chains (v2, v3) with different parameterisations. The final converged chain (v4; 19 free parameters; $R-1 = 0.019$; 20,800 accepted samples) yields $\sigma_8 = 0.806 \pm 0.005$, $S_8 = 0.805 \pm 0.005$, $S_{8,\text{cl}} = 0.814 \pm 0.006$, and $b = 0.188 \pm 0.023$.
4. The $\varepsilon \rightarrow 0$ result has a clear physical explanation: the SCM’s enhanced HMF overpredicts cluster counts by +8–19%, and the data minimise this by driving ε to the Λ CDM limit. With ε suppressed, both σ_8 (reduced by 0.021, 2.2σ) and the mass bias ($b = 0.188$, 1.4σ below the prior centre) cooperate to reduce the theoretical count prediction.
5. $S_{8,\text{cl}} = 0.814 \pm 0.006$ sits $+2.0\sigma$ above the Planck SZ self-calibration (0.762 ± 0.025) and -2.0σ below the Planck CMB prediction (0.844 ± 0.014), while the weak-lensing $S_8 = 0.805 \pm 0.005$ is in 0.8σ agreement with DES Y1 (reduced from 1.6σ at Step 12).

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